SEVEN OPEN UNIT EQUILATERAL TRIANGLES DO NOT COVER A REGULAR UNIT HEXAGON

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ABSTRACT. We prove that the regular hexagon of side length one cannot be covered by seven open equilateral triangles of side length one. The center and the six vertices canonically label the seven covering triangles. A finite classification routes every hypothetical cover to one of three methods: trace-length deficit, area loss, or a finite equilateral-enclosure obstruction. The finite-enclosure method is organized here as a reusable local-capacity toolkit, two universal nonzero-gap terminals, exceptional T3-like and Vd placements, and a separate zero-gap nine-point theorem. Exact symbolic verification is used only for the two mixed support-arc intersections in the nine-point terminal.

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1. Introduction and Proof Outline

1.1. The theorem and the canonical labeling. Let

$$O = (0,0), \qquad V_i = \left(\cos\frac{i\pi}{3}, \sin\frac{i\pi}{3}\right), \qquad H = \operatorname{conv}\{V_0,\dots,V_5\},$$

where indices lie in $\mathbb{Z}/6\mathbb{Z}$. Write

$$e_{i,i+1} = [V_i, V_{i+1}], \qquad r_i = [O, V_i], \qquad M_i = \frac{1}{2}V_i,$$

and put

$$S = \partial H \cup \bigcup_{i=0}^{5} r_i.$$

Then

$$\mathcal{H}^1(\partial H) = 6, \qquad \mathcal{H}^1\left(\bigcup_{i=0}^{5} r_i\right) = 6, \qquad \mathcal{H}^1(S) = 12.$$

Theorem 1.1 (Main theorem)**.** *The hexagon H cannot be covered by seven open equilateral triangles of side length one.*

Corollary 1.2 (Soifer's Hexagon Problem)**.** *For every $L > 1$, the regular hexagon $H_L = LH$ cannot be covered by seven closed unit equilateral triangles.*

The seven points $O, V_0, \dots, V_5$ are pairwise at distance at least one, whereas any two points in an open unit equilateral triangle are at distance strictly less than one. Hence the seven covering triangles admit a unique labeling

$$U_C, U_0, \dots, U_5$$

such that

$$O \in U_C, \qquad V_i \in U_i.$$

We call U_C the *C triangle* and U_i the *V triangle at V_i*. Set

$$T_C = \overline{U_C}, \qquad T_i = \overline{U_i}.$$

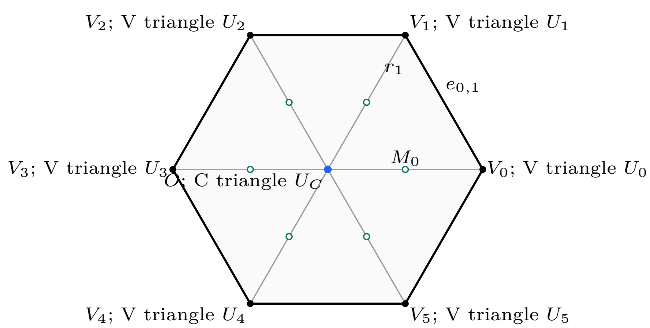
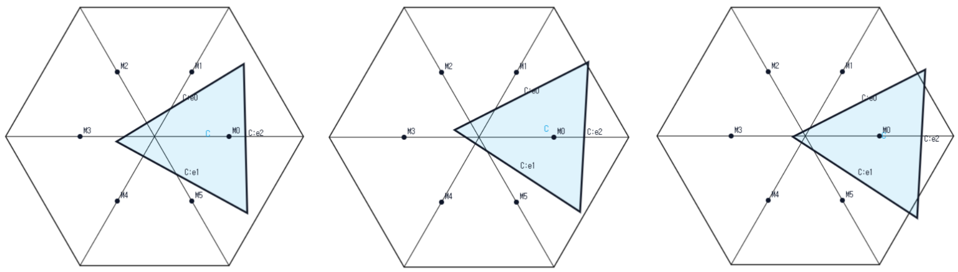


FIGURE 1. The center, vertices, radial segments, and radial midpoints of H.

1.2. **Finite classifications.** The center type is the number of boundary edges on which T_C has a positive-length trace. The only possibilities are $0, 1, 2$, denoted by $\mathrm{CE0}, \mathrm{CE1}, \mathrm{CE2}$.



(a) CE0 (b) CE1 (c) CE2

FIGURE 2. Examples of the three C triangle classes. From left to right, the C triangle has positive-length trace on zero, one, and two edges of ∂H; point contacts do not count. The screenshots are illustrative and not to scale; no proof depends on visual inspection.

For the V triangle T_i, let $o(T_i)$ be the number of its triangle vertices outside H and put

$$n(T_i) = \left|\left\{j \in \{i-1, i+1\} : \mathcal{H}^1(T_i \cap r_j) > 0\right\}\right|,$$

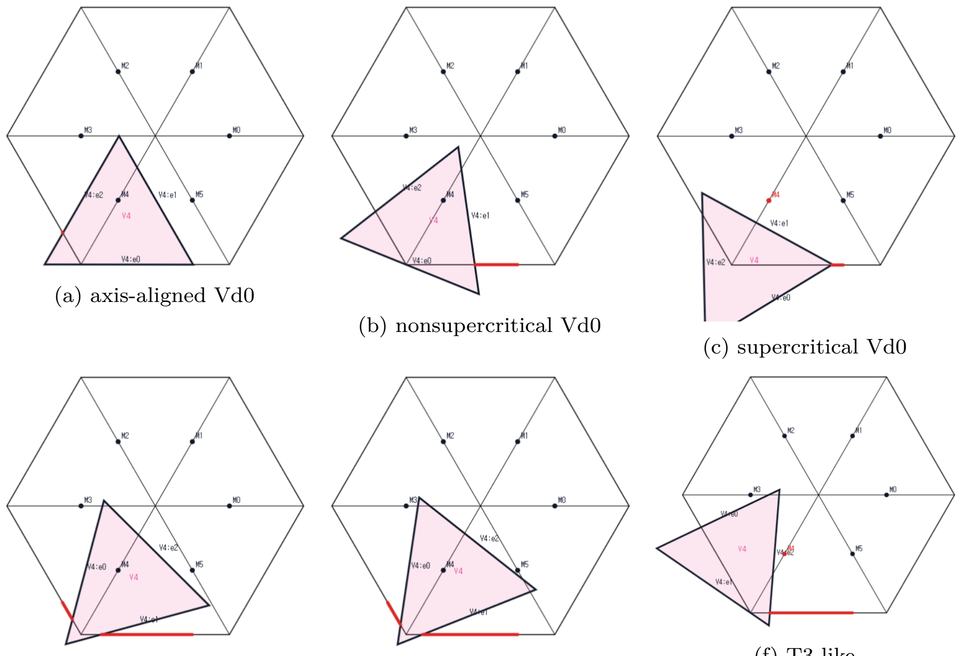
with indices modulo 6. The raw pair $(3, 0)$ says that all three vertices of T_i lie outside H; it does not say that T_i is disjoint from H, since $V_i \in \operatorname{int}(T_i) \cap H$. Proposition 2.5 replaces every such triangle by a same-orientation translate with exactly the same trace in H, still containing V_i in its interior, and with at most two vertices outside H. We make this exact-trace normalization before defining any reach. The normalized exhaustive types used below are

type	$(o(T_i), n(T_i))$
Vd0	$(1, 0), (2, 0)$
Vd1	$(1, 1)$
Vd2	$(1, 2)$
T3-like	$(2, 1)$.

Let A_i, B_i, C_i be the maximal reaches of T_i toward V_{i-1}, V_{i+1}, O, respectively. The V triangle T_i is *supercritical* when $A_i + B_i > 1$, and

$$N_+ = |\{i : A_i + B_i > 1\}|.$$

Lowercase (a_i, b_i, c_i) always denotes selected lower bounds for these reaches and never the maxima defining N_+.



(a) axis-aligned Vd0 (b) nonsupercritical Vd0 (c) supercritical Vd0

(d) Vd1 (e) Vd2 (f) T3-like

FIGURE 3. Examples of the normalized V triangle types. The top row shows an axis-aligned Vd0, a nonsupercritical Vd0 with $A_i + B_i \leq 1$, and a supercritical Vd0 with $A_i + B_i > 1$. The bottom row shows Vd1, Vd2, and T3-like. The screenshots are illustrative and not to scale; no proof depends on visual inspection.

On $e_{i,i+1}$, parametrized from V_i to V_{i+1}, the two incident open traces have the form $[0, B_i)$ and $(1 - A_{i+1}, 1]$. Their nonempty complement

$$[B_i, 1 - A_{i+1}]$$

is a *boundary gap*; equality gives a singleton gap, which remains uncovered. Let N_{gap} be the number of positive center traces that contain a boundary gap.

For the case tables, put

$$d = |\{i : T_i \text{ is Vd1 or Vd2}\}|, \qquad t = |\{i : T_i \text{ is T3-like}\}|,$$

and let $N_{\text{sp}} = d + t$. The common trace calculation proves that the number of V triangles that are either supercritical or meet an adjacent radial segment in positive length is

$$N_+ + N_{\text{sp}}.$$

In particular, every CE1/CE2 configuration with $N_+ + N_{\text{sp}} \geq 3$ is impossible by a trace-length bound on S.

1.3. **Exhaustive routing.** Lemma 2.8 proves the fundamental equivalence

$$N_{\text{gap}} = 0 \iff U_0, \ldots, U_5 \text{ cover } \partial H.$$

Thus Table 1 first dispatches every zero-gap case without reference to the center class. The nonzero-gap part is read from top to bottom; the first applicable row is the unique route for a hypothetical cover. The full classification proof is Proposition 2.14.

ID	N_{gap}	N_+	vertex-type refinement	center class	method
R0	0	0	all (d,t)	CE0/CE1/CE2	1
R1	0	1	$d \geq 1$	CE0/CE1/CE2	1
R2	0	1	$d = 0,\ t \geq 1$	CE0/CE1/CE2	2
R3	0	1	$(d,t) = (0,0)$	CE0/CE1/CE2	3
R4	0	≥ 2	all (d,t)	CE0/CE1/CE2	2
R5	≥ 1	≥ 2	all (d,t)	CE1/CE2	1
R6	≥ 1	0 or 1	$N_+ + d + t \geq 3$	CE1/CE2	1
R7	≥ 1	0	$(d,t) = (0,0)$	CE1/CE2	3
R8	≥ 1	0	$d \geq 1,\ d + t \leq 2$	CE1/CE2	1
R9	≥ 1	0	$d = 0,\ t \in \{1,2\}$	CE1/CE2	3
R10	≥ 1	1	$(d,t) = (0,0)$	CE1/CE2	3
R11	≥ 1	1	$(d,t) = (0,1)$	CE1/CE2	3
R12	≥ 1	1	$(d,t) = (1,0)$	CE1	1
R13	≥ 1	1	$(d,t) = (1,0)$	CE2	1+3

Table 1. The exhaustive routing with stable case IDs. Method 1 is a trace-length argument, Method 2 sums area loss, and Method 3 forces a finite set into the C triangle and proves an equilateral-enclosure obstruction. The nonzero-gap rows are refined into case cards NG0–NG7 in Section 5.14; the zero-gap row R3 is handled in Section 6.

1.4. **Organization of the proof.** Section 2 proves the structural classification, strict handoff selection, midpoint facts, and the signed center normal form. Section 3 gives Method 1, the trace-length argument. Section 4 gives Method 2, the local and cyclic area-loss argument.

Section 5 begins Method 3 with a reader contract, the exact local-capacity interfaces, endpoint and compactness lemmas, and the two universal nonzero-gap terminals: complementary gap and CE2 short ray. It then treats the all-Vd0, T3-like, and one-Vd placements one case card at a time and ends with a nonzero-gap dispatch table. Section 6 is a separate chapter for the center-independent nine-point obstruction in routing row R3. Section 7 closes the proof from the routing table.

Appendix A contains the scalar derivations and selector audits removed from the main narrative. Appendix B contains the nonzero-gap trace-exact envelope atlas. Appendix C contains the exact symbolic certificate for the two mixed support-arc intersections in the zero-gap theorem.

2. Structural Reduction and Common Geometry

This section supplies the complete finite reduction used by every later method. It proves the open–closed scaling equivalence, the center and vertex classifications, strict boundary handoffs, midpoint facts, and exhaustive routing. It then records the signed center normal form used in both the trace and finite-enclosure arguments. Boundary reaches are used only inside direct geometric inequalities; no global composition of reach maps is part of the proof. Each result is stated and proved once.

2.1. **Structural Reduction to Finite Cases.** Throughout, indices are taken in $\mathbb{Z}/6\mathbb{Z}$. Put

$$O = (0,0), \qquad V_i = \left(\cos\frac{i\pi}{3}, \sin\frac{i\pi}{3}\right), \qquad H = \operatorname{conv}\{V_0, \dots, V_5\}.$$

Thus H is the regular hexagon of side length one. We use the notation

$$e_{i,i+1} = [V_i, V_{i+1}], \qquad r_i = [O, V_i], \qquad M_i = \frac{1}{2}V_i,$$

and, when needed,

$$D = \bigcup_{i=0}^{5} r_i, \qquad S = \partial H \cup D, \qquad S_{1/2} = \partial H \cup \{O, M_0, \dots, M_5\}.$$

The arms r_i meet only at O, up to sets of one-dimensional measure zero, and consequently

$$\mathcal{H}^1(\partial H) = 6, \qquad \mathcal{H}^1(D) = 6, \qquad \mathcal{H}^1(S) = 12. \tag{1}$$

For $L > 0$, $H_L := LH$ denotes the concentric regular hexagon of side length L. All symmetries used below belong to the dihedral group D_6; applying a symmetry always transports the indices, orientations, and incoming and outgoing labels together.

2.1.1. *Open, closed, and scaled formulations.*

Proposition 2.1 (Compactness and scaling equivalence)**.** *The following assertions are equivalent.*

(1) *The set H is covered by seven open equilateral triangles of side length 1.*
(2) *For some $\varepsilon > 0$, the set H is covered by seven closed equilateral triangles of side length $1 - \varepsilon$.*
(3) *For some $L > 1$, the set H_L is covered by seven closed equilateral triangles of side length 1.*

Proof. Suppose that open unit triangles $U_1, \dots, U_7$ cover H. Define

$$m_j(p) = \operatorname{dist}(p, \mathbb{R}^2 \setminus U_j), \qquad m(p) = \max_{1 \le j \le 7} m_j(p).$$

The continuous function m is positive on the compact set H, so $\eta := \min_{p \in H} m(p) > 0$. Choose

$$0 < \rho < \min\left\{\eta, \frac{\sqrt{3}}{6}\right\}.$$

Moving each side of U_j inward through distance ρ gives a closed equilateral triangle K_j of side $1 - 2\sqrt{3}\rho$. Every $p \in H$ has $m_j(p) \ge \eta > \rho$ for some j, and hence belongs to K_j. This proves (1)$\Rightarrow$(2), with $\varepsilon = 2\sqrt{3}\rho$.

Conversely, enlarging a closed triangle of side $1 - \varepsilon$ about its incenter to an open triangle of side 1 preserves containment, proving (2)$\Rightarrow$(1). Finally, dilation by $(1 - \varepsilon)^{-1}$ identifies (2) with (3), with

$$L = (1 - \varepsilon)^{-1} > 1.$$

$\square$

We conduct the geometric argument in the open side-one model. Denote the original open triangles by U_C and U_i for $0 \le i \le 5$, and put

$$T_C = \overline{U_C}, \qquad T_i = \overline{U_i}.$$

The distinction between U_i and T_i will be maintained: closure may add endpoints or tangencies, although it does not change area or one-dimensional trace measure.

Lemma 2.2 (Distinct triangles)**.** *The seven triangles can be labelled so that*

$$O \in U_C, \qquad V_i \in U_i \quad (0 \le i \le 5),$$

and these seven triangles are distinct. Consequently

$$O \in \operatorname{int}(T_C), \qquad V_i \in \operatorname{int}(T_i).$$

Proof. The seven distinguished points $O, V_0, \dots, V_5$ are pairwise at distance at least 1. Any two points of an open unit equilateral triangle are at distance strictly less than its diameter 1. Thus one open triangle contains at most one distinguished point. Selecting a covering triangle for each of the seven points gives an injection from seven points to seven triangles, hence a bijection and the asserted labelling. □

2.1.2. *The center classification.* For a closed C triangle define

$$N_E(T_C) = \left|\{i : T_C \cap e_{i,i+1} \text{ contains a positive-length interval}\}\right|.$$

Point contacts, endpoint contacts, and tangencies do not contribute. We say that T_C has type CE0, CE1, or CE2 according as $N_E(T_C)$ is 0, 1, or 2.

Proposition 2.3 (Exhaustive center types)**.** *Every C triangle has exactly one of the types CE0, CE1, and CE2. Equivalently,*

$$N_E(T_C) \le 2.$$

Proof. Among any three edges of the six-cycle ∂H, two have no common endpoint. It is therefore enough to show that relative-interior points of two nonadjacent edges are more than unit distance apart. Up to symmetry there are two cases. For $0 < t, s < 1$, put

$$x = (1-t)V_0 + tV_1.$$

If $y = (1-s)V_2 + sV_3$, direct calculation gives

$$\|x-y\|^2 - 1 = (1-t)(2-t) + s(s+t) > 0.$$

If instead $y = (1-s)V_3 + sV_4$ and $u = t+s$, then

$$\|x-y\|^2 - 1 = (u-1)^2 + 2 > 0.$$

A unit equilateral triangle has diameter 1. It therefore cannot contain relative-interior points on two nonadjacent hexagon edges. Three positive-length edge traces would supply such a pair, a contradiction. □

For the CE1/CE2 types, Proposition 2.12 below records the additional fact that T_C contains exactly one radial midpoint.

2.1.3. *Vertex types and exact-trace normalizations.* For the closure T_i of an original open V triangle U_i, let

$$o(T_i) = |\{\text{vertices of } T_i \text{ outside } H\}|$$

and put

$$n(T_i) = \left|\left\{j \in \{i-1, i+1\} : \mathcal{H}^1(T_i \cap r_j) > 0\right\}\right|.$$

Lemma 2.4 (Local wedge)**.** *Every original V triangle has $1 \le o \le 3$. If $\mathcal{H}^1(T_i \cap r_j) > 0$ for some $j \in \{i-1, i+1\}$, at least one of its vertices belongs to H; if both $\mathcal{H}^1(T_i \cap r_{i-1}) > 0$ and $\mathcal{H}^1(T_i \cap r_{i+1}) > 0$, at least two of its vertices belong to H.*

Proof. Normalize to $i = 0$ and write

$$X = V_0 + x(V_1 - V_0) + y(V_5 - V_0).$$

Then $V_0 = (0, 0)$,

$$\|(x, y)\|^2 = x^2 + y^2 - xy,$$

and, within the closed unit ball about V_0, membership in H is equivalent to membership in the wedge

$$W = \{(x, y) : x \ge 0, y \ge 0\}.$$

Indeed,

$$x^2 + y^2 - xy = \left(y - \frac{x}{2}\right)^2 + \frac{3x^2}{4} = (x-y)^2 + xy$$

shows that $x, y \ge 0$ and norm at most one imply the remaining hexagon inequalities. The two relevant radial segments are

$$r_1 = \{(1, y) : 0 \le y \le 1\}, \qquad r_5 = \{(x, 1) : 0 \le x \le 1\}.$$

Suppose $T_0 \cap r_1$ has positive length, and let m be the largest x-coordinate of a vertex of T_0. If $m > 1$, a maximizing vertex (m, y) lies in the unit ball and hence satisfies $0 < y < 1$; it belongs to H. If $m = 1$, positive-length contact with the supporting line $x = 1$ forces a side of T_0 to lie there, and both endpoints satisfy $1 + y^2 - y \le 1$, hence belong to H. The same argument applies to r_5. If both r_1 and r_5 are met but only one triangle vertex belonged to H, that vertex would have to have both $x > 1$ and $y > 1$. This is impossible because $(x-y)^2 + xy > 1$.

Finally, if all three triangle vertices belonged to the convex set H, then $T_0 \subset H$. This is incompatible with the extreme point V_0 lying in the interior of T_0. Hence $o \ge 1$. $\square$

We use the following orientation normal forms for both translations below. Set

$$D_t = 1 - t + t^2, \qquad z = \sqrt{D_t}, \qquad 0 \le t \le 1.$$

After relabelling the vertices of the equilateral triangle, every orientation belongs to one of the two families

	first side vector	second side vector
I	$z^{-1}(1, t)$	$z^{-1}(1-t, 1)$
II	$z^{-1}(t, -(1-t))$	$z^{-1}(1, t)$.

In Type I write

$$U = \alpha + y - tx, \qquad V = \beta + x - (1-t)y, \qquad T = \{U \ge 0, V \ge 0, U + V \le z\}, \tag{2}$$

and in Type II write

$$U = \alpha + (1-t)x + ty, \qquad V = \beta + tx - y, \qquad T = \{U \ge 0, V \ge 0, U + V \le z\}. \tag{3}$$

In either form, $V_0 \in \operatorname{int}(T)$ is equivalent to

$$\alpha > 0, \qquad \beta > 0, \qquad \alpha + \beta < z. \tag{4}$$

For later use, the Type-II vertex in the wedge is

$$Q = \left(\frac{z - \alpha - t\beta}{D_t}, \frac{t(z-\alpha) + (1-t)\beta}{D_t} \right). \tag{5}$$

Both coordinates are positive under (4).

Proposition 2.5 (Exact-trace normalization of raw $(3,0)$ roles)**.** *Let U be an original open V_i-triangle, put $T = \overline{U}$, and suppose $(o(T), n(T)) = (3,0)$. There is a translate $\widehat{U}$ of U, with closure $\widehat{T}$, the same orientation, and the same side length, such that*

$$V_i \in \widehat{U}, \qquad (o(\widehat{T}), n(\widehat{T})) = (2,0), \tag{6}$$

and both the closed and open traces on the hexagon are unchanged:

$$T \cap H = \widehat{T} \cap H, \qquad U \cap H = \widehat{U} \cap H. \tag{7}$$

Consequently the three maximal reaches on the two incident edges and the own radial segment are unchanged. In the notation introduced below, the values A_i, B_i, C_i, the inequality $A_i + B_i > 1$, and hence N_+ are all preserved.

Proof. Normalize to $i = 0$ and use the common forms above. A Type-II triangle cannot have $o = 3$: the point Q in (5) lies in W, and because T contains the origin and has diameter one, the local wedge lemma puts Q in H. Thus T has Type I form.

The three Type-I vertices are

$$\begin{aligned} Q_0 &= \left(-\frac{(1-t)\alpha + \beta}{D_t}, -\frac{\alpha + t\beta}{D_t} \right), \\ Q_1 &= \left(\frac{z - (1-t)\alpha - \beta}{D_t}, \frac{tz - \alpha - t\beta}{D_t} \right), \\ Q_2 &= \left(\frac{(1-t)z - (1-t)\alpha - \beta}{D_t}, \frac{z - \alpha - t\beta}{D_t} \right). \end{aligned} \tag{8}$$

The vertex Q_0 has two negative coordinates, the first coordinate of Q_1 is positive, and the second coordinate of Q_2 is positive. Hence $o = 3$ is equivalent to

$$\alpha + t\beta > tz, \qquad (1-t)\alpha + \beta > (1-t)z. \tag{9}$$

These strict inequalities force $0 < t < 1$.

Put

$$s = \alpha + \beta, \qquad L = z - s > 0.$$

The two inequalities in (9) become

$$\alpha > \frac{tL}{1-t}, \qquad \beta > \frac{(1-t)L}{t}. \tag{10}$$

For every $(x, y) \in W$ satisfying

$$(1-t)x + ty \le L, \tag{11}$$

the minimum of U is attained at $(L/(1-t), 0)$ and the minimum of V at $(0, L/t)$. Thus (10) makes both U and V strictly positive throughout the triangle in (11). It follows that

$$T \cap W = \{(x, y) \in W : (1-t)x + ty \le L\} . \tag{12}$$

Keep t and s fixed and replace the affine constants by

$$\widehat{\alpha} = \frac{tL}{1-t}, \qquad \widehat{\beta} = s - \widehat{\alpha}. \tag{13}$$

The first inequality in (10) gives $0 < \widehat{\alpha} < \alpha$ and $\widehat{\beta} > \beta > 0$; also $\widehat{\alpha} + \widehat{\beta} = s < z$. Since the linear map defining U, V in (2) is nonsingular, changing only these affine constants gives a translate $\widehat{T}$ of T. In particular, (4) shows that the origin remains in $\mathrm{int}(\widehat{T})$.

For the translated triangle, the inequality $\widehat{U} + \widehat{V} \le z$ is still exactly (11). On that triangle $\widehat{U} \ge 0$, with equality only at $(L/(1-t), 0)$, while $\widehat{V} > 0$ by the second inequality in (10) and $\widehat{\beta} > \beta$. Therefore

$$\widehat{T} \cap W = T \cap W,$$

and their interiors have the same trace on W: the only new equality point of $\widehat{U}$ already lies on the common boundary $(1-t)x + ty = L$. Both triangles contain the origin and have diameter one, so the local wedge lemma identifies their intersections with H with their intersections with W. This proves both equalities in (7).

Equation (13) makes the second coordinate of Q_1 in (8) zero. Thus $Q_1 \in H$. The vertex Q_0 remains outside H, and the strict second inequality in (10), strengthened by $\widehat{\beta} > \beta$, keeps Q_2 outside H. Hence $o(\widehat{T}) = 2$. The local wedge lemma gives $n(T) = 0$ from $o(T) = 3$, and the exact trace equality preserves n, so $n(\widehat{T}) = 0$. Finally, all three reach segments lie in H, and (7) preserves their exact open and closed traces. This proves every assertion. □

Apply Proposition 2.5 simultaneously to every raw $(3, 0)$ vertex role and then reuse the symbols U_i, T_i for the translated roles. The rolewise equalities (7) preserve the entire open cover, every closed trace, the condition $V_i \in U_i$, all actual reaches, and N_+. This simultaneous exact-trace normalization is the convention throughout the rest of the paper.

For the normalized roles, we use the names

type	(o, n)
Vd0	$(1, 0), (2, 0)$
Vd1	$(1, 1)$
Vd2	$(1, 2)$
T3-like	$(2, 1)$.

Proposition 2.6 (Exhaustive normalized vertex types)**.** *Every normalized V triangle has exactly one of the four types Vd0, Vd1, Vd2, and T3-like.*

Proof. We have $n \in \{0, 1, 2\}$. Lemma 2.4 gives $o \le 2$ when $n \ge 1$, and $o \le 1$ when $n = 2$. Together with $o \ge 1$, the possibilities for a raw role are

n	possible o
0	$1, 2, 3$
1	$1, 2$
2	1.

The exact-trace normalization replaces the sole extra possibility $(3,0)$ by $(2,0)$ and leaves all other pairs unchanged. The remaining pairs are precisely the four named classes. $\square$

The next normalization is useful for trace estimates, but its final warning is essential.

Proposition 2.7 (T3-like closed-trace domination). *Let T be the closure of an original T3-like V_i-triangle. There is a translate $\widehat{T}$ of T, with the same orientation and side length, such that*

$$V_i \in \mathrm{relint}(\textit{a side of } \widehat{T}), \qquad T \cap H \subsetneq \widehat{T} \cap H,$$

and $\widehat{T}$ remains of type $(o,n) = (2,1)$. Every old open interior point in $H \setminus \{V_i\}$ remains an interior point of $\widehat{T}$. The point V_i itself becomes a boundary point, so $\widehat{T}$ is a closed-trace majorant and is not a replacement open triangle.

Proof. Use the wedge coordinates of Lemma 2.4 and the common orientation normal forms above. In Type I, direct inversion shows that if exactly two triangle vertices are outside W, the sole wedge vertex has both coordinates strictly below 1: in the two possible cases its coordinates are bounded respectively by $(1-t)/z$ and t/z, with the endpoint strictness supplied by $\alpha > 0$ or $\beta > 0$. The support argument in Lemma 2.4 then excludes positive-length contact with either $x = 1$ or $y = 1$. Thus Type I cannot be T3-like.

The triangle therefore has Type II form. Reflecting in $x = y$ if necessary, assume $\mathcal{H}^1(T_0 \cap r_1) > 0$. The two outside vertices have respectively a negative x- and a negative y-coordinate, while the unique wedge vertex is the point Q in (5). The endpoint cases $t = 0, 1$ give no positive adjacent interval, so $0 < t < 1$. The exact axis reaches are

$$A = \beta, \qquad B = z - \alpha - \beta,$$

and T3-like support is equivalent to

$$Q_x > 1, \qquad Q_y \le 1.$$

Define $\widehat{T}$ by replacing α with 0 and keeping t, β fixed. Since the linear map $(x,y) \mapsto (U,V)$ is nonsingular, this changes only the translation. At the origin, $\widehat{U} = 0$ and $0 < \widehat{V} = \beta < z$, so V_0 is in the relative interior of a side. For $(x,y) \in W$,

$$\widehat{U} = (1-t)x + ty \ge 0, \qquad \widehat{V} = V, \qquad \widehat{U} + \widehat{V} = U + V - \alpha.$$

Thus $T \cap W \subset \widehat{T} \cap W$. The old x-axis reach is B and the new one is $B + \alpha$; because $B < 1$ and $\alpha > 0$, the inclusion is strict inside H.

The two outside vertices remain outside, whereas V_0 and the new wedge vertex lie in H, so $o = 2$. Moreover $\widehat{Q}_x > Q_x > 1$. From $Q_x > 1$ one gets $t\beta < z - \alpha - D_t$, and hence

$$tD_t(1 - \widehat{Q}_y) > D_t(1 - z) + (1-t)\alpha > 0.$$

Thus $\widehat{Q}_y < 1$ and $n = 1$. Finally, for every nonzero point of W, $\widehat{U} > 0$; the displayed slack relations show that every old interior point other than the origin remains interior. $\square$

2.1.4. *Reach coordinates, supercritical V triangles, and gaps.* For the V triangle U_i under the simultaneous normalization convention, define its maximal incoming, outgoing, and radial reaches by

$$\begin{aligned} A(U_i) &= \sup_{\substack{t\in[0,1] \\ V_i+t(V_{i-1}-V_i)\in U_i}} t, \\ B(U_i) &= \sup_{\substack{t\in[0,1] \\ V_i+t(V_{i+1}-V_i)\in U_i}} t, \\ C(U_i) &= \sup_{\substack{t\in[0,1] \\ V_i+t(O-V_i)\in U_i}} t. \end{aligned}$$

Closure does not change the suprema, so

$$A(U_i) = A(T_i), \qquad B(U_i) = B(T_i), \qquad C(U_i) = C(T_i).$$

For compact indexed formulas, put

$$A_i := A(U_i) = A(T_i), \qquad B_i := B(U_i) = B(T_i), \qquad C_i := C(U_i) = C(T_i).$$

The closure T_i contains the corresponding endpoints. We reserve lowercase (a_i, b_i, c_i) for selected or prescribed reach lower bounds whenever actual and selected values occur together. A realizing triangle then satisfies

$$A_i \geq a_i, \qquad B_i \geq b_i, \qquad C_i \geq c_i.$$

An actual V triangle is supercritical if $A_i + B_i > 1$, and

$$N_+ = |\{i : A_i + B_i > 1\}|\,. \tag{14}$$

This definition never counts a smaller reach pair in place of the actual V triangle.

Parametrize $e_{i,i+1}$ by

$$X_i(x) = V_i + x(V_{i+1} - V_i), \qquad 0 \leq x \leq 1.$$

The two incident open traces are

$$U_i \cap e_{i,i+1} = [0, B_i), \qquad U_{i+1} \cap e_{i,i+1} = (1 - A_{i+1}, 1]$$

in this coordinate. A *boundary gap* on this edge is the nonempty closed complement between these two traces,

$$[B_i, 1 - A_{i+1}] \qquad \text{when } B_i \leq 1 - A_{i+1}.$$

In particular, equality gives a singleton gap. This convention retains the actual open traces; a point missing from both open triangles is not erased merely because both closures contain it.

Lemma 2.8 (Exhaustiveness of the gap split)**.** *On each edge the complement of the two incident V triangle traces is either empty or one closed interval; a singleton interval counts as a gap. Every edge having a gap has positive-length intersection with the open center triangle. Consequently a* CE0 *center permits no gaps, a* CE1 *center permits at most one, and a* CE2 *center permits at most two. Moreover, under a hypothetical cover,*

$$N_{\text{gap}} = 0 \quad \Longleftrightarrow \quad U_0, \ldots, U_5 \text{ cover } \partial H.$$

Proof. The two incident traces are intervals issuing from the two endpoints, so their complement is exactly the interval displayed above when nonempty. A nonincident V triangle cannot contain a relative-interior point of the edge: that point is more than unit distance from the triangle's distinguished vertex. Thus every point of a gap lies in U_C. Because U_C is open, its intersection with that edge contains a relative neighborhood and hence has positive length. Different gap edges therefore give different edges counted by $N_E(T_C)$. Proposition 2.3 gives the three stated bounds, including singleton gaps. Since every boundary gap lies in a positive center trace, $N_{\text{gap}} = 0$ is equivalent to the absence of all boundary gaps, which is exactly coverage of ∂H by the six open V triangle traces. □

Lemma 2.9 (T3-like V triangles are nonsupercritical)**.** *Every original T3-like V triangle satisfies*

$$A_i + B_i \leq 1. \tag{15}$$

In particular, it is not counted by N_+.

Proof. The proof of Proposition 2.7 showed that a T3-like triangle has Type II orientation with

$$A_i = \beta, \qquad B_i = z - \alpha - \beta, \qquad z = \sqrt{1 - t + t^2} \leq 1.$$

Thus $A_i + B_i = z - \alpha \leq 1$. The same conclusion holds after reflection. □

We state the precise passage from maximal reaches to strict handoff lower bounds. Its hypothesis that the six V triangles themselves cover the boundary is part of the statement.

Proposition 2.10 (Strict boundary handoffs)**.** *Assume that $U_0, \dots, U_5$ cover ∂H. On each edge choose*

$$\xi_i \in (1 - A_{i+1}, B_i)$$

and define

$$a_i = 1 - \xi_{i-1}, \qquad b_i = \xi_i.$$

Then $0 < a_i < A_i$, $0 < b_i < B_i$, the two selected anchors lie in U_i, and

$$a_i^2 + a_i b_i + b_i^2 < 1.$$

Moreover:

1. *if exactly one actual V triangle, indexed by p, is supercritical, every strict selection has exactly one selected supercritical V triangle, also indexed by p;*
2. *if at least two actual V triangles are supercritical, some simultaneous strict selection has at least two selected supercritical V triangles.*

Proof. Because V_i is interior to a diameter-one closed triangle,

$$0 < A_i < 1, \qquad 0 < B_i < 1.$$

No nonincident V triangle contains a relative-interior point of $e_{i,i+1}$, since such a point is more than unit distance from its distinguished vertex. The two displayed incident traces therefore cover the edge and, being relatively open and nonempty, overlap. Hence each interval $(1 - A_{i+1}, B_i)$ is nonempty. The anchor and distance assertions follow from the choice and from the fact that two points of U_i are at distance strictly less than one.

Put $\ell_i = 1 - A_{i+1}$ and $u_i = B_i$. Then

$$A_i + B_i > 1 \iff \ell_{i-1} < u_i, \qquad a_i + b_i > 1 \iff \xi_{i-1} < \xi_i. \tag{16}$$

If actual V triangle i is nonsupercritical, then

$$\xi_i < u_i \leq \ell_{i-1} < \xi_{i-1},$$

so every selected version of it is strictly subcritical. Also

$$\sum_{i=0}^{5}(a_i + b_i) = \sum_{i=0}^{5}(1 - \xi_{i-1} + \xi_i) = 6. \tag{17}$$

If p is the sole actual supercritical index, the other five selected sums are strictly below one; equation (17) forces the pth sum above one. This proves (1).

For (2), first take two nonadjacent supercritical indices p, q. For each $r \in \{p, q\}$ choose

$$\ell_{r-1} < z_r < u_r,$$

then choose

$$\xi_{r-1} \in (\ell_{r-1}, \min\{u_{r-1}, z_r\}), \quad \xi_r \in (\max\{\ell_r, z_r\}, u_r).$$

The two pairs of variables are disjoint and give ascents at p, q. Any set of at least three indices on a six-cycle contains two nonadjacent indices. It remains to consider exactly two adjacent supercritical indices $p, p{+}1$. For the other four nonsupercritical V triangles, successive inequalities $\ell_i < u_i \leq \ell_{i-1}$ give

$$\ell_{p-1} < \ell_{p+1}.$$

Together with supercriticality at $p, p+1$, this implies

$$\max\{\ell_{p-1}, \ell_p\} < \min\{u_p, u_{p+1}\}.$$

Choose ξ_p between these bounds and then choose $\xi_{p-1} < \xi_p < \xi_{p+1}$ in their respective overlap intervals. By (16), both T_p and T_{p+1} are selected supercritical. □

2.1.5. *Midpoint forcing.* We first need a local fact for V triangles. Take unit vectors

$$u = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \qquad v = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right).$$

Lemma 2.11 (Self-midpoint obstruction). *Let a closed unit triangle contain*

$$0, \qquad au, \qquad bv, \qquad \frac{u+v}{2}.$$

Then $a + b \leq 1$. Consequently, for every V triangle,

$$M_i \in T_i \quad \Longrightarrow \quad A_i + B_i \leq 1.$$

If the two demanded boundary points and the midpoint are contained with a positive margin, the conclusion is strict.

Proof. Suppose $a + b > 1$ and set

$$K = \operatorname{conv}\left\{0, au, \frac{u+v}{2}, bv\right\}.$$

For outward unit normals n_0, n_1, n_2 separated by $120°$, the least equilateral triangle with those normals and containing K has side length

$$\frac{2}{\sqrt{3}} \sum_{j=0}^{2} h_K(n_j).$$

The finite-caliper theorem, Theorem 6.1, reduces its minimum to an outward normal of one of the four edges of K.

For the edge from 0 to au, direct projection gives

$$S_{0A} = \max\left\{\frac{\sqrt{3}}{4}, \frac{\sqrt{3}a}{2}\right\} + \frac{\sqrt{3}b}{2} > \frac{\sqrt{3}}{2};$$

if $a \geq 1/2$ this is $(\sqrt{3}/2)(a+b)$, and if $a < 1/2$ then $b > 1/2$. The edge through bv is symmetric. For the edge from au to $(u+v)/2$, put $D_a = \sqrt{4a^2 - 2a + 1}$. Projection gives

$$S_{AM} = \begin{cases} \dfrac{\sqrt{3}(a+b)}{2D_a}, & a \leq \frac{1}{2}, \\ \dfrac{\sqrt{3}(2a^2+b)}{2D_a}, & a > \frac{1}{2}. \end{cases}$$

The first value is strictly above $\sqrt{3}/2$ because $D_a \leq 1$. In the second case, $b > 1-a$ and

$$(2a^2 - a + 1)^2 - D_a^2 = a^2(2a-1)^2 \geq 0,$$

so again $S_{AM} > \sqrt{3}/2$. The fourth edge is symmetric. Every enclosing equilateral triangle therefore has side greater than one, a contradiction.

If equality $a + b = 1$ occurred while all relevant points had positive margin, both boundary reach lower bounds could be increased slightly; the closed conclusion would then be violated. $\square$

The corresponding center fact is exact and uses the interior condition at O.

Proposition 2.12 (CE1/CE2 exactly-one-midpoint property)**.** *Let T_C be a closed unit equilateral triangle with $O \in \operatorname{int}(T_C)$. If T_C is CE1 or CE2, equivalently if it has positive-length intersection with a boundary edge of H, then*

$$|T_C \cap \{M_0, \dots, M_5\}| = 1.$$

Proof. Normalize the positive trace by a dihedral symmetry. Proposition 2.15 gives a single signed side-slack model for both CE1 and CE2 and proves directly that its midpoint set is $\{M_0\}$. Undoing the normalization proves the assertion. $\square$

Remark 2.13. The hypothesis $O \in \operatorname{int}(T_C)$ cannot be weakened to $O \in T_C$: the triangle $\operatorname{conv}\{O, V_0, V_1\}$ has a boundary-edge overlap and contains both M_0 and M_1.

Proposition 2.14 (Exhaustive structural reduction)**.** *Every hypothetical cover by seven open unit equilateral triangles admits the distinct center and V triangles used in Table 1, and its maximal reaches determine exactly one row of that table.*

Proof. Lemma 2.2 gives the seven triangles. Propositions 2.3 and 2.6 give the exhaustive and mutually exclusive center and vertex types. The maximal reaches determine N_+, while the actual open boundary traces determine N_{gap}.

First suppose $N_{\mathrm{gap}} = 0$. Lemma 2.8 makes the six V triangles boundary-complete, independently of the center class. If $N_+ = 0$ or $N_+ \geq 2$, the first or fifth zero-gap row applies. If $N_+ = 1$, the vertex classification gives exactly one of $d \geq 1$, $d = 0, t \geq 1$, and $(d,t) = (0,0)$. These are the remaining three zero-gap rows.

Now suppose $N_{\mathrm{gap}} \geq 1$. Lemma 2.8 forces the center to be CE1 or CE2, and Proposition 2.12 gives its unique radial midpoint. If $N_+ \geq 2$, Lemma 3.5 supplies a distinct positive-support V triangle, so $N_{\mathrm{sp}} \geq 1$ and Theorem 3.14 applies. For $N_+ \in \{0,1\}$, the same theorem removes $N_+ + d + t \geq 3$. Every survivor therefore has $N_+ = 0$ with $d + t \leq 2$, or $N_+ = 1$ with $d + t \leq 1$. In the former case the exhaustive alternatives are $(d,t) = (0,0)$, $d \geq 1$, and $d = 0$ with $t \in \{1,2\}$. In the latter case they are $(0,0)$, $(0,1)$, and $(1,0)$. The final alternative separates by center type: CE1 is the length row and CE2 is the hybrid placement row. These gap-first alternatives are disjoint and exhaustive. □

2.2. **Signed CE1/CE2 Reductions.** This section gives the signed center model used by both later strategies. Its boundary traces and six radial exits are inserted directly into the finite-witness arguments of Section 5.

2.2.1. *One signed center normal form.*

Proposition 2.15 (Signed CE1/CE2 center normal form)**.** *Let T_C be a closed unit equilateral triangle with $O \in \operatorname{int}(T_C)$ and a positive-length boundary trace. Normalize such a trace to $\mathrm{e}_{0,1}$ and use affine coordinates*

$$X = V_0 + b(V_1 - V_0) + a(V_5 - V_0).$$

There are parameters

$$0 < R < 1, \qquad W = 1 - R,$$
$$E = \sqrt{1 - RW}, \qquad \eta = 1 - E, \qquad P = E(1 - E),$$

and nonnegative center slacks α, δ such that, with

$$k = \eta + \alpha + \delta,$$

the C triangle is

$$\boxed{\begin{aligned} F_0 &= R + \alpha - a + Wb, \\ F_1 &= Rb + Wa - k, \\ F_2 &= W + \delta - b + Ra, \end{aligned}} \qquad T_C = \{F_0, F_1, F_2 \geq 0\}. \tag{18}$$

Define

$$\Delta_R = P - \alpha - W\delta, \qquad \Delta_L = P - R\alpha - \delta. \tag{19}$$

Then

$$T_C \cap \mathrm{e}_{0,1} = \left[\frac{k}{R}, W + \delta\right], \qquad \mathcal{H}^1(T_C \cap \mathrm{e}_{0,1}) = \frac{\Delta_R}{R}, \tag{20}$$

and the possible companion trace is

$$T_C \cap \mathrm{e}_{5,0} = \left[\frac{k}{W}, R + \alpha\right] \quad \text{when } \Delta_L > 0, \qquad \mathcal{H}^1(T_C \cap \mathrm{e}_{5,0}) = \frac{[\Delta_L]_+}{W}. \tag{21}$$

The normalization requires $\Delta_R > 0$, and

$$T_C \text{ is CE1} \iff \Delta_L \leq 0, \qquad T_C \text{ is CE2} \iff \Delta_L > 0. \tag{22}$$

The six center exits, measured from O, are

$$\boxed{\begin{aligned} &d_0^C = E - \alpha - \delta, && d_1^C = \frac{\delta}{R}, && d_2^C = \delta, \\ &d_3^C = \min\left\{\frac{\alpha}{R}, \frac{\delta}{W}\right\}, && d_4^C = \alpha, && d_5^C = \frac{\alpha}{W}. \end{aligned}} \tag{23}$$

Moreover,

$$T_C \cap \{M_0, \ldots, M_5\} = \{M_0\}. \tag{24}$$

For closures of original open C triangles one has $\alpha, \delta > 0$.

Proof. Write

$$T_C \cap e_{0,1} = [s, t], \qquad 0 \leq s < t \leq 1.$$

The two sides through these endpoints have a positive slope ratio. Reflect across the perpendicular bisector of $e_{0,1}$ if necessary and call the ratio $R \leq 1$; put $W = 1 - R$. Direct equations for the three side lines give

$$F_1 = Rb + Wa - Rs, \qquad F_2 = -b + Ra + t, \qquad F_0 = Wb - a + E + Rs - t,$$

where $E = \sqrt{1 - RW}$ is forced by the unit side-length relation. If $R = 1$, then $F_0(O) = s - t < 0$, contrary to $O \in \operatorname{int}(T_C)$; hence $0 < R < 1$.

Set

$$\alpha = F_0(O), \qquad \delta = F_2(O).$$

Then

$$t = W + \delta, \qquad Rs = \eta + \alpha + \delta = k,$$

and substitution gives (18). The gradients are the three unit-equilateral side-normal directions and

$$F_0 + F_1 + F_2 = E,$$

so the three inequalities define the same unit equilateral triangle.

On $e_{0,1}$ one has $a = 0$. The active inequalities are

$$Rb \geq k, \qquad b \leq W + \delta,$$

which gives (20); its length is

$$W + \delta - \frac{k}{R} = \frac{P - \alpha - W\delta}{R}.$$

On $e_{5,0}$ one has $b = 0$, and the active inequalities are

$$Wa \geq k, \qquad a \leq R + \alpha.$$

Their signed interval length is

$$R + \alpha - \frac{k}{W} = \frac{P - R\alpha - \delta}{W},$$

which proves (21).

The positive right trace gives

$$\delta < \frac{P}{W} = \frac{ER}{1 + E} < \frac{R}{2}.$$

On $e_{1,2}$, parameterized by $b = 1 + q$, $a = q$, the slack F_2 is

$$\delta - R - Wq < 0.$$

Thus a second positive trace cannot occur on $e_{1,2}$. The CE classification says that a C triangle has at most two positive boundary traces and that two such traces are adjacent. Hence $e_{5,0}$ is the only possible companion edge, and its signed length proves (22). Equality $\Delta_L = 0$ is only a point contact and is therefore CE1.

For the radial exits, substitute the six ray parametrizations in (18). On r_1 the decreasing slack is $\delta - Rq$; on r_2 it is $\delta - q$; on r_3 the two decreasing slacks are $\alpha - Rq$ and $\delta - Wq$. Reflection gives r_4, r_5, and on r_0 the decreasing slack is $E - \alpha - \delta - q$. This proves (23).

Finally,

$$W(\alpha + \delta) \le \alpha + W\delta < P,$$

so

$$E - \alpha - \delta > E - \frac{P}{W} = \frac{E(E-R)}{W} > \frac{1}{2},$$

where the last inequality is equivalent to $2E(E-R) - W = (E-R)^2 > 0$. Together with

$$\alpha < P < \min\left\{\frac{R}{2}, \frac{W}{2}\right\}, \qquad \delta < \frac{R}{2},$$

the exits in (23) satisfy

$$d_0^C > \frac{1}{2}, \qquad d_i^C < \frac{1}{2} \quad (i = 1, \dots, 5).$$

Thus the exact midpoint set is $\{M_0\}$. For the closure of an original open C triangle, all three center slacks are positive; hence $\alpha, \delta > 0$. □

Remark 2.16 (Affine-chart variables). If $\kappa_j = F_j(O)$ denotes the auxiliary side slack in the CE1 affine chart, take

$$\lambda = R, \qquad s = \frac{k}{R}, \qquad t = W + \delta, \qquad \kappa_0 = \alpha, \qquad \kappa_2 = \delta.$$

For CE2, take

$$x = \frac{k}{W}, \qquad \nu = R + \alpha, \qquad y = \frac{k}{R}, \qquad \upsilon = W + \delta.$$

If $S = x + y$ and $D = \sqrt{x^2 + xy + y^2}$, then

$$S = \frac{k}{RW}, \qquad D = \frac{Ek}{RW},$$

and the old coupling equation is

$$(\nu + \upsilon)S - xy = D.$$

Thus the CE1 and CE2 formula lists are two coordinate readings of Proposition 2.15.

Lemma 2.17 (CE2 total slack)**.** *In the CE2 sign range, with $T = \alpha + \delta$,*

$$T < \frac{\eta}{E}. \tag{25}$$

Proof. The CE2 inequalities are

$$\alpha + W\delta < P, \qquad R\alpha + \delta < P.$$

Multiply them by W and R and add. Since $W + R^2 = W^2 + R = E^2$, this gives $E^2 T < P = E\eta$. □

2.2.2. *Boundary contribution and diameter transfer.*

Corollary 2.18 (Common center-boundary formula)**.** *The full center-boundary contribution is*

$$L_{\partial H}(T_C) = \frac{\Delta_R}{R} + \frac{[\Delta_L]_+}{W}. \tag{26}$$

In CE1,

$$L_{\partial H}(T_C) \le P \le \frac{\sqrt{3}}{2} - \frac{3}{4}.$$

In CE2,

$$L_{\partial H}(T_C) = \frac{E}{1+E} - \frac{E^2(\alpha+\delta)}{RW} < \frac{1}{2}.$$

Proof. Equation (26) is the sum of (20) and (21). If $\Delta_L \le 0$, then $R\alpha + \delta \ge P$. Multiplying by W and using $\alpha \ge RW\alpha$ gives

$$\alpha + W\delta \ge WP.$$

Thus $\Delta_R \le RP$, so the contribution is at most P. Since $E \ge \sqrt{3}/2$ and $E(1-E)$ decreases on this interval, the CE1 cap follows.

If $\Delta_L > 0$, direct addition gives

$$\begin{aligned} L_{\partial H}(T_C) &= \frac{P-\alpha-W\delta}{R} + \frac{P-R\alpha-\delta}{W} \\ &= \frac{P-E^2(\alpha+\delta)}{RW}. \end{aligned}$$

Here $W + R^2 = W^2 + R = E^2$ and $P/(RW) = E/(1+E)$, so the last expression is strictly below $1/2$. $\square$

Lemma 2.19 (Adjacent-edge diameter transfer)**.** *For $0 \le q \le 1$, the zero-radial forward envelope is*

$$M_0(q) = \frac{-q + \sqrt{4-3q^2}}{2}.$$

Here the argument-free symbol M_0 continues to denote the radial midpoint; $M_0(q)$ denotes the envelope map. If a unit equilateral triangle reaches parameters q, b on two adjacent boundary edges, then

$$b \le M_0(q).$$

The map M_0 is strictly decreasing. For $q > 0$,

$$M_0(q) < 1 - \frac{q}{2}, \qquad 1 - M_0(q) > \frac{q}{2}.$$

Proof. The squared distance between the two boundary points is $q^2 + qb + b^2$, so it is at most one. Solving the quadratic in b gives the first assertion. Differentiating the displayed formula gives $M_0'(q) = (-1 - 3q/\sqrt{4-3q^2})/2 < 0$. For $q > 0$, $\sqrt{4-3q^2} < 2$, which gives both strict linear comparisons. $\square$

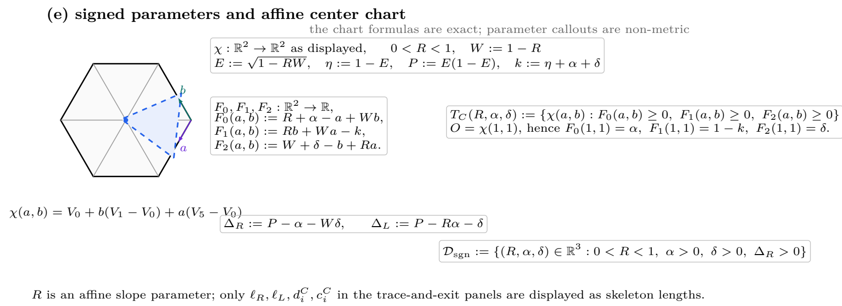
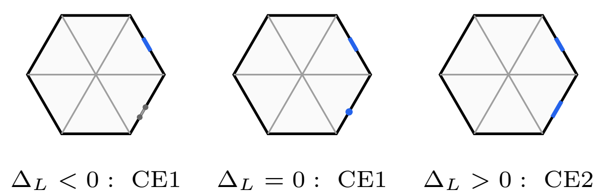


FIGURE 4. The affine center chart and the exact dependency structure of the signed parameters.

(f) companion-surplus sign on $e_{5,0}$



candidate endpoint coordinates on $e_{5,0}$ from V_0: k/W, $R + \alpha$

$$\ell_L = \frac{[\Delta_L]_+}{W}, \qquad \ell_R = \frac{\Delta_R}{R} > 0 \text{ on the normalized active edge } e_{0,1}.$$

FIGURE 5. The exact sign test for the candidate companion trace, including the CE1 point-contact case.

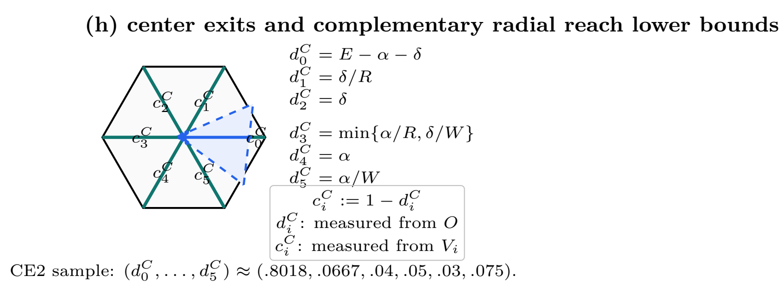
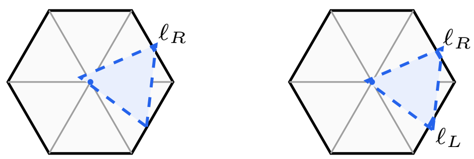


FIGURE 6. The six exact center exits d_i^C and complementary radial lower bounds $c_i^C = 1 - d_i^C$.

(i) boundary-contribution bounds on the skeleton



CE1 : $L_{\partial H}(T_C) = \ell_R \le P \le \frac{\sqrt{3}}{2} - \frac{3}{4}$

CE2 : $L_{\partial H}(T_C) = \ell_R + \ell_L < \frac{1}{2}$

FIGURE 7. The exact CE1 and CE2 center-boundary contributions and their general bounds.

3. Trace-Length Bounds

The first mechanism compares the total one-dimensional trace available from the seven triangles with the length of either the perimeter ∂H or the full skeleton S. The local trace estimates, the master perimeter deficit, the conditional skeleton theorem, and every routed application are proved in this section.

3.1. Detailed Trace-Length Estimates.

This section develops the boundary and conditional-skeleton trace bounds used by the length mechanism.

For a closed triangle T and a rectifiable target E, write

$$L_E(T) = \mathcal{H}^1(T \cap E).$$

The passage from an original open triangle U to its closure $T = \overline{U}$ can only enlarge the trace. Hence a cover of E by the open triangles implies

$$\mathcal{H}^1(E) \le L_E(T_C) + \sum_{i=0}^{5} L_E(T_i). \tag{27}$$

All estimates in this section concern full traces. They therefore remain valid for any assigned measurable subsets of those traces.

The strategy uses only the perimeter ∂H and the full skeleton S, whose measures were recorded in (1). Every skeleton estimate below is conditional on its stated center and V triangle hypotheses; no unconditional noncoverage assertion for S is made.

FIGURE 8. The two one-dimensional targets. The full skeleton is used only under the hypotheses of the corresponding trace budget.

3.1.1. *Conditional skeleton trace caps.* We now develop the skeleton estimate used only when the center is CE1 or CE2. The first lemma is the center contribution.

Lemma 3.1 (Center skeleton cap)**.** *If T_C is a CE1 or CE2 C triangle and contains exactly one radial midpoint, then*

$$L_S(T_C) < \frac{3}{2}.$$

Proof. Normalize the unique midpoint as M_0 and the positive boundary trace to $e_{0,1}$. Use the signed variables of Proposition 2.15; in particular

$$W = 1 - R, \qquad E = \sqrt{1 - RW}, \qquad \eta = 1 - E, \qquad P = E(1 - E).$$

The positive trace condition is

$$\alpha + W\delta < P. \tag{28}$$

The two possible boundary contributions are

$$\ell_{01} = W + \delta - \frac{\eta + \alpha + \delta}{R}, \qquad \ell_{50} = \frac{[P - (R\alpha + \delta)]_+}{W}.$$

The six radial contributions are bounded by

i	0	1	2	3	4	5
$\mathcal{H}^1(T_C \cap r_i)$	$E - \alpha - \delta$	δ/R	δ	$\min\{\alpha/R, \delta/W\}$	α	$\alpha/W.$

Adding gives

$$L_S(T_C) \le K_R + \mathcal{E}(\alpha, \delta),$$

where

$$K_R = W + E - \frac{\eta}{R}$$

and

$$\mathcal{E}(\alpha, \delta) = -\frac{\alpha}{R} + \frac{\alpha}{W} + \delta + \min\left\{\frac{\alpha}{R}, \frac{\delta}{W}\right\} + \frac{[P - (R\alpha + \delta)]_+}{W}.$$

Splitting only according to the minimum and the positive part gives

$$\mathcal{E}(\alpha, \delta) \le \frac{P}{W}. \tag{29}$$

For example, if $\alpha/R \le \delta/W$, the expression without the positive part is $\alpha/W + \delta$; when the positive part is active, its excess over P/W is $\alpha - R\delta/W \le 0$. The reflected order gives excess $\delta - W\alpha/R \le 0$. If the positive part vanishes, (28) gives the same bound strictly.

Put $A = 1 + R - R^2$. Then

$$\frac{3}{2} - K_R - \frac{P}{W} = \frac{1 + A - 2EA}{2RW}.$$

Both sides of $1 + A > 2EA$ are positive, and

$$(1 + A)^2 - 4A^2E^2 = R^2W^2(5 + 4R - 4R^2) > 0.$$

Thus $K_R + P/W < 3/2$. □

Lemma 3.2 (Positive-support skeleton cap)**.** *Let T_i be the closure of an original open V triangle, so $V_i \in \mathrm{int}(T_i)$. If*

$$\mathcal{H}^1(T_i \cap r_j) > 0 \quad \textit{for some } j \in \{i-1, i+1\},$$

then

$$L_S(T_i) < \frac{3}{2}.$$

Proof. Normalize to $i = 0$ and translate the hexagon by V_0:

$$\widetilde{H} = V_0 + H, \qquad \widetilde{O} = V_0, \qquad W_j = V_0 + V_j.$$

Then $W_3 = O$, $W_2 = V_1$, and $W_4 = V_5$. The five original skeleton components on which a unit triangle containing V_0 can have positive-length trace are

$$[V_0, V_1],\ [V_0, V_5],\ [O, V_1],\ [O, V_0],\ [O, V_5].$$

All five belong to the skeleton $\widetilde{S}$ of $\widetilde{H}$. Since $V_0 = \widetilde{O}$ is interior to T_0, the triangle is a legitimate center triangle for $\widetilde{H}$. Its positive trace on r_1 or r_5 is a positive boundary-edge trace of $\widetilde{H}$, so it is CE1 or CE2 by Proposition 2.3. Proposition 2.12 and Lemma 3.1 therefore give

$$L_{\widetilde{S}}(T_0) < \frac{3}{2}.$$

A nonlocal component of S is more than unit distance from the interior point V_0, apart from zero-measure endpoints. Hence $L_S(T_0) \le L_{\widetilde{S}}(T_0)$. □

Lemma 3.3 (No-support skeleton cap)**.** *If a V triangle has $A_i + B_i \le 1$ and*

$$\mathcal{H}^1(T_i \cap r_{i-1}) = \mathcal{H}^1(T_i \cap r_{i+1}) = 0,$$

then

$$L_S(T_i) \le 2.$$

Proof. Its boundary contribution is $A_i + B_i \le 1$ by (32). For $P = sV_j \in r_j \setminus \{O\}$,

$$\|V_i - P\|^2 = 1 + s^2 - 2s \cos \frac{(j-i)\pi}{3} > 1$$

when $j \notin \{i-1, i, i+1\}$. The traces on r_{i-1} and r_{i+1} have zero length by hypothesis, so only r_i can contribute positive length. This adds at most one. □

For the remaining cap we use one specialized part of the exact local equilateral enclosure criterion. In local directions u, v meeting at $120°$, suppose a closed unit triangle contains

$$0, \quad au, \quad bv, \quad c(u+v),$$

with $0 \le a \le b \le 1$ and $a + b > 1$. The support-line calculation gives the selected supercritical cell

$$\begin{aligned} &a^2 + ab + b^2 \le 1, \qquad c \le \frac{1}{2}, \\ &(a^2 - 1)c^2 + (2ab^2 + b)c + (b^4 - b^2) \le 0, \end{aligned} \tag{30}$$

where c lies on the connected component containing $c = 0$. To see the selector directly, the quadratic is negative at $c = 0$, positive at $c = 1/2$, and opens downward. For the middle assertion, four times its value at $1/2$ is increasing in a and, at $a = 1 - b$, equals $b^2(2b-1)^2$; it is therefore positive when $a + b > 1$. Hence $c \le 1/2$ selects its smaller-root component. Formula (30) follows by fixing in the support sum the three active contacts $\{B\}$, $\{B, c(u+v)\}$, and $\{A\}$, and squaring the positive unit-side equation.

Lemma 3.4 (Supercritical skeleton cap)**.** *If $A_i + B_i > 1$, then*

$$L_S(T_i) < \frac{3}{2}.$$

Proof. By Proposition 2.6, a V triangle is Vd0, Vd1, Vd2, or T3-like. Proposition 3.6 makes every Vd1/Vd2 V triangle nonsupercritical, and Lemma 2.9 does the same for T3-like V triangles. Hence a supercritical triangle is Vd0 and, by definition, $\mathcal{H}^1(T_i \cap r_{i-1}) = \mathcal{H}^1(T_i \cap r_{i+1}) = 0$. Thus, after normalizing and writing $a = A_i$, $b = B_i$, the skeleton trace has length $a + b + c$, where c is the reach on r_i. Reflect so that $a \le b$, and put

$$s = a + b, \qquad \delta = b - a, \qquad c_0 = \frac{3}{2} - s.$$

The diameter constraint gives

$$1 < s \le \frac{2}{\sqrt{3}}, \qquad 0 \le \delta \le \sqrt{4 - 3s^2}.$$

Let $P(c)$ denote the quadratic in (30). Substitution and multiplication by 16 give

$$\begin{aligned} 16P(c_0) = Q(s,\delta) = {} & \delta^4 + 8\delta^3 s - 6\delta^3 + 14\delta^2 s^2 - 18\delta^2 s + 5\delta^2 \\ & - 8\delta s^3 + 30\delta s^2 - 34\delta s + 12\delta \\ & + s^4 - 6s^3 - 19s^2 + 60s - 36. \end{aligned}$$

Its derivative factors as

$$\frac{\partial Q}{\partial \delta} = 2(2\delta + 4s - 3)\big(\delta^2 + \delta(4s-3) + (s-1)(2-s)\big) \ge 0.$$

Therefore

$$Q(s,\delta) \ge Q(s,0) = (s-1)(s^3 - 5s^2 - 24s + 36) > 0.$$

Indeed, the cubic is decreasing on $[1, 2/\sqrt{3}]$ and at its right endpoint equals $88/3 - 136\sqrt{3}/9 > 0$. Hence $P(c_0) > 0$. Since (30) selects the smaller-root component, $P(c) \le 0$ forces $c < c_0$. Thus $a + b + c < 3/2$. □

Lemma 3.5 (A positive-support adjacent exceptional triangle is forced)**.** *Suppose T_C is CE1 or CE2 and, after symmetry, $T_C \cap \{M_0, \ldots, M_5\} = \{M_0\}$. If at least two V triangles are supercritical, then for a supercritical index $s \ne 0$ there is some $j \in \{s-1, s+1\}$ such that*

$$M_s \in U_j, \qquad \mathcal{H}^1(T_j \cap r_s) > 0.$$

Proof. Choose a supercritical index $s \ne 0$. Lemma 2.11 gives $M_s \notin T_s$, and $M_s \notin T_C$ by the unique-midpoint property. In the original open cover, M_s is therefore contained by some other open V triangle U_j. Diameter one restricts j to $\{s-1, s+1\}$. Because U_j is open and M_s is an interior point of r_s, one has $\mathcal{H}^1(T_j \cap r_s) > 0$. □

3.1.2. *Vd1/Vd2 corner estimates.* We next isolate the much stronger trace loss caused by Vd1 and Vd2 geometry. For the following two results use the temporary coordinates

$$X = V_0 + x(V_5 - V_0) + y(V_1 - V_0).$$

Thus $V_0 = (0,0)$, $V_5 = (1,0)$, $V_1 = (0,1)$, $O = (1,1)$, and $\|(x,y)\|^2 = x^2 + y^2 - xy$.

Proposition 3.6 (Vd1/Vd2 corner normal form)**.** *Let T be the closure of an original Vd1 or Vd2 triangle at V_0, and let a, b be its exact reaches toward V_5, V_1. There is a unique $t > 0$ such that, with $d = \sqrt{t^2+t+1}$,*

$$T = \left\{ (x,y) : \begin{array}{r} x - (t+1)y \le a, \\ ty - (t+1)x \le tb, \\ tx + y \le d - a - tb \end{array} \right\}.$$

The raw traces, in coordinates measured from the indicated outer vertex, are

$$\begin{array}{c|c} r_0 & 0 \le s \le \dfrac{d-a-tb}{t+1} \\ r_1 & \dfrac{t(1-b)}{t+1} \le s \le \dfrac{d-a-tb-1}{t} \\ r_5 & \dfrac{1-a}{t+1} \le s \le d - a - tb - t. \end{array}$$

Intersecting with $[0,1]$ gives the actual traces. Moreover,

$$a + b < \frac{1}{2}. \tag{31}$$

Proof. A Vd1/Vd2 triangle has one vertex W outside H and two vertices U, Z in H. Since V_0 is an interior convex combination of them, both coordinates of W are negative. The two axis endpoints $(a,0), (0,b)$ lie on distinct sides incident with W. Put $p = U - W$, $q = Z - W$. Both vectors have positive coordinates and satisfy

$$\|p\| = \|q\| = \|p - q\| = 1.$$

The order in which the two sides cross the positive axes selects the $60°$ rotation

$$q = (p_x - p_y, p_x), \qquad p_x > p_y > 0.$$

Writing $t = (p_x - p_y)/p_y$ gives uniquely

$$p = \frac{(t+1, 1)}{d}, \qquad q = \frac{(t, t+1)}{d}.$$

The two side lines through the axis endpoints and the parallel third side are exactly those in (3.6). Substitution of $(s,s), (s,1), (1,s)$ gives (3.6).

If the r_1 trace has positive length, its lower endpoint is smaller than its upper endpoint. Clearing denominators gives

$$(t+1)a + tb < (t+1)(d-1) - t^2.$$

Since the left side is at least $t(a+b)$,

$$a + b < \frac{(t+1)(d-1) - t^2}{t} < \frac{1}{2}.$$

For the last inequality, both sides are positive and

$$\left(t^2 + \frac{3t}{2} + 1\right)^2 - (t+1)^2(t^2+t+1) = \frac{t^2}{4} > 0.$$

Reflection proves the same conclusion when the supported arm is r_5. □

Lemma 3.7 (Vd2 adjacent-midpoint cap)**.** *If an open role U_i has Vd2 closure $T_i = \overline{U_i}$ and T_i contains M_{i-1} or M_{i+1}, then its exact boundary reaches satisfy*

$$a + b < \frac{1}{3}.$$

Proof. In Proposition 3.6, reflection sends (t, a, b, M_1, M_5) to $(1/t, b, a, M_5, M_1)$, so assume $t \geq 1$ and put $U = a + tb$.

If $M_5 = (1, 1/2)$ is contained, the third half-plane in (3.6) gives

$$a + b \leq U \leq d - t - \frac{1}{2} \leq \sqrt{3} - \frac{3}{2} < \frac{1}{3};$$

the middle expression decreases for $t \geq 1$.

If $M_1 = (1/2, 1)$ is contained, the second and third half-planes give

$$b \geq \frac{t-1}{2t}, \qquad a + b \leq S(t) := d - t - \frac{1}{2t}.$$

Positive-length contact with r_5 gives, by (3.6),

$$a + b < R(t) := \frac{2(t+1)d - 3t^2 - t - 2}{2t}.$$

Direct differentiation shows that S is increasing and R decreasing on $[1, \infty)$. For example, the sign calculation for S' reduces to

$$(2t+1)^2 t^4 - (t^2 + t + 1)(2t^2 - 1)^2 = (t+1)(t^3 + 3t^2 - 1) > 0,$$

while $R' < 0$ follows from $d > t + 1/2$. Splitting at $t = 4/3$ yields

$$1 \leq t \leq \frac{4}{3}: \quad a + b \leq S(4/3) = \frac{8\sqrt{37} - 41}{24} < \frac{1}{3},$$
$$t \geq \frac{4}{3}: \quad a + b < R(4/3) = \frac{7\sqrt{37} - 39}{12} < \frac{1}{3}.$$

The final inequalities follow respectively from $8\sqrt{37} < 49$ and $7\sqrt{37} < 43$. □

3.1.3. *Boundary trace caps.* The signed center normal form gives the CE1 and CE2 center contributions from one formula; no separate maximal-interval calculation is needed here.

Theorem 3.8 (Boundary trace table)**.** *For closures of original open triangles, the following bounds hold:*

triangle	*hypothesis*	$L_{\partial H}$
T_C	CE0	0
T_C	CE1	$\leq \frac{\sqrt{3}}{2} - \frac{3}{4}$
T_C	CE2	$< \frac{1}{2}$
T_i	Vd0, $A_i + B_i \leq 1$	≤ 1
T_i	Vd0, $A_i + B_i > 1$	$\leq \frac{2}{\sqrt{3}}$
T_i	Vd1/Vd2	$< \frac{1}{2}$
T_i	*T3-like*	$< 1.$

Proof. For every V triangle,

$$L_{\partial H}(T_i) = A_i + B_i. \tag{32}$$

Indeed, the two incident traces are initial intervals of these lengths. A relative-interior point of any nonincident edge is more than unit distance from V_i, whereas V_i is interior to the diameter-one triangle T_i. This excludes any other positive-length boundary trace.

The CE0 C-triangle row is the definition. The other two C-triangle rows are the class specializations of the common signed formula in Corollary 2.18.

For a nonsupercritical Vd0 V triangle, (32) is enough. For a supercritical V triangle, its incident endpoints are at squared distance $A_i^2 + A_iB_i + B_i^2 \le 1$, and hence

$$\frac{3}{4}(A_i + B_i)^2 \le 1.$$

This proves the $2/\sqrt{3}$ cap. The Vd1/Vd2 V triangle is (31). Finally, the Type II calculation in Proposition 2.7 gives for an original T3-like triangle

$$A_i + B_i = z - \alpha < z < 1$$

because $0 < t < 1$ and $\alpha > 0$. □

3.1.4. *Final length contradictions.*

Proposition 3.9 (Vd2 adjacent-midpoint branch)**.** *Assume a CE2, $N_+ = 1$ configuration has exactly one Vd1/Vd2 triangle, that triangle is T_i of type Vd2 and contains M_{i-1} or M_{i+1}, and all other V triangles are Vd0. Then the boundary cannot be covered.*

Proof. This is item (5) of Corollary 3.15, using Lemma 3.7. □

Proposition 3.10 (Branches closed by Method 1)**.** *Direct length-sum obstructions close the following routing entries and hybrid subcase:*

1. *$N_{\text{gap}} = 0$, $N_+ = 0$, for any center and V triangle types;*
2. *$N_{\text{gap}} = 0$, $N_+ = 1$, with a Vd1/Vd2 triangle;*
3. *CE1/CE2 with $N_+ + N_{\text{sp}} \ge 3$;*
4. *CE1/CE2, $N_+ = 0$, with a Vd1/Vd2 triangle;*
5. *CE1, $N_+ = 1$, with exactly one Vd1/Vd2 triangle and no T3-like triangle;*
6. *the CE2 hybrid subcase in which the Vd2 triangle T_i contains M_{i-1} or M_{i+1}.*

Proof. For item (1), Lemma 2.8 says that the six open V triangles cover ∂H. The two incident open V traces therefore overlap on every edge, so $1 < B_i + A_{i+1}$. Summing contradicts $N_+ = 0$.

For item (2), use only those six boundary-complete V triangles. The unique supercritical Vd0 triangle contributes at most $2/\sqrt{3}$, a Vd1/Vd2 triangle contributes strictly less than $1/2$, and each of the other four V triangles contributes at most 1. Their total boundary contribution is strictly less than

$$\frac{2}{\sqrt{3}} + \frac{1}{2} + 4 < 6,$$

contradicting their coverage of the length-6 perimeter.

Items (4) and (5) are items (2) and (3) of Corollary 3.15; each is one application of the master perimeter deficit. Item (3) is Theorem 3.14. Item (6) is Proposition 3.9. This is only the indicated subcase of the exceptional CE2 V triangle; its remaining placements are handled by the boundary-reach argument. □

3.2. **Common perimeter and skeleton budgets.** The signed center formula also removes repeated length arithmetic. We first state a generic perimeter deficit, then give the direct CE1/CE2 skeleton count used by the routing table.

Lemma 3.11 (Master perimeter deficit)**.** *Put*

$$\theta = \frac{2}{\sqrt{3}} - 1.$$

Suppose a branch has a center contribution L_C, n supercritical Vd0 V triangles, $m \geq 1$ distinguished nonsupercritical V triangles with strict boundary caps $q_1, \dots, q_m < 1$, and $6 - n - m$ remaining V triangles whose boundary contributions are at most 1. If

$$L_C + n\theta \leq \sum_{j=1}^{m} (1 - q_j), \tag{33}$$

then the seven triangles do not cover ∂H.

Proof. The supercritical Vd0 cap is $2/\sqrt{3}$. Hence the total boundary contribution is strictly less than

$$\begin{aligned} L_C + n\frac{2}{\sqrt{3}} + \sum_{j=1}^{m} q_j + (6 - n - m) &= 6 + L_C + n\theta - \sum_{j=1}^{m}(1 - q_j) \\ &\leq 6. \end{aligned}$$

The strictness follows from $m \geq 1$ and the strict distinguished-V triangle caps. A cover of the perimeter would give the opposite weak inequality by subadditivity. □

Lemma 3.12 (Small center slacks in the one-Vd1/Vd2 branch)**.** *Assume there is exactly one supercritical Vd0 V triangle, exactly one Vd1/Vd2 V triangle, and four nonsupercritical Vd0 V triangles. Then every candidate surviving the perimeter budget is CE2 and satisfies*

$$\boxed{\alpha + \delta < \frac{1}{24}, \qquad \alpha + \delta < \frac{\min\{R, W\}}{6}.} \tag{34}$$

Proof. The Vd1/Vd2 boundary cap is strict and equals 1/2. If the perimeter were covered, Lemma 3.11 would force

$$L_C + \theta > \frac{1}{2}.$$

Put

$$\kappa = \frac{1}{2} - \theta = \frac{3}{2} - \frac{2}{\sqrt{3}}.$$

Thus $L_C > \kappa$. The CE1 cap in Corollary 2.18 gives

$$L_C + \theta \leq \left(\frac{\sqrt{3}}{2} - \frac{3}{4}\right) + \left(\frac{2}{\sqrt{3}} - 1\right) < \frac{1}{2},$$

so every survivor is CE2.

Set $T = \alpha + \delta$. The CE2 formula gives

$$T < M(E) := \frac{RW}{E^2}\left(\frac{E}{1 + E} - \kappa\right).$$

Since $RW = 1 - E^2$,

$$M(E) = \frac{1-E}{E^2}\big(E - \kappa(1+E)\big),$$

and

$$M'(E) = -\frac{E-2\kappa}{E^3} < 0 \qquad \left(\frac{\sqrt{3}}{2} \le E < 1\right).$$

Here $2\kappa < \sqrt{3}/2$, because it is equivalent to $6\sqrt{3} < 11$, whose square is $108 < 121$. Therefore

$$T < M\left(\frac{\sqrt{3}}{2}\right) = \frac{8\sqrt{3}}{9} - \frac{3}{2} < \frac{1}{24}.$$

The last inequality is equivalent to $64\sqrt{3} < 111$, and follows after squaring from $12288 < 12321$.

For the second estimate, use $E/(1+E) < 1/2$:

$$T < \frac{RW}{E^2}\left(\frac{1}{2} - \kappa\right) = \frac{RW}{E^2}\theta.$$

Since

$$E^2 = W + R^2 = R + W^2,$$

one has $RW/E^2 \le R$ and $RW/E^2 \le W$. Also $\theta < 1/6$, equivalently $12 < 7\sqrt{3}$, whose square is $144 < 147$. This proves (34). □

Lemma 3.13 (CE2 initial endpoints dominate the total slack)**.** *Assume* $\Delta_R > 0$ *and* $\Delta_L > 0$*. Put*

$$T = \alpha + \delta, \qquad x = \frac{\eta + T}{W}, \qquad y = \frac{\eta + T}{R}.$$

Then

$$\boxed{T < \frac{1}{2}\min\{x, y\}.} \tag{35}$$

Proof. Equation (25) gives $T < \eta/E$. Moreover

$$E^2 - (2R-1)^2 = 3RW > 0,$$

so $|2R-1| < E$ and therefore $|2R-1|T < \eta$. Hence

$$x - 2T = \frac{\eta - (1-2R)T}{W} > 0,$$

and

$$y - 2T = \frac{\eta - (2R-1)T}{R} > 0.$$

This is (35). □

Theorem 3.14 (Direct CE1/CE2 skeleton count)**.** *If a CE1/CE2 branch satisfies* $T_C \cap \{M_0, \ldots, M_5\} = \{M_0\}$ *and*

$$N_+ + N_{\mathrm{sp}} \ge 3, \tag{36}$$

then the seven triangles do not cover the full skeleton S*.*

Proof. Every Vd1, Vd2, or T3-like V triangle has positive support on an adjacent radial segment and is nonsupercritical: use Proposition 3.6 and Lemma 2.9. Conversely, every supercritical V triangle is Vd0 and has no such support. The two classes counted by N_+ and N_{sp} are therefore disjoint.

The center skeleton cap is strictly below 3/2. Each of the $N_+ + N_{\mathrm{sp}}$ distinguished V triangles also contributes strictly less than 3/2, by the supercritical or positive-support cap. Every remaining V triangle is nonsupercritical Vd0 and contributes at most 2. Consequently

$$\begin{aligned} L_S(T_C) + \sum_{i=0}^{5} L_S(T_i) &< \frac{3}{2} + \frac{3}{2}(N_+ + N_{\mathrm{sp}}) + 2(6 - N_+ - N_{\mathrm{sp}}) \\ &= \frac{27 - N_+ - N_{\mathrm{sp}}}{2} \le 12. \end{aligned}$$

This contradicts subadditivity for a cover of the length-12 skeleton. □

Corollary 3.15 (Length branches closed by the common budgets)**.** *The following branches are impossible.*

1. *CE0, $N_+ = 1$, and at least one Vd1/Vd2 V triangle;*
2. *$N_+ = 0$ and at least one Vd1/Vd2 V triangle, for either CE1 or CE2;*
3. *CE1, $N_+ = 1$, and at least one Vd1/Vd2 V triangle;*
4. *CE2, $N_+ = 1$, and at least two Vd1/Vd2 V triangles;*
5. *CE2, $N_+ = 1$, with a Vd2 triangle T_i containing M_{i-1} or M_{i+1};*
6. *CE1 or CE2 with $N_+ + N_{\mathrm{sp}} \ge 3$;*
7. *CE1 or CE2 with $N_+ \ge 2$.*

Proof. For item 1, take $L_C = 0$, one supercritical cap $2/\sqrt{3}$, and one strict Vd1/Vd2 cap 1/2 in Lemma 3.11; the needed inequality is

$$\frac{2}{\sqrt{3}} - 1 < \frac{1}{2}.$$

For item 2, the center contribution is strictly less than 1/2 and the chosen Vd1/Vd2 V triangle has strict cap 1/2; the other five V triangles contribute at most 1 each. This is Lemma 3.11 with $n = 0$ and $q_1 = 1/2$.

For item 3, the unique supercritical V triangle is Vd0, while the CE1 center cap, the Vd1/Vd2 cap, and the four remaining unit caps give

$$\left(\frac{\sqrt{3}}{2} - \frac{3}{4}\right) + \frac{1}{2} + \frac{2}{\sqrt{3}} + 4 < 6.$$

For item 4, the corresponding CE2 sum is strictly less than

$$\frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{2}{\sqrt{3}} + 3 < 6.$$

For item 5, the M_{i-1}/M_{i+1} Vd2 cap is 1/3, and the total is strictly less than

$$\frac{1}{2} + \frac{1}{3} + \frac{2}{\sqrt{3}} + 4 < 6.$$

Item 6 is Theorem 3.14.

Finally suppose $N_+ \ge 2$. By Proposition 2.12, normalize so that

$$T_C \cap \{M_0, \dots, M_5\} = \{M_0\}.$$

Choose a supercritical index $s \neq 0$. Its own midpoint M_s is not covered by T_s, and it is not covered by T_C. Diameter locality leaves only U_{s-1}, U_s, U_{s+1} as possible V triangles containing M_s. Thus $M_s \in U_j$ for some $j \in \{s-1, s+1\}$ and, by openness, $\mathcal{H}^1(T_j \cap r_s) > 0$. Thus T_j is counted by N_{sp} and is distinct from the two supercritical V triangles. Hence $N_+ + N_{\mathrm{sp}} \geq 3$, and Theorem 3.14 applies. $\square$

4. Area-Loss Estimate

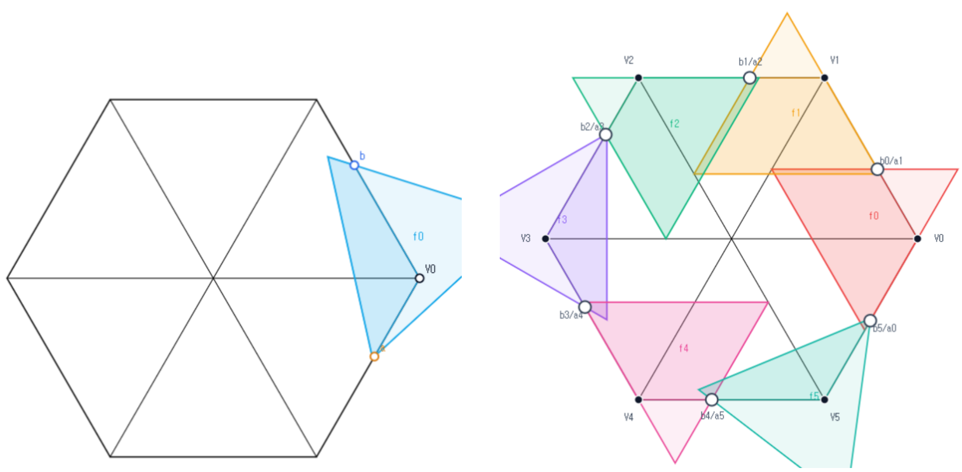
The second strategy measures the normalized area forced outside H by a V triangle with prescribed boundary reaches. The local square-loss and T3-like estimates are proved first, then summed around the six-cycle to close the two center-independent zero-gap area branches.

4.1. Detailed Area-Loss Estimates. This section proves the local inequalities and cyclic estimates used in the area-loss argument.

Let

$$A_\triangle = \frac{\sqrt{3}}{4}$$

be the area of a unit equilateral triangle. Throughout this section areas are divided by $A_\triangle$. Thus a unit triangle has normalized area one and H has normalized area six.



(a) local retained area (b) global retained areas

Figure 9. Schematic views of Method 3, not to scale. The local panel shows one V triangle with boundary reaches a, b, and the global panel shows the six cyclic triangles. Only the darker portions represent area retained inside H; the proof uses the inequalities below, not the picture.

4.1.1. *The local square-loss inequalities.* Fix a vertex V_i. Let P_a and P_b be the points at distances a and b from V_i on its incoming and forward boundary edges. For a closed unit equilateral triangle T containing V_i, P_a, P_b, put

$$\mathcal{L}(T) = \frac{\operatorname{area}(T \setminus H)}{A_\triangle}.$$

Equivalently, if

$$f(a,b)=\max_T \frac{\operatorname{area}(T\cap H)}{A_\triangle}, \qquad \mathcal{L}(a,b)=1-f(a,b),$$

then $\mathcal{L}(a,b)$ is the least possible normalized loss. Reflection in the radial axis interchanges a and b, and increasing either reach lower bound can only decrease f.

Lemma 4.1 (Local wedge reduction). *After rotating indices to $i=0$, write*

$$X=V_0+x(V_1-V_0)+y(V_5-V_0), \qquad W=\{(x,y):x\ge 0,\ y\ge 0\}.$$

If a closed unit equilateral triangle T contains V_0, then

$$T\cap H=T\cap W.$$

Normalized Euclidean area is twice the corresponding (x,y)-coordinate area.

Proof. The two basis vectors have length one and angle 120°, so

$$\|(x,y)\|^2=x^2+y^2-xy, \qquad |\det(V_1-V_0,V_5-V_0)|=\frac{\sqrt{3}}{2}.$$

The hexagon is

$$H=\{0\le x,y\le 2,\ |x-y|\le 1\}\subset W.$$

Conversely, if $(x,y)\in T\cap W$, then the diameter-one condition gives $x^2+y^2-xy\le 1$. Hence

$$|x-y|^2\le x^2+y^2-xy\le 1, \qquad \frac{3}{4}x^2\le x^2+y^2-xy,$$

and the same estimate holds for y. Thus $|x-y|\le 1$ and $0\le x,y\le 2/\sqrt{3}<2$, proving $(x,y)\in H$. The determinant identity proves the area assertion. $\square$

Theorem 4.2 (Unconditional local square loss). *For every feasible pair $0\le a,b\le 1$ and every realizing closed unit equilateral triangle T,*

$$\mathcal{L}(T)\ge \min(a,b)^2. \tag{37}$$

If $a+b>1$, then

$$\mathcal{L}(T)\ge \max(a,b)^2. \tag{38}$$

Consequently the same inequalities hold for $\mathcal{L}(a,b)=1-f(a,b)$.

Proof. In the coordinates of Lemma 4.1, physical rotation through 60° is $\mathsf{R}(x,y)=(x-y,x)$. Put

$$0\le t\le 1, \qquad D=1-t+t^2, \qquad z=\sqrt{D}.$$

After choosing one of the six directed sides, every orientation has one of the following two forms:

$$\begin{array}{c|cc} & d & \mathsf{R}d \\ \hline \mathrm{I} & z^{-1}(1,t) & z^{-1}(1-t,1) \\ \mathrm{II} & z^{-1}(t,-(1-t)) & z^{-1}(1,t). \end{array} \tag{39}$$

Indeed these forms cover, respectively, the physical direction sectors $[120°,180°]$ and $[60°,120°]$; their endpoints cover the common boundary orientations.

For Type I introduce

$$U=\alpha+y-tx, \qquad V=\beta+x-(1-t)y.$$

Then T is the simplex $U, V \geq 0$, $U + V \leq z$. Containment of $(0,0), (0,a), (b,0)$ gives

$$\alpha \geq tb, \quad \beta \geq (1-t)a, \quad \alpha + \beta + (1-t)b \leq z, \quad \alpha + \beta + ta \leq z. \tag{40}$$

The two inequalities defining W become

$$(1-t)U + V \geq (1-t)\alpha + \beta, \qquad U + tV \geq \alpha + t\beta.$$

For fixed t, lowering α or β therefore enlarges $T \cap W$. The outside loss is minimized at $\alpha = tb$, $\beta = (1-t)a$. At that placement the portion outside W has vertices, in the (U, V)-plane,

$$(0,0), \quad (a + tb, 0), \quad (tb, (1-t)a), \quad (0, b + (1-t)a).$$

Since the Jacobian of $(x, y) \mapsto (U, V)$ has absolute value D,

$$\mathcal{L}(T) \geq \mathcal{L}_{\mathrm{I}}(a, b; t) = \frac{(1-t)a^2 + tb^2 + 2t(1-t)ab}{D}. \tag{41}$$

If $a \leq b$, then

$$D(\mathcal{L}_{\mathrm{I}} - a^2) = t\{b^2 - a^2 + (1-t)(a^2 + 2ab)\} \geq 0;$$

the reflected factorization proves $\mathcal{L}_{\mathrm{I}} \geq b^2$ when $b \leq a$. This proves (37) in Type I.

Suppose now that $a + b > 1$ and $a \leq b$. Set

$$r = \frac{a}{b}, \qquad t_0 = \frac{1 - r^2}{1 + 2r}.$$

Since $b > 1/(1+r)$, feasibility in (41) and (40) excludes $0 \leq t < t_0$: for $0 < t < t_0$,

$$\left(\frac{1 + r(1-t)}{1+r}\right)^2 - D = \frac{t(1+2r)(t_0 - t)}{(1+r)^2} > 0,$$

so $b + (1-t)a > z$, while $t = 0$ would give $a + b \leq 1$. Hence $t \geq t_0$, and

$$D(\mathcal{L}_{\mathrm{I}} - b^2) = (1-t)b^2\{r^2 - 1 + t(2r+1)\} \geq 0.$$

If $b \leq a$, put $r = b/a$ and $t_1 = (r^2 + 2r)/(1 + 2r)$. The identical reflected calculation excludes $t > t_1$ and gives

$$D(\mathcal{L}_{\mathrm{I}} - a^2) = ta^2\{r^2 + 2r - t(1+2r)\} \geq 0.$$

Thus (38) holds in Type I.

For Type II put

$$U = \alpha + (1-t)x + ty, \qquad V = \beta + tx - y.$$

Again T is the simplex $U, V \geq 0$, $U + V \leq z$. Its exact reaches on the positive y- and x-axes are

$$A = \beta, \qquad B = z - \alpha - \beta, \qquad A + B \leq z \leq 1. \tag{42}$$

In particular Type II is impossible when $a + b > 1$. Its wedge intersection is the quadrilateral with vertices

$$(0,0), \quad (B, 0), \quad \left(\frac{B + (1-t)A}{D}, \frac{A + tB}{D}\right), \quad (0, A).$$

Hence

$$\mathcal{L}(T) = \mathcal{L}_{\mathrm{II}}(A, B; t) = 1 - \frac{(1-t)A^2 + tB^2 + 2AB}{D}. \tag{43}$$

Let $S = A + B$. If $A \le B$, write $A = rS$, $B = (1-r)S$ with $0 \le r \le 1/2$. Since $S^2 \le D$, and with

$$C = (1-t)r^2 + t(1-r)^2 + 2r(1-r),$$

we have $\mathcal{L}_{\mathrm{II}} \ge 1 - C$. The exact factorization

$$1 - C - r^2 D = (1-t)(1 - 2r + tr^2) \ge 0$$

gives $\mathcal{L}_{\mathrm{II}} \ge r^2 D \ge A^2$. Reflection gives $\mathcal{L}_{\mathrm{II}} \ge B^2$ when $B \le A$. Since $A \ge a$ and $B \ge b$, this proves (37). The two orientation types are exhaustive, so the theorem follows. □

4.1.2. *The direct* T3-like *loss.* The next theorem uses the classification data (o, n) from Section 2.1: o counts triangle vertices outside H, and $n = n(T_i)$ is

$$\left|\left\{j \in \{i-1, i+1\} : \mathcal{H}^1(T_i \cap r_j) > 0\right\}\right|.$$

Thus a T3-like triangle has $(o, n) = (2, 1)$.

Theorem 4.3 (Direct T3-like loss)**.** *Let T be a closed unit V_i-triangle of* T3-like *type, and let a, b be lower bounds for its adjacent boundary reaches. Then*

$$a + b \le 1. \tag{44}$$

Moreover, for $0 \le m \le 1/2$,

$$a, b \ge m \quad \Longrightarrow \quad \mathcal{L}(T) \ge 2m - 4m^2. \tag{45}$$

Proof. The first assertion is Lemma 2.9. For the loss bound apply the closed-trace domination in Proposition 2.7. It gives a translated Type-II triangle $\widehat{T}$ of the same area with

$$T \cap H \subset \widehat{T} \cap H, \qquad \mathcal{L}(T) \ge \mathcal{L}(\widehat{T}).$$

In the notation of that proposition, $0 < t < 1$, $D = 1 - t + t^2$, $z = \sqrt{D}$, and after reflection the translated majorant has exact reaches

$$A = \beta, \qquad B = z - \beta$$

and unique wedge vertex

$$Q = \left(\frac{B + (1-t)A}{D}, \frac{A + tB}{D}\right), \qquad Q_x > 1, \quad Q_y \le 1.$$

Thus $z - tA > D$, so

$$A < L := \frac{z - D}{t} = \frac{z(1-z)}{t}. \tag{46}$$

The exact normalized loss is

$$\mathcal{L}(\widehat{T}) = \mathcal{L}(z, A) = \frac{tA^2 + (1-t)(z-A)^2}{D}.$$

This decreases for $A < L < (1-t)z$, and therefore

$$\mathcal{L}(T) \ge \mathcal{L}(\widehat{T}) > \mathcal{L}(z, L). \tag{47}$$

Set $r = (1-z)/t$. Then $0 < r < 1/2$, and eliminating t, z gives

$$t = \frac{1-2r}{1-r^2}, \qquad z = \frac{r^2 - r + 1}{1 - r^2},$$

$$L = \alpha_{\mathrm{lim}}(r) := \frac{r(r^2 - r + 1)}{1 - r^2}, \qquad z - L = \beta_{\mathrm{lim}}(r) := \frac{r^2 - r + 1}{1 + r}.$$

The limiting normalized inside area and loss are

$$F_0(r) = \frac{(r^2-r+1)^2}{1-r^2}, \qquad \mathcal{L}_0(r) = 1 - F_0(r).$$

Since the exact reach A is at least the required reach lower bound $a \geq m$, (46) gives $\alpha_{\lim}(r) > m$. Direct simplification yields

$$\mathcal{L}_0(r) - \{2\alpha_{\lim}(r) - 4\alpha_{\lim}(r)^2\} = \frac{r^2(5r^4 - 8r^3 + 13r^2 - 8r + 2)}{(1-r^2)^2} \geq 0,$$

because the last polynomial is at least $9r^2 - 8r + 2 \geq 2/9$ on $[0, 1/2]$. Also $\alpha_{\lim}(r) \leq r$, and

$$\mathcal{L}_0(r) - \frac{1}{4} = \frac{(1-2r)(2r^3 - 3r^2 + 6r - 1)}{4(1-r^2)} \geq 0 \quad \text{when } \alpha_{\lim}(r) \geq \frac{1}{4};$$

the cubic is increasing and has value $11/32$ at $r = 1/4$.

Finally $\psi(u) = 2u - 4u^2$ is increasing on $[0, 1/4]$ and never exceeds $1/4$ on $[0, 1/2]$. If $\alpha_{\lim}(r) \leq 1/4$, then $\mathcal{L}_0(r) \geq \psi(\alpha_{\lim}(r)) \geq \psi(m)$; if $\alpha_{\lim}(r) \geq 1/4$, then $\mathcal{L}_0(r) \geq 1/4 \geq \psi(m)$. Together with (47), this proves (45). The reflected crossing is identical. □

4.1.3. *The cyclic area certificates.* The selected handoffs of Proposition 2.10 have the form

$$(a_i, b_i) = (1 - \xi_{i-1}, \xi_i), \qquad 0 < \xi_i < 1,$$

with cyclic indices. V triangle i is selected supercritical exactly when $\xi_i > \xi_{i-1}$.

Lemma 4.4 (Cyclic area-loss certificate)**.** *Let the six actual V triangles realize selected handoffs*

$$(a_i, b_i) = (1 - \xi_{i-1}, \xi_i), \qquad 0 < \xi_i < 1.$$

Their total normalized loss is strictly greater than one if either

(1) *at least two selected V triangles are supercritical; or*
(2) *exactly one selected V triangle is supercritical, every triangle is* Vd0 *or* T3-like*, and at least one triangle is* T3-like*.*

Proof. Put $m = \min_i \xi_i$ and $M = \max_i \xi_i$. The reflection $y_i = 1 - \xi_{-i-1}$ swaps each V triangle's two coordinates and preserves both alternatives and the total loss. Hence assume $m \leq 1 - M$. Every selected V triangle coordinate is then at least m, so (37) gives loss at least m^2 for every V triangle.

In case (1), choose an ascent out of a minimum plateau, $\xi_{p-1} = m < \xi_p$. Its V triangle contains the coordinate $1 - m$ and loses at least $(1-m)^2$ by (38). A second ascent has one coordinate strictly larger than $1/2$ and hence loses strictly more than $1/4$. The other four V triangles lose at least m^2 each, so the total loss is strictly larger than

$$4m^2 + (1-m)^2 + \frac{1}{4} = 5\left(m - \frac{1}{5}\right)^2 + \frac{21}{20} > 1.$$

In case (2), rotate so that the unique ascent leaves the minimum: $\xi_5 = m < \xi_0 = M$. The supercritical V triangle loses at least $(1-m)^2$. Lemma 2.9 shows that a T3-like V triangle is different from it, and Theorem 4.3 gives that V triangle loss at least $2m - 4m^2$. The remaining four V triangles lose at least m^2 each. Their total loss is at least

$$(1-m)^2 + (2m - 4m^2) + 4m^2 = 1 + m^2 > 1.$$

□

Proposition 4.5 (Branches closed by area)**.** *The following branches are impossible:*

(1) $N_{\text{gap}} = 0$, $N_+ \geq 2$, *with arbitrary center and V triangle types;*
(2) $N_{\text{gap}} = 0$, $N_+ = 1$, *no triangle is* Vd1 *or* Vd2, *and at least one triangle is* T3-like.

Proof. By Lemma 2.8, $N_{\text{gap}} = 0$ says exactly that the six V triangles cover ∂H, so Proposition 2.10 applies independently of the center class.

If $N_+ \geq 2$, its at-least-two clause selects handoffs with at least two supercritical V triangles. Lemma 4.4 makes the six vertex triangles' total normalized inside area strictly less than five. The center triangle contributes at most one, so the total is strictly less than six.

If $N_+ = 1$, the exact-one clause preserves the unique actual supercritical index for every strict selection. The exhaustive vertex-type hypothesis gives case (2) of Lemma 4.4; the six V triangles contribute less than five normalized area, and the C triangle contributes at most one. Again the sum is strictly below the normalized area six of H. □

5. Finite-Enclosure Toolkit and Nonzero-Gap Obstructions

Method 3 has one contract. Local boundary and radial demands first produce a point or compact set missed by all six open V roles. Coverage then forces the object into the open C triangle. The case closes either because its least enclosing equilateral triangle has side at least one or because one forced point lies beyond the C-triangle exit on the same ray.

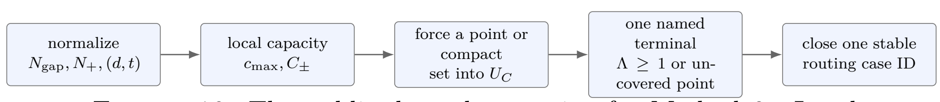


Figure 10. The public dependency spine for Method 3. Local capacity and witness generation are proved once; every combinatorial row invokes one named terminal.

The recurring notation is:

$$\Lambda(K), \quad c_{\max}(a,b), \quad C_\pm(a,b), \quad D_i = (1-\Gamma_i)V_i, \quad J(p,q), \quad K_{410}.$$

Here $\Lambda(K)$ is the least enclosing equilateral side, $c_{\max}$ is the own-ray capacity, $C_\pm$ are the permitted neighboring-ray capacities, D_i is a type-aware radial witness, $J(p,q)$ is the complementary boundary gap, and K_{410} is the anisotropic one-gap witness retaining all six actual radial endpoints.

The chapter is intentionally ordered by logical dependency rather than by case number: endpoint rules, local capacities, witness generators, universal terminals, exceptional cases, and finally the case-dispatch table.

5.1. Endpoint, compactness, and continuity.

The arguments above repeatedly pass between open role traces, closed maximal reaches, and compact witness sets. We record the precise endpoint statements used in those passages.

Lemma 5.1 (Open trace endpoint)**.** *Let U be an open equilateral triangle and $T = \overline{U}$. If $T \cap [V, O]$ is the closed interval from V to a point P, then $P \notin U$. If Q is a noncentral point of a radial arm and $Q \in U$, then U has positive-length intersection with that arm.*

Proof. The first assertion follows because P is a boundary point of the closed convex intersection. For the second, a small Euclidean ball about Q is contained in U; its intersection with the arm is a relative open interval. □

Lemma 5.2 (Compact set in an open unit triangle)**.** *If a compact set K is contained in an open unit equilateral triangle, then*

$$\Lambda(K) < 1.$$

Proof. The distance from K to each of the three side lines is positive. Move the three side lines inward by a common smaller distance. Their intersection is a closed equilateral triangle of side strictly below one and still contains K. □

Lemma 5.3 (Continuity of the enclosure number)**.** *For nonempty compact sets K, L,*

$$|\Lambda(K) - \Lambda(L)| \leq \frac{3}{h} d_H(K, L),$$

where d_H is the Hausdorff distance.

Proof. For every unit normal,

$$|h_K(n) - h_L(n)| \leq d_H(K, L).$$

Apply the support formula and take the minimum over orientations in both directions. □

These lemmas justify all singleton-gap limits and every use of a radial trace endpoint as a center-forced point.

5.2. **Trace-exact AB envelopes at an actual gap.** The endpoints of a displayed V-gap are actual maximal reaches, not selected lower bounds. If the gap on $e_{i,i+1}$ is

$$J_i = X_i([\ell, r]), \qquad X_i(t) = V_i + t(V_{i+1} - V_i),$$

then

$$\boxed{\ell = B_i, \qquad r = 1 - A_{i+1}.} \tag{6.24a}$$

Consequently the two incident open traces are exactly

$$U_i \cap e_{i,i+1} = X_i([0, \ell)), \qquad U_{i+1} \cap e_{i,i+1} = X_i((r, 1]), \tag{6.24b}$$

and the whole closed interval J_i, including a singleton, is missed by the six open V roles and therefore lies in U_C.

For visualization and local overapproximation it is useful to condition the ordinary AB-union on this actual endpoint. If S is a closed unit equilateral triangle with $V_i \in \text{int} S$, let $A_i(S), B_i(S)$ be its actual maximal backward and forward reaches. Define

$$\mathcal{E}_i^{\rightarrow}(a \mid b) = \bigcup \left\{ S \cap \mathcal{C}_i : A_i(S) \geq a,\ B_i(S) = b \right\}, \tag{6.24c}$$

$$\mathcal{E}_i^{\leftarrow}(a \mid b) = \bigcup \left\{ S \cap \mathcal{C}_i : A_i(S) = a,\ B_i(S) \geq b \right\}, \tag{6.24d}$$

where the union ranges over such closed unit triangles and $\mathcal{C}_i$ is the corner cone at V_i. We call these sets the *trace-exact AB envelopes*. They are source-conditioned subunions of the ordinary AB-set, and their relevant closed edge sections are exactly

$$\mathcal{E}_i^{\rightarrow}(a \mid b) \cap e_{i,i+1} = X_i([0,b]),$$

$$\mathcal{E}_{i+1}^{\leftarrow}(1-r \mid b') \cap e_{i,i+1} = X_i([r,1]). \tag{6.24e}$$

Thus the actual incident roles are locally contained in

$$\mathcal{E}_i^{\rightarrow}(a_i \mid \ell), \qquad \mathcal{E}_{i+1}^{\leftarrow}(1-r \mid b_{i+1}), \tag{6.24f}$$

while the interval between the two exact traces is center-forced.

Proposition 5.4 (Trace-exact AB-envelope interface)**.** *At an actual gap $J_i = X_i([\ell,r])$, the endpoint identities are $\ell = B_i$ and $r = 1 - A_{i+1}$. The source-conditioned envelopes have the exact edge sections in* (6.24e)*, contain the actual incident roles as in* (6.24f)*, and satisfy*

$$\mathcal{E}_i^{\rightarrow}(a \mid b), \quad \mathcal{E}_i^{\leftarrow}(a \mid b) \subseteq \mathcal{K}_i(a,b).$$

Consequently the ordinary capacities $c_{\max}, C_+, C_-$ remain valid upper bounds while the full closed interval between the two actual open traces is center-forced.

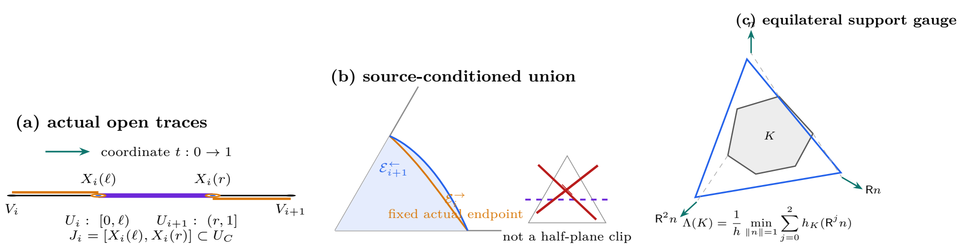


FIGURE 11. Three conventions used throughout the finite-enclosure argument. (a) A hollow outer marker records nonmembership in an open V trace, while the filled center marker records selection of the same endpoint as a center-forced witness. (b) A trace-exact envelope is obtained by restricting the source family to the prescribed actual maximal reach and then taking its union; it is not an affine half-plane clipping. (c) Equilateral enclosure is measured by three support directions separated by 120°. Incidences and interval orders are exact; lengths are schematic.

Proof. The word "trace-exact" is essential. These sets are not obtained by intersecting an ordinary AB-union with an affine half-plane through the gap endpoint: a source triangle whose edge trace stops at that endpoint can cross such a line elsewhere. The valid operation restricts the source family to triangles having the prescribed actual maximal reach and then takes their union. The valid operation restricts the source family to triangles having the prescribed actual maximal reach and then takes their union. This gives the edge sections and the inclusions into the ordinary AB-sets directly. The endpoint identities follow from the two actual open incident traces. The full gap interval is therefore missed by all six V roles and belongs to U_C.

□

5.3. **Universal enclosure and own-ray capacity.** Let R denote counterclockwise rotation through $2\pi/3$, and put $h = \sqrt{3}/2$. For a nonempty compact set K, let

$$h_K(n) = \max_{x \in K} \langle x, n \rangle.$$

Proposition 5.5 (Equilateral enclosure gauge)**.** *The least side length of a closed equilateral triangle containing K is*

$$\Lambda(K) = \frac{1}{h} \min_{\|n\|=1} \sum_{j=0}^{2} h_K(\mathsf{R}^j n). \tag{6.1}$$

If K is a polygon, a minimizing normal can be chosen perpendicular to a hull edge.

Proof. For fixed outward normals $n, \mathsf{R}n, \mathsf{R}^2 n$, the least containing triangle has those three support numbers, and its side is their sum divided by h. On an open angular cell with fixed support points, the support sum is $A\cos\theta + B\sin\theta$ and has second derivative equal to its negative. Thus an interior critical point is a maximum, and a minimum occurs at a support tie, equivalently at a hull-edge normal. □

Take unit vectors u, v meeting at $120°$, so that $u+v$ is the inward radial direction, and put

$$K(a,b,c) = \operatorname{conv}\{0, au, bv, c(u+v)\}.$$

The demand triple (a,b,c) is *admissible* when $K(a,b,c)$ lies in a closed unit equilateral triangle.

Proposition 5.6 (Exact local admissible set)**.** *Put*

$$s = a+b, \quad d = a^2+ab+b^2, \quad m = \min\{a,b\}, \quad M = \max\{a,b\},$$
$$q = s^4 - s^2 + ab,$$

and

$$\begin{aligned} F_L &= c^4 - c^2 + mc - m^2, \\ F_T &= (s^2-1)c^2 + Mc - M^2, \\ F_S &= (m^2-1)c^2 + (2mM^2 + M)c + M^4 - M^2. \end{aligned}$$

Then the admissible set is the union of

$$\begin{aligned} \mathcal{A}_L &= \{d \le 1,\ s \le 1,\ q \le 0,\ F_L \le 0\}, \\ \mathcal{A}_T &= \{d \le 1,\ s \le 1,\ q \ge 0,\ F_T \le 0,\ c \le 2M\}, \\ \mathcal{A}_S &= \{d \le 1,\ s > 1,\ F_S \le 0,\ c \le 1/2\}. \end{aligned}$$

Consequently the maximal admissible radial demand is

$$c_{\max}(a,b) = \begin{cases} 1, & a=b=0, \\ C_L(m), & s \le 1,\ q \le 0, \quad (a,b) \ne (0,0), \\ \dfrac{2M}{1+\sqrt{4s^2-3}}, & s \le 1,\ q > 0, \\ \dfrac{M(2mM+1) - M\sqrt{4d-3}}{2(1-m^2)}, & s > 1, \end{cases} \tag{6.2}$$

where $C_L(m)$ is the selected root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

The admissible set is coordinatewise down-closed.

Proof. If $cs \le ab$, the radial point lies in $\operatorname{conv}\{0, au, bv\}$ and the least side is the diameter $\sqrt{d}$. Otherwise the convex hull has four edges. Proposition 5.5 reduces the minimum to the four edge normals. The corresponding side lengths are

$$L_{0A} = b + \max\{a, c\}, \qquad L_{B0} = a + \max\{b, c\},$$

$$L_{AC} = \begin{cases} (a^2 + bc)/\sqrt{a^2 - ac + c^2}, & a \ge c, \\ c(a + b)/\sqrt{a^2 - ac + c^2}, & a < c \le a + b, \\ c^2/\sqrt{a^2 - ac + c^2}, & c > a + b, \end{cases}$$

and L_{CB} is obtained by interchanging a, b. Squaring their positive unit-frontier equations gives F_L, F_T, F_S. The inactive support conditions give respectively $q \le 0, q \ge 0, s > 1$. In the T cell the selector $c \le 2M$ removes the spurious large root; in the S cell $c \le 1/2$ selects the geometric component. Solving the selected equations gives (6.2). Convexity of a containing triangle proves down-closedness. □

The zero-radial diameter envelope is

$$M_0(x) = \frac{-x + \sqrt{4 - 3x^2}}{2}. \tag{6.3}$$

For

$$0 < e < 1 - \frac{\sqrt{3}}{2},$$

define the selected low root

$$\epsilon(e) = \frac{1 - e}{2}\left(1 - \sqrt{4(1 - e)^2 - 3}\right). \tag{6.4}$$

It satisfies

$$e < \epsilon(e), \qquad \epsilon(e) < \frac{2e}{1 - 2e}, \tag{6.5}$$

and, for $e \le 1/8$,

$$\epsilon(e) \le 2e + 5e^2. \tag{6.6}$$

These follow by substituting the proposed upper bounds in the selected quadratic and checking its sign.

Lemma 5.7 (Direct high-radial threshold)**.** *Let a nonsupercritical V triangle have incoming boundary reach A and outgoing boundary reach B. If*

$$C \ge 1 - e, \qquad A > \epsilon(e),$$

then

$$\boxed{B \le \epsilon(e).} \tag{6.7}$$

Proof. The condition $C > 1/2$ forces $A + B \le 1$. In the exact cells of Proposition 5.6, the only selected component with $A > \epsilon(e)$ has $B \le \epsilon(e)$; equivalently the two roots of the selected T-cell quadratic separate the feasible fiber. The selector $c \le 2\max\{A, B\}$ removes the formal high root. □

Lemma 5.8 (Strict-supercritical outgoing envelope)**.** *For $0 \le c \le 1/2$, put*

$$M_c^{\sup} = \frac{c + \sqrt{c^2 - 8c + 4}}{2}. \tag{6.8}$$

The value is a supremum and is not attained. Hence every actual strictly supercritical V triangle with own-radial reach at least c has outgoing boundary reach strictly below $M_c^{\sup}$.

Proof. The support-cell elimination and the nonattainment at $a + b = 1$ are given in Appendix A.1.2. □

Lemma 5.9 (Selected-arc chord bounds)**.** *Let $0 < e < 1/2$. On the selected Q_+ component for radial demand $1 - e$, let p be the incoming boundary reach and let q be the following incoming reach forced across a center-free edge. Then*

$$q - p = 1 - \sqrt{1 - x + x^2},$$

where

$$q = e + (1 - e)x, \qquad p = e + (1 - e)x - 1 + \sqrt{1 - x + x^2}. \tag{6.9}$$

The selected arc is increasing and concave. In particular, on every interval used below,

$$q \ge p + \lambda(p - e) \tag{6.10}$$

for the chord slope λ joining the appropriate selected endpoints. The CE1 applications may take

$$\lambda > 1 - 4e \quad (e = \alpha), \qquad \lambda > 1 - 5e \quad (e = \alpha/R). \tag{6.11}$$

Proof. The selected T-cell equation reduces, after putting $x = (q - e)/(1 - e)$, to

$$x(1 - x) = (q - p)(2 - q + p).$$

The selected component takes the smaller root, giving (6.9). Strict concavity follows by differentiation. A concave graph lies above its chords. For the first slope, (6.5) gives

$$\frac{e}{\epsilon(e) - e} > 1 - 4e.$$

For the second, use (6.6) when $e \le 1/8$ and $\epsilon(e) < (1 - e)/2$ otherwise; the resulting slope is greater than $1 - 5e$ throughout $0 < e < 1 - \sqrt{3}/2$. □

5.4. **Exact neighboring-ray capacity.** The T3-like and Vd placements require the exact capacity of a V triangle on a neighboring radial arm. Normalize at V_0, let $A_a \in e_{5,0}$ and $B_b \in e_{0,1}$ have reaches a, b, and let

$$X_c = (1 - c)V_1 \in r_1.$$

Define $C_+(a, b)$ as the maximal c for which V_0, A_a, B_b, X_c lie in a closed unit equilateral triangle. Put $C_-(a, b) = C_+(b, a)$.

For $0 \le a \le 1/2$, let $p(a)$ be the unique root in $[a, 1]$ of

$$p^3 - (a + 2)p^2 + 2(a + 1)p - 1 = 0, \tag{6.24a}$$

and define

$$\sigma(a) = 1 - p(a), \qquad \tau(a) = 1 - a - (p(a) - a)(1 - p(a)).$$

Proposition 5.10 (Neighboring-ray formula)**.**

$$C_+(a,b) = \begin{cases} \textit{undefined}, & a+b>1, \\ 1-b, & a+b\le 1,\ a>1/2, \\ 1-b, & 0\le a\le 1/2,\ 0\le b\le \sigma(a), \\ p(a), & 0\le a\le 1/2,\ \sigma(a)\le b\le \tau(a), \\ a+\dfrac{1}{2}-\sqrt{(a+b)^2-\dfrac{3}{4}}, & 0\le a\le 1/2,\ \tau(a)<b\le 1-a. \end{cases} \tag{6.24b}$$

At $b=\tau(a)$ the plateau value $p(a)$ is selected.

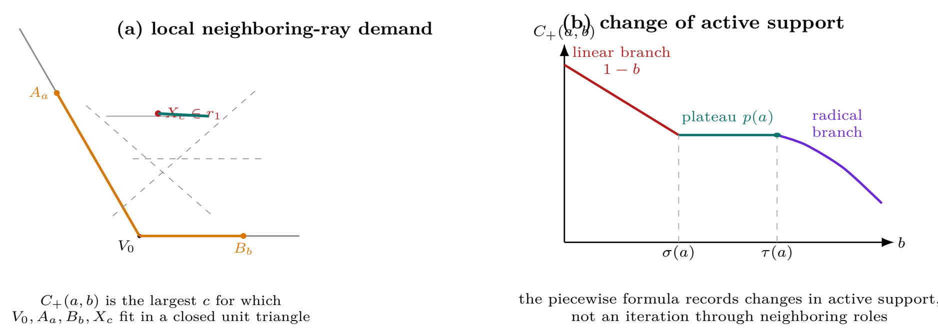


FIGURE 12. Geometry behind the neighboring-ray capacity $C_+(a,b)$. The local problem asks how far a triangle anchored at V_0 and the two boundary points A_a, B_b can reach on the neighboring arm r_1. As b varies with a fixed, the active support pattern changes from the linear branch to a plateau and then to the radical branch, with transition points $\sigma(a)$ and $\tau(a)$. The graph is schematic; the transition order, endpoint selection, and branch labels are exact.

Proof. Use the cone basis

$$E_- = V_5 - V_0, \qquad E_+ = V_1 - V_0.$$

In these coordinates

$$V_0=(0,0), \quad A=(a,0), \quad B=(0,b), \quad X=(c,1),$$

and the metric is u^2+v^2-uv.

Let R, S satisfy $R^2+S^2-RS=1$, and define the support coordinates

$$U(u,v) = (R-S)u + Sv,$$

$$V(u,v) = Su - Rv, \qquad W(u,v) = Ru + (S-R)v.$$

For this orientation, the least enclosing equilateral side is

$$\max_Y W(Y) - \min_Y U(Y) - \min_Y V(Y), \tag{6.24c}$$

where Y ranges over V_0, A, B, X. At a maximal feasible c, one support side contains a hull edge. The non-dominated possibilities are:

(6.24d)

supported edges or contacts	candidate
V_0A and BX	$c = 1 - b,$
AX supported, B inactive	$c = p(a),$
AX supported, B active	$c = a + \dfrac{1}{2} - \sqrt{(a+b)^2 - \dfrac{3}{4}}.$

We verify the three patterns and their selectors.

For the first pattern take $R = S = 1$. The containing triangle is

$$U \geq 0, \qquad V \geq -b, \qquad W \leq c,$$

with side $b + c$. It contains A exactly when $a \leq c$. Hence the unit value is $c = 1 - b$, feasible precisely when $a + b \leq 1$.

For the second pattern put $d = c - a$. The side through A, X gives $R = Sd$, and the unit condition gives $S^2(d^2 - d + 1) = 1$. With B inactive, the opposite support is $W(V_0) = 0$ and $W(X) = -1$. Eliminating S gives

$$(c-a)\big(c^3 - (a+2)c^2 + 2(a+1)c - 1\big) = 0.$$

The nondegenerate root is $c = p(a)$. For $0 \leq a \leq 1/2$ it is unique because the cubic is strictly increasing on $[a, 1]$. The remaining containment condition is

$$b \leq 1 - p(a)(1 + a - p(a)) = \tau(a).$$

This branch dominates $1 - b$ exactly when $b \geq \sigma(a)$.

For the third pattern the active supports give

$$S = -\frac{1}{a+b}, \qquad d^2 - d + 1 = (a+b)^2.$$

The selected root $0 \leq d \leq 1/2$ is

$$d = \frac{1 - \sqrt{4(a+b)^2 - 3}}{2},$$

which gives the radical candidate in (6.24b). The inactive support inequalities reduce to

$$0 \leq a \leq \frac{1}{2}, \qquad \tau(a) \leq b \leq 1 - a.$$

On this interval the radical candidate dominates $1 - b$ because, with $s = a + b \leq 1$,

$$s - \frac{1}{2} \geq \sqrt{s^2 - \frac{3}{4}}.$$

The three selected intervals are ordered since

$$\tau(a) - \sigma(a) = (p(a) - a)p(a) \geq 0, \qquad \tau(a) \leq 1 - a.$$

If $a + b > 1$, none of the three support patterns is feasible, including $c = 0$, so the capacity is undefined. Reflection gives C_-. $\square$

A consequence used repeatedly below is

$$\boxed{C_+(p,q), C_-(p,q) \leq 1 - \min\{p,q\}} \qquad (p + q \leq 1). \tag{6.24e}$$

Indeed this is immediate on the linear branch; on the plateau branch the cubic gives $p(a) \leq 1 - a$, and the radical branch is no larger than its selected plateau at the transition.

5.5. **High-radial output interfaces.** For a nonsupercritical triangle with incoming reach at least a and radial reach at least c, let

$$\mathcal{B}_c(a) = \max\{b : (a,b,c) \text{ is admissible and } a+b \le 1\}.$$

For $1/2 < c \le \sqrt{3}/2$, put

$$h(c) = \frac{2c}{1+\sqrt{16c^2-3}},$$

and for $\sqrt{3}/2 < c < 1$ put

$$\ell(c) = \frac{c}{2}\left(1-\sqrt{4c^2-3}\right), \qquad r(c) = c - \ell(c).$$

Also set

$$Q_-(a,c) = \frac{\sqrt{a^2-ac+c^2}}{c} - a,$$

$$Q_+(a,c) = \frac{2(1-a^2)c^2}{2ac^2+c+\sqrt{(2ac^2+c)^2-4(1-c^2)(1-a^2)c^2}}.$$

Proposition 5.11 (Four direct high-radial outputs)**.** *For $1/2 < c \le \sqrt{3}/2$,*

$$\mathcal{B}_c(a) = \begin{cases} 1-a, & 0 \le a \le 1-c, \\ Q_+(a,c), & 1-c < a < h(c), \\ h(c), & a = h(c), \\ Q_-(a,c), & h(c) < a < c, \\ 1-a, & c \le a \le 1. \end{cases} \tag{6.66}$$

For $\sqrt{3}/2 < c < 1$,

$$\mathcal{B}_c(a) = \begin{cases} 1-a, & 0 \le a \le 1-c, \\ Q_+(a,c), & 1-c < a \le \ell(c), \\ \ell(c), & \ell(c) < a < r(c), \\ Q_-(a,c), & r(c) \le a < c, \\ 1-a, & c \le a \le 1. \end{cases} \tag{6.67}$$

At $a = \ell(c)$, the selected value is $r(c)$; immediately to the right it is $\ell(c)$.

Proof. The exact support-cell derivation, including the genuine downward jump at $a = \ell(c)$, is given in Appendix A.1. □

Lemma 5.12 (Quarter radial envelope)**.** *If*

$$0 \le h_0 \le p, \qquad p + h_0 < 1,$$

then

$$\boxed{c_{\max}(p,h_0) \le 1 - \frac{h_0}{4}.} \tag{6.69}$$

The inequality is strict when $h_0 > 0$.

Proof. See Appendix A.2, where the L and T cells are checked separately. □

Lemma 5.13 (Vd corner radial margins)**.** *In the exact Vd1/Vd2 corner form, let a, b be the boundary reaches, let $t > 0$, and put $d = \sqrt{t^2 + t + 1}$. The own-radial reach c_0 and the far endpoint u_+ on a supported forward adjacent arm satisfy*

$$c_0 = \frac{d - a - tb}{t + 1} < 1 - \min\{a, b\}, \qquad u_+ = \frac{d - a - tb - 1}{t} < 1 - b.$$

The reflected inequalities hold for the backward adjacent arm.

Proof. Since $d < t + 1$,

$$c_0 < 1 - \frac{a + tb}{t + 1} \leq 1 - \min\{a, b\}, \qquad u_+ < 1 - b - \frac{a}{t} < 1 - b.$$

□

5.6. **Radial witnesses.** Suppose $T_i = \overline{U_i}$ contains selected boundary anchors at reaches a_i, b_i. Its exact own-ray capacity is $c_{\max}(a_i, b_i)$. If the preceding role has positive support on r_i, let its exact neighboring-ray capacity be $C_+(a_{i-1}, b_{i-1})$; define the reflected capacity C_- for the following role. Terms not permitted by the actual V type are omitted. Put

$$\Gamma_i = \max\{c_{\max}(a_i, b_i), C_+(a_{i-1}, b_{i-1}), C_-(a_{i+1}, b_{i+1})\}$$

and

$$D_i = (1 - \Gamma_i)V_i. \tag{6.25}$$

Lemma 5.14 (Type-aware radial forcing)**.** *Every D_i is missed by all six open V roles. Hence, under a hypothetical cover,*

$$\boxed{D_i \in U_C.}$$

Proof. If $D_i \in U_i$, openness increases the own-radial demand beyond $c_{\max}(a_i, b_i)$. If an adjacent role contains D_i, openness either creates forbidden positive adjacent support or increases the exact neighboring capacity beyond C_+ or C_-. For the three nonlocal roles, writing $D_i = dV_i$ gives

$$\|dV_i - V_{i\pm 2}\|^2 = 1 + d + d^2 > 1, \qquad \|dV_i - V_{i+3}\| = 1 + d > 1.$$

A unit equilateral triangle has diameter one. Thus every V role misses D_i, and coverage forces it into U_C. □

Lemma 5.15 (Common-pair domination)**.** *Let $p, q \geq 0$ and $p + q \leq 1$. Put*

$$c_* = c_{\max}(p, q), \qquad m = \min\{p, q\}.$$

Then

$$c_* \geq 1 - m, \qquad C_+(p, q), C_-(p, q) \leq 1 - m \leq c_*. \tag{6.26}$$

Consequently any actual local pair dominating (p, q) has all permitted own- and neighboring-ray capacities at most c_*.

Proof. Assume $q = m \leq p$. The demand $(p, q, 1 - q)$ has one exact support side of length $q + \max\{p, 1 - q\} = 1$, so it is admissible; hence $c_* \geq 1 - q$. The exact neighboring-ray formula has linear value $1 - q$, a plateau no larger than $1 - q$, and a final radical branch no larger than its plateau. Thus $C_+(p, q) \leq 1 - q$. Reflection treats the other order and C_-. The last assertion follows from coordinatewise antitonicity. □

5.7. **Common pairs supplied by actual gaps.**

Lemma 5.16 (One-gap common boundary pair)**.** *Assume there is exactly one V-gap*

$$J = [B_0, 1 - A_1] = [\ell, r] \subset e_{0,1}$$

and every V role on the remaining center-free boundary path is nonsupercritical. Put

$$p = 1 - r, \qquad q = \ell.$$

Then every local boundary pair used in the one-gap argument dominates (p, q) coordinatewise.

Proof. Choose open-overlap coordinates x_i on the other five edges. The nonsupercritical inequalities give

$$x_1 \le r, \qquad x_i \le x_{i-1}\ (2 \le i \le 5), \qquad x_5 \ge \ell.$$

The pair at T_1 is (p, x_1); the pair at T_i, $2 \le i \le 5$, is $(1 - x_{i-1}, x_i)$; and the pair at T_0 is $(1 - x_5, q)$. Each coordinate is at least the corresponding coordinate of (p, q). Singleton overlaps follow by the endpoint-continuity lemma. □

Lemma 5.17 (Two-gap CE2 common boundary pair)**.** *In signed CE2 coordinates, suppose both center intervals contain V-gaps. Put*

$$p = W - \alpha, \qquad q = R - \delta.$$

If $T_1, \ldots, T_5$ are nonsupercritical, then every one of these five roles dominates (p, q) coordinatewise.

Proof. The two incident endpoint roles give $A_1 \ge q$ and $B_5 \ge p$. Choose handoffs $x_1, \ldots, x_4$ on the four intervening center-free edges. Nonsupercriticality gives

$$p < x_4 < x_3 < x_2 < x_1 < 1 - q.$$

Reading the incoming and outgoing pair of each of $T_1, \ldots, T_5$ proves the coordinatewise domination. □

5.8. **Universal terminal G: complementary gap.**

Lemma 5.18 (Disk-plus-point formula)**.** *Let $\mathcal{D}_\eta$ be the disk of radius η about O, and let $\|G\| = \rho$. Then*

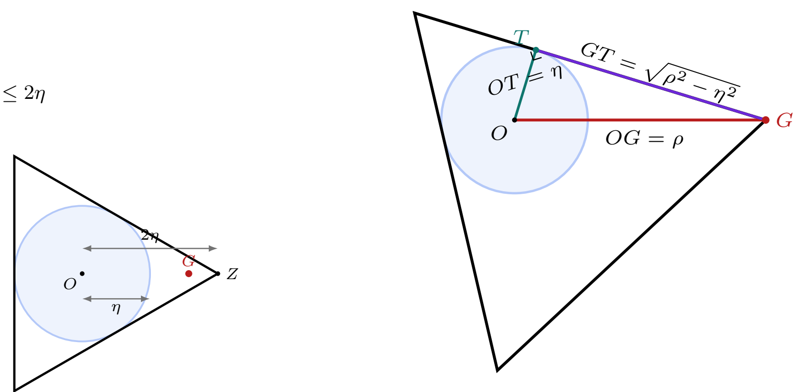
$$\Lambda(\mathcal{D}_\eta \cup \{G\}) = \begin{cases} 3\eta/h, & \rho \le 2\eta, \\[1ex] \dfrac{3\eta + \sqrt{3(\rho^2 - \eta^2)}}{2h}, & \rho > 2\eta. \end{cases} \tag{6.27}$$

Proof. For a unit normal n, write $s_j = \langle G, \mathsf{R}^j n \rangle$. Then $s_0 + s_1 + s_2 = 0$ and $\sum s_j^2 = 3\rho^2/2$. The support sum is $\sum \max\{\eta, s_j\}$. If $\rho \le 2\eta$, orient the point projections as $\rho/2, \rho/2, -\rho$. If $\rho > 2\eta$, an angular minimum occurs when one point projection equals η. The other active projection is

$$\frac{-\eta + \sqrt{3(\rho^2 - \eta^2)}}{2}.$$

Substitution in Proposition 5.5 gives (6.27). □

(a) $\rho \le 2\eta$ **(b)** $\rho > 2\eta$



$G \in [O, Z]$ and the disk-only minimizer still contains G

$\Lambda = \dfrac{3\eta}{h}$

one side through G is tangent to $\mathcal{D}_\eta$

$\Lambda = \dfrac{3\eta + \sqrt{3(\rho^2 - \eta^2)}}{2h}$

FIGURE 13. The two minimizing configurations in the disk-plus-point formula. For $\rho \le 2\eta$, the centered equilateral triangle tangent to $\mathcal{D}_\eta$ already contains G. For $\rho > 2\eta$, the point G becomes a vertex of the minimizing triangle and one incident side is tangent to the disk. The tangent length $GT = \sqrt{\rho^2 - \eta^2}$ is the geometric source of the radical term.

For $p + q \le 1$, let

$$J(p, q) = [X(q), X(1 - p)] \subset e_{0,1}, \qquad X(t) = V_0 + t(V_1 - V_0),$$

and put

$$c_* = c_{\max}(p, q), \qquad d = 1 - c_*.$$

Theorem 5.19 (Complementary-gap enclosure)**.**

$$\boxed{\Lambda\left(\mathcal{D}_{hd} \cup J(p, q)\right) \ge 1.} \tag{6.28}$$

Proof. Put $m = \min\{p, q\}$. The farther endpoint of $J(p, q)$ has squared norm

$$\rho^2 = 1 - m + m^2.$$

By Lemma 5.18, it is enough to prove

$$3c_*(1 - c_*) \ge m(1 - m). \tag{6.29}$$

Lemma 5.15 gives $c_* \ge 1 - m \ge 1/2$. If $c_* \le h$, then the left side of (6.29) is at least $3h(1 - h) > 1/4$, while $m(1 - m) \le 1/4$.

Assume $c_* > h$. In the exact local support formula, let $x = \cos|t - \pi/6|$. The pure radial support gives $x \le h/c_*$. The two mixed supports imply

$$mx + \frac{c_*}{2}\left(x + \sqrt{3}\sqrt{1 - x^2}\right) \le h.$$

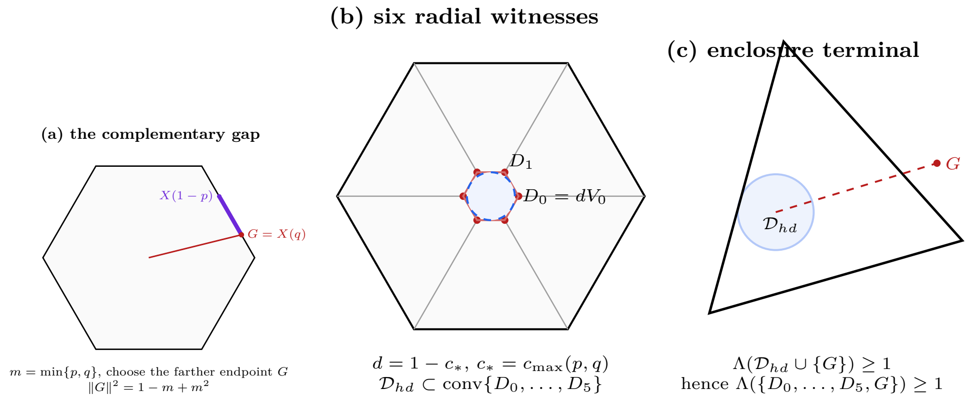


FIGURE 14. The complementary-gap enclosure mechanism. The farther endpoint G of $J(p,q)$ is center-forced. Common-pair domination produces the six center-forced points $D_i = (1-c_*)V_i$, whose convex hull contains the disk $\mathcal{D}_{h(1-c_*)}$. The disk-plus-point formula then forces enclosing side length at least one. The disk is not an additional witness; it is contained in the convex hull of the six finite radial witnesses.

The left side is concave in x. Its value at $x = h$ is greater than h unless $c_* = 1 - m$, in which case (6.29) is immediate. Hence its value at $x = h/c_*$ is at most h, giving

$$m \leq \zeta(c_*) := c_* \left(\frac{1}{2} - \sqrt{c_*^2 - \frac{3}{4}} \right).$$

A direct calculation gives

$$\begin{aligned} &3c_*(1-c_*) - \zeta(c_*)(1-\zeta(c_*)) \\ &= \frac{c_*(1-c_*)}{2} \left(5 - 2c_* - 2c_*^2 + 2\sqrt{c_*^2 - \frac{3}{4}} \right) \geq 0. \end{aligned}$$

Since $0 \leq m \leq \zeta(c_*) < 1/2$, this proves (6.29). $\square$

5.9. **Universal terminal S: CE2 short ray.**

Theorem 5.20 (CE2 short-ray obstruction)**.** *In the strict signed CE2 domain put*

$$p = W - \alpha, \qquad q = R - \delta, \qquad e = \min\{\alpha, \delta\}.$$

Then

$$\boxed{c_{\max}(p,q) < 1 - e.} \tag{6.30}$$

Consequently, whenever both CE2 traces contain V-gaps and the five roles $T_1, \dots, T_5$ realize the common pair (p,q), one of the radial witnesses on r_2, r_4 lies beyond the corresponding center exit.

Proof. The CE2 inequalities are

$$\alpha + W\delta < P, \qquad R\alpha + \delta < P.$$

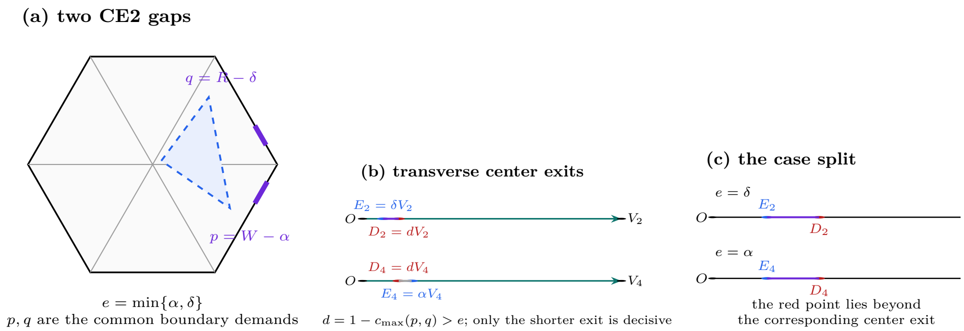


FIGURE 15. The CE2 short-ray terminal. The two boundary gaps determine the common pair (p, q) and the shorter center residual $e = \min\{\alpha, \delta\}$. The theorem proves $d = 1 - c_{\max}(p, q) > e$. Consequently the radial witness on the shorter of r_2 and r_4 lies strictly beyond the corresponding center exit and is uncovered. The displayed numerical ordering is illustrative; the labelled case split is exact.

Using $RW = \eta + P$, they give

$$p, q > \eta + e. \tag{6.31}$$

If $M = \max\{R, W\}$, then

$$e < \frac{P}{1 + M}, \qquad \frac{\eta}{e} > \frac{1 + M}{E} \geq \sqrt{3}.$$

Hence

$$p, q > (1 + \sqrt{3})e. \tag{6.32}$$

Moreover

$$0 < e < e_* := \frac{2\sqrt{3} - 3}{6}. \tag{6.33}$$

Put $c = 1 - e$ and

$$f_c(t) = c^4 - c^2 + ct - t^2.$$

Its roots are

$$t_\pm = \frac{c}{2}\left(1 \pm \sqrt{4c^2 - 3}\right).$$

At $t_0 = (1 + \sqrt{3})e$,

$$f_{1-e}(t_0) = eB(e),$$

where

$$B(e) = e^3 - 4e^2 - 3\sqrt{3}e + \sqrt{3} - 1.$$

The function B decreases on $[0, e_*]$ and

$$B(e_*) = \frac{302\sqrt{3} - 501}{72} > 0.$$

Thus $t_- < (1 + \sqrt{3})e < p, q$. Since $p + q \leq 1 - 2e = c - e$, each of p, q is also below t_+. Therefore

$$f_c(p) > 0, \qquad f_c(q) > 0. \tag{6.34}$$

At radial demand $c = 1 - e$, the four exact support-contact side lengths of $K(p, q, c)$ are all greater than one: two are $p + c, q + c$, and the other two exceed one exactly by (6.34). Thus c is not admissible, proving (6.30).

If $e = \delta$, the point $(1 - c_{\max}(p, q))V_2$ lies farther than δ from O, while the center exit on r_2 is δ. If $e = \alpha$, the analogous point on r_4 lies beyond the exit α. □

Uses: Lemmas 5.16, 5.14, and 5.15.
Closes: Every one-gap $N_+ = 0$ all-Vd0/T3-like row.

Uses: Lemma 5.17 and the signed CE2 exit formula.
Closes: Every two-gap all-Vd0/T3-like row with $N_+ \in \{0, 1\}$.

5.10. No supercritical V role.

Proposition 5.21 (One-gap common-disk closure)**.** *Assume $N_{\text{gap}} = 1$, $N_+ = 0$, no V role is Vd1 or Vd2, and at most two V roles are T3-like. Then the cover is impossible.*

Proof. Normalize the gap to $J = [X(\ell), X(r)]$ and put $p = 1 - r$, $q = \ell$, $c_* = c_{\max}(p, q)$. Lemma 5.16 gives the common pair. Type-aware radial forcing and common-pair domination put all six points $(1 - c_*)V_i$ in U_C, even when the one or two T3-like roles contribute their permitted neighboring term. Their convex hull contains $\mathcal{D}_{h(1-c_*)}$, while the entire complementary gap is in U_C. Theorem 5.19 gives $\Lambda \geq 1$, contradicting Lemma 5.2. □

Case NG0. Hypotheses: $N_{\text{gap}} = 1$, $N_+ = 0$, no Vd1/Vd2, $t \leq 2$.
Forced set or point: $\mathcal{D}_{h(1-c_*)} \cup J(p, q)$.
Terminal: Theorem 5.19.
Closes: Routing rows R7 and R9 with one gap.

Proposition 5.22 (Two-gap common-pair closure)**.** *Assume CE2, both center traces contain V-gaps, and the five intervening roles $T_1, \ldots, T_5$ are nonsupercritical Vd0 or T3-like roles. Then the cover is impossible. The conclusion is independent of the criticality of T_0.*

Proof. Lemma 5.17 supplies $(p, q) = (W - \alpha, R - \delta)$ on $T_1, \ldots, T_5$. Common-pair domination controls every permitted T3-like neighboring term. The type-aware witnesses D_2, D_4 are therefore missed by all V roles, and Theorem 5.20 puts one strictly beyond the corresponding C exit. That point is uncovered. □

Case NG1. Hypotheses: CE2, two gaps, $N_+ \in \{0, 1\}$, intervening roles Vd0 or T3-like.
Forced set or point: D_2, D_4 and the two actual gaps.
Terminal: Theorem 5.20.
Closes: The two-gap portions of routing rows R7, R9, R10, and R11.

Corollary 5.23 (All-Vd0 and T3-like $N_+ = 0$ gap closures)**.** *Every nonzero-gap state with $N_+ = 0$, no Vd1/Vd2 role, and at most two T3-like roles is impossible.*

Proof. There is one or two gaps by the CE classification. Apply Proposition 5.21 or Proposition 5.22. □

5.11. **One gap, one supercritical role, and all V roles Vd0.** Let the gap be $[X(\ell), X(r)]$, and choose strict handoffs

$$x_5 < x_4 < x_3 < x_2 < x_1 < r, \qquad x_5 < \ell.$$

The selected pairs are

$$(a_0, b_0) = (1 - x_5, \ell), \qquad (a_1, b_1) = (1 - r, x_1),$$
$$(a_i, b_i) = (1 - x_{i-1}, x_i), \qquad 2 \le i \le 5.$$

Let C_i be the actual own-radial reach of T_i and put

$$P_i = (1 - C_i)V_i.$$

Lemma 5.24 (The anisotropic set is center-forced)**.** *Every P_i is missed by all six open V roles and lies in U_C. If an open unit triangle U'_C contains*

$$K_{410} = \{O, M_0, X(\ell), X(r), P_0, \dots, P_5\},$$

then U'_C together with the fixed six V roles covers the full skeleton.

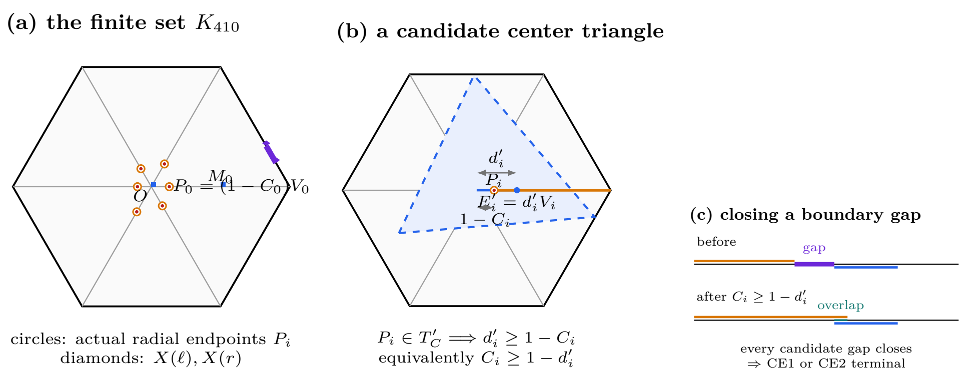


FIGURE 16. The actual-reach mechanism behind the anisotropic set K_{410}. The six points $P_i = (1 - C_i)V_i$ retain the actual own-radial reaches rather than a common relaxed radius. If a candidate center triangle T'_C contains K_{410}, its exit d'_i on every radial arm satisfies $d'_i \ge 1 - C_i$, equivalently $C_i \ge 1 - d'_i$. The corresponding V trace then meets the center contribution and closes the associated boundary gap, forcing the candidate into the CE1 or CE2 terminal used in the proof. Lengths are schematic; the order relations are exact.

Proof. The point P_i is the closed own-trace endpoint and is not in U_i by Lemma 5.1. Adjacent roles are Vd0 and have no positive interval on r_i; nonlocal roles are excluded by diameter. Thus $P_i \in U_C$. Convexity of U'_C covers $[O, P_i]$ and the full gap segment, while U_i covers every point of r_i short of P_i and the six V roles cover the rest of the perimeter. ☐

Lemma 5.25 (Common terminal upper bound at the supercritical role)**.** *Give a candidate U'_C containing K_{410} the signed center variables and put*

$$k = \eta + \alpha + \delta, \qquad Q = \frac{k}{2R}.$$

Then the unique supercritical role T_0 satisfies

$$\boxed{A_0 < 1 - Q.}$$

Proof. The active center trace gives $B_0 \geq k/R$, and containment of P_0 gives $C_0 \geq k$. Discarding the radial point only weakens the diameter condition, so

$$A_0 \leq M_0(k/R) < 1 - \frac{k}{2R} = 1 - Q.$$

□

Proposition 5.26 (CE1 direct reverse-path certificate)**.** *Use the signed CE1 center variables*

$$0 < R < 1, \qquad w = 1 - R, \qquad E = \sqrt{1 - Rw},$$
$$\eta = 1 - E, \qquad P = E(1 - E),$$

and write

$$A = \alpha, \qquad D = \delta.$$

Assume

$$A > 0, \qquad D > 0, \qquad A + wD < P, \qquad D + RA \geq P. \tag{6.12}$$

Put

$$X = R - D, \qquad m = \frac{A}{R}, \qquad h_0 = \frac{\eta + A + D}{2R}. \tag{6.13}$$

Let T_0 be the unique supercritical role and $T_1, \dots, T_5$ nonsupercritical Vd0 *roles in a one-gap skeleton cover. Then*

$$\boxed{A_0 > 1 - h_0.} \tag{6.14}$$

Proof. The direct path $T_5 \to T_4 \to T_3 \to T_2 \to T_1$, its two selected-chord steps, and the final low-root contradiction are proved in Appendix A.3. No reach map is introduced or composed. □

Proposition 5.27 (CE2 anisotropic threshold terminal)**.** *No CE2 open unit triangle contains K_{410}.*

Proof. Put $X = R - \delta$. Appendix A.5.2 proves

$$X > \epsilon(\alpha), \qquad \min\{\epsilon(\alpha), \epsilon(\delta)\} < Q.$$

The five nonsupercritical roles satisfy

$$A_1 \leq A_2 \leq A_3 \leq A_4 \leq A_5 \leq A_0.$$

If $\epsilon(\alpha) < Q$, the direct threshold at T_4 gives $B_4 \leq \epsilon(\alpha)$ and hence $A_5 > 1 - Q$. Otherwise $\epsilon(\delta) < Q$ and the threshold at T_2 gives $B_2 \leq \epsilon(\delta)$ and hence $A_3 > 1 - Q$. Both conclusions contradict Lemma 5.25. □

Proposition 5.28 (CE1 anisotropic reverse-path terminal)**.** *No CE1 open unit triangle contains K_{410}.*

Proof. Proposition 5.26 gives $A_0 > 1 - Q$, whereas Lemma 5.25 gives $A_0 < 1 - Q$. □

Proposition 5.29 (Anisotropic one-gap enclosure)**.** *The compact set K_{410} has*

$$\boxed{\Lambda(K_{410}) \geq 1.}$$

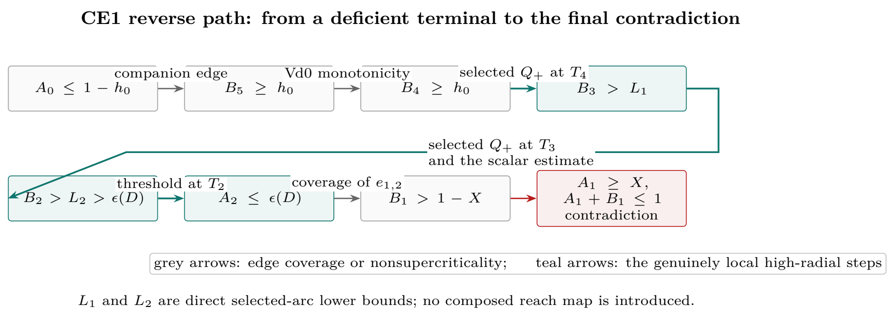


FIGURE 17. Logical structure of the CE1 direct reverse-path certificate. The proof starts from the negation of the desired terminal bound at T_0, moves through the actual boundary reaches, uses two selected-arc estimates and one high-radial threshold, and returns to T_1 with incompatible incoming and outgoing demands.

Proof. An open unit triangle containing K_{410} has a positive boundary trace and is CE1 or CE2. Propositions 5.28 and 5.27 exclude the two alternatives. Lemma 5.2 converts this noncontainment into $\Lambda(K_{410}) \ge 1$. □

Case NG2. Hypotheses: One gap, $N_+ = 1$, all V roles Vd0.
Forced set or point: $K_{410} = \{O, M_0, X(\ell), X(r), P_0, \dots, P_5\}$.
Terminal: CE1 reverse path or CE2 T_4/T_2 threshold dichotomy.
Closes: Routing row R10 with one gap.

5.12. Exactly one T3-like role.

Lemma 5.30 (The T3-like O-side endpoint is center-forced)**.** *Normalize the unique T3-like role to T_0, its supported arm to r_1, and the unique supercritical role to T_1. If the T3-like interval measured from V_1 toward O is $[c, u]$ with $c \le 1/2 \le u$, then*

$$P_T = (1-u)V_1 \in U_C.$$

Proof. The O-side endpoint is not in the open T3-like role. The adjacent supercritical role contains V_1 but misses M_1, so convexity excludes the O-side of M_1. The four remaining roles are Vd0; nonlocal roles fail by diameter. Hence every V role misses P_T. □

Proposition 5.31 (One-gap T3-like rescuer terminal)**.** *Assume $N_+ = 1$, exactly one V role is T3-like, no role is Vd1/Vd2, and there is exactly one V-gap. Then the skeleton is not covered.*

Proof. Use the normalization of Lemma 5.30. Let a be the T3-like boundary reach on $e_{5,0}$. The translated normal form has parameters

$$1 \le D \le \frac{2}{\sqrt{3}}, \quad E_0 = \sqrt{4 - 3D^2}, \quad R_0 = \frac{D + E_0}{2},$$

and

$$c = \frac{D(1+a) - 1}{R_0}, \qquad u = 1 - \frac{R_0}{D} + a.$$

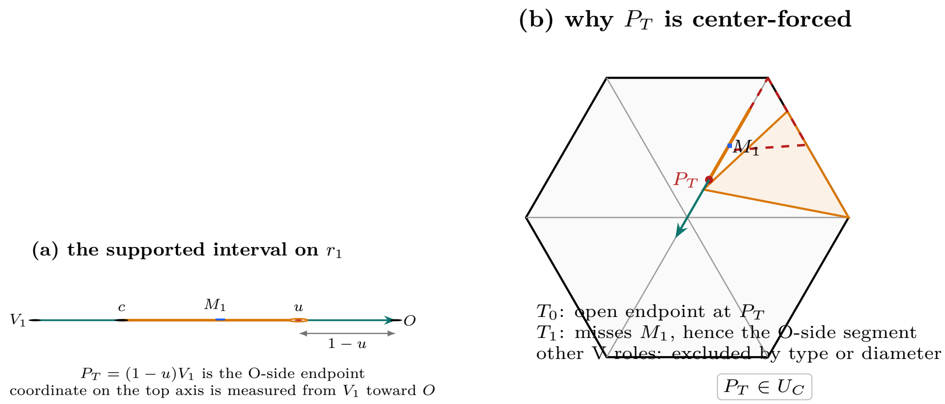


FIGURE 18. Construction of the T3-like rescuer endpoint. The interval $[c, u]$ is parametrized from V_1 toward O, so its O-side endpoint is $P_T = (1 - u)V_1$. The T3-like open role omits its endpoint, the adjacent supercritical role contains V_1 but misses M_1 and therefore misses the O-side segment by convexity, and all remaining V roles are excluded. Thus P_T is forced into the C triangle.

Put $x = aD/R_0$ and $\theta = (D - 1)/R_0$. The midpoint constraints give

$$\frac{1 - 4\theta + \theta^2}{2(1 - 2\theta)} \le x \le \frac{1}{2} - \theta.$$

The elementary quadratic check recorded in Appendix A.4 yields

$$a \le 1 - M_c^{\sup}, \qquad \frac{a}{a + 1 - u} \le 1 - M_c^{\sup}.$$

Together with the center-forced point P_T, these inequalities show that the far boundary reach forced on T_5 is at least $M_c^{\sup}$. The adjacent strictly supercritical role has $B_1 < M_c^{\sup}$ by Lemma 5.8. Adding the four actual edge-cover inequalities gives

$$\sum_{i=2}^{5} (A_i + B_i) > 4,$$

contrary to nonsupercriticality of the four ordinary roles. □

Proposition 5.32 (Two-gap T3-like terminal)**.** *Under the same vertex-type hypotheses, a two-gap state is impossible.*

Proof. Apply Proposition 5.22; the T3-like neighboring capacity is bounded by the common own-ray capacity. □

Corollary 5.33 (Direct T3-like rescuer witness)**.** *Assume $N_+ = 1$, exactly one V triangle is T3-like, no V triangle is Vd1/Vd2, and at least one V-gap is present. Then the skeleton is not covered.*

Proof. Use Proposition 5.31 or Proposition 5.32 according to the gap rank. □

Case NG3. Hypotheses: $N_+ = 1$, exactly one T3-like role, no Vd1/Vd2.
Forced set or point: P_T plus a four-role path in the one-gap case; D_2, D_4 in the two-gap case.

Terminal: Propositions 5.31 and 5.32.
Closes: Routing row R11.

5.13. **Exactly one Vd1/Vd2 role in CE2.** Throughout this subsection let T_σ be the unique supercritical role and T_τ the unique Vd1/Vd2 role. They are distinct. If a further role has positive adjacent support, Method 1 applies through Theorem 3.14; hence the case propositions below assume every other role is Vd0.

Proposition 5.34 (Adjacent one-Vd radial separation)**.** *Assume $\sigma = 0$ and, after reflection, $\tau = 1$. Then a point of r_2 is missed by all seven roles.*

Proof. Let ρ_R, ρ_L be the two residual boundary reaches after removing the supercritical trace and the two center intervals. The Vd cap gives $\rho_R + \rho_L < 1/2$, and Appendix A.5.4 proves $4\delta < \rho_L$. The Vd supported far endpoint satisfies $u_{1\to 2} < 1 - \rho_L < 1 - \delta$. Boundary coverage at the ordinary role T_2 gives $A_2 > 1/2 + \rho_R$ and $B_2 \geq \rho_L$; Lemma 5.12 gives

$$C_2 \leq 1 - \rho_L/4 < 1 - \delta.$$

Thus neither local V role reaches the center interval whose vertex-side entry on r_2 is $1 - \delta$. All other roles are excluded by Vd0 locality and diameter. □

Proposition 5.35 (Nonadjacent one-Vd radial separation)**.** *Assume $\sigma = 0$ and $\tau \in \{2, 3, 4\}$. Then the point*

$$D_\tau = \min\{\rho_R, \rho_L\} V_\tau$$

is missed by all seven roles.

Proof. Appendix A.5.5 proves

$$\alpha + \delta < \min\{\rho_R, \rho_L\}.$$

Every relevant C exit is at most $\alpha + \delta$, while Lemma 5.13 puts the Vd own trace strictly short of D_τ. The neighboring roles are Vd0 and the nonlocal roles fail by diameter. □

Proposition 5.36 (Vd1 neighboring-midpoint rescuer)**.** *Assume $\tau = 0$, T_0 is Vd1, and after reflection $\sigma = 1$. Then the four ordinary path roles have total boundary demand greater than four, a contradiction.*

Proof. Write the supported interval on r_1 as $[c, u]$ and put $P = (1 - u)V_1$. Exactly as in Lemma 5.30, the point P is center-forced. Appendix A.5.6 proves

$$a \leq 1 - M_c^{\mathrm{sup}}, \qquad \frac{a}{a + 1 - u} \leq 1 - M_c^{\mathrm{sup}}.$$

The center cannot hide the far boundary tail; its terminal reach on T_5 is at least M_c^{sup}, whereas the adjacent supercritical role has $B_1 < M_c^{\mathrm{sup}}$. Summing the four actual edge-cover inequalities yields a boundary-sum total greater than four. □

Proposition 5.37 (Two-chart replacement and output-gap rerouting)**.** *Assume $\sigma \neq 0$, $\tau \neq 0$, and the exceptional role is Vd1. The corrected two-chart replacement produces six nonsupercritical Vd0 roles preserving the covered skeleton. After recomputing the output gap rank, every possible output is impossible.*

Proof. Midpoint rescue makes the two distinguished roles adjacent. Perform the replacement in the two correct vertex charts; no preservation of the input gap rank is asserted. If the output has rank zero, six nonsupercritical Vd0 roles cover the perimeter, contradicting the strict boundary-overlap sum. If its rank is one or two, apply Proposition 5.23. □

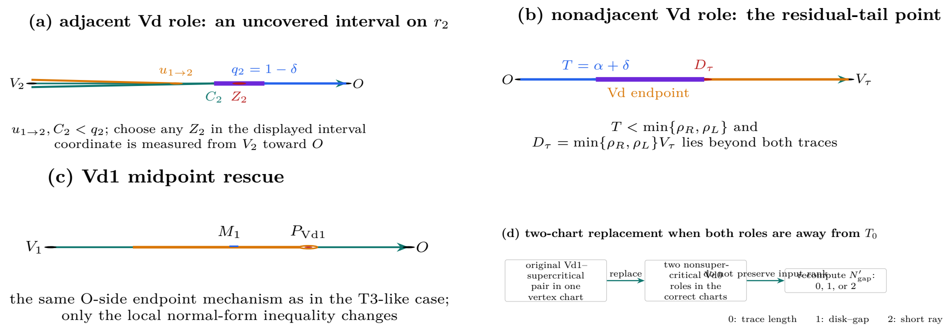


FIGURE 19. The four geometric terminals in the one-Vd placement assembly. In the adjacent placement the radial coordinate is measured from V_2 toward O; the two local V endpoints lie before the center interval entry $q_2 = 1 - \delta$, leaving a literal uncovered interval on r_2. The nonadjacent placement produces the point $D_\tau = \min\{\rho_R, \rho_L\}V_\tau$ beyond both the Vd and center traces; the Vd1 midpoint rescue uses the same O-side endpoint construction as the T3-like case; and the remaining placement is reduced by a two-chart replacement followed by recomputation of the output gap rank. Lengths are schematic; all labelled order relations are the proof relations.

Corollary 5.38 (Direct one-Vd placement assembly)**.** *Assume CE2, $N_+ = 1$, exactly one Vd1/Vd2 role, and at least one V-gap. Then the skeleton is not covered.*

Proof. If a further role has positive adjacent support, use Method 1. Otherwise all other roles are Vd0. If $\sigma = 0$, apply Proposition 5.34 or 5.35. If $\tau = 0$, a Vd2 role is closed by Proposition 3.9, while a Vd1 role is closed by Proposition 5.36. If neither index is zero, Vd2 again uses the Method 1 terminal and Vd1 uses Proposition 5.37. These placements are exhaustive. □

> **Case NG4. Hypotheses:** CE2, $N_+ = 1$, exactly one Vd1/Vd2 role.
> **Forced set or point:** A literal radial point, a Vd1 O-side endpoint, or a replacement output.
> **Terminal:** Four placement propositions plus the Vd2 Method 1 terminal.
> **Closes:** Routing row R13.

5.14. **Nonzero-gap case dispatch.**

Table 2: Reader-oriented nonzero-gap Method 3 dispatch. Each row names its forced object and one terminal theorem.

ID	hypotheses	forced object	terminal
NG0	one gap, $N_+ = 0$, all Vd0/T3-like	common radial disk and actual gap	complementary-gap theorem
NG1	two gaps, $N_+ \in \{0, 1\}$, all Vd0/T3-like	D_2, D_4 and both gaps	CE2 short-ray theorem
NG2	one gap, $N_+ = 1$, all Vd0	K_{410} with six actual radial endpoints	CE1 reverse path or CE2 threshold dichotomy
NG3	one gap, $N_+ = 1$, exactly one T3-like	O-side endpoint P_T and four-role path	T3 rescuer path budget
NG4	two gaps, $N_+ = 1$, exactly one T3-like	D_2, D_4 and both gaps	CE2 short-ray theorem
NG5	one-Vd, adjacent to supercritical T_0	point on r_2 beyond both local traces	adjacent radial separation
NG6	one-Vd, nonadjacent to supercritical T_0	D_τ	nonadjacent radial separation
NG7	Vd based at T_0 or pair away from T_0	Vd1 endpoint or replacement output	rescuer, Method 1, or output-gap rerouting

Proposition 5.39 (All nonzero-gap finite-enclosure rows)**.** *Every nonzero-gap row of Table 1 assigned to Method 3 is impossible.*

Proof. Rows R7 and R9 are Corollary 5.23. Row R10 uses Proposition 5.29 for one gap and Proposition 5.22 for two gaps. Row R11 is Corollary 5.33. The finite-enclosure portion of row R13 is Corollary 5.38; its Method 1 exits are recorded explicitly there. □

6. Zero-Gap Nine-Point Obstruction

This chapter treats routing row R3 separately from the actual-gap toolkit. Assume the six V roles are Vd0, cover the full boundary, and have exactly one actual supercritical role. Six radial points and three asymmetric frontier points are forced directly into the C triangle. Their parameter-dependent coordinates are proved below; the figure records the original repository nine-point schematic and does not replace any calculation.

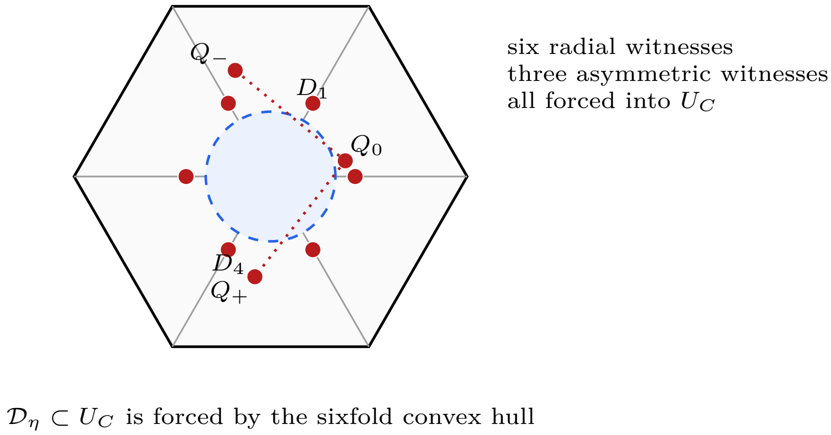


FIGURE 20. The original repository nine-point witness figure: six radial points D_i and three asymmetric frontier points Q_-, Q_0, Q_+. The three asymmetric positions vary with the exact handoff pair; the proof uses their formulas, not the schematic coordinates.

Case ZG0. **Hypotheses:** $N_{\text{gap}} = 0$, $N_+ = 1$, all six V roles Vd0; any CE0/CE1/CE2 center.
Forced set or point: $D_0, \ldots, D_5, Q_-, Q_0, Q_+$.
Terminal: The support-arc enclosure theorem and compact-open shrink.
Closes: Routing row R3.

6.1. **Finite calipers and exact-one handoffs.**

6.2. **The finite caliper criterion.** We first state the finite caliper criterion used below. It enters the proof through the exact statement and verification argument given here; neither a plot nor an unrecorded numerical convention is used.

Let R and J denote counterclockwise rotations through $2\pi/3$ and $\pi/2$, respectively. For a nonempty finite set $S \subset \mathbb{R}^2$ and a vector N, put

$$W_S(N) = \sum_{k=0}^{2} \max_{z \in S} \langle z, \mathsf{R}^k N \rangle. \tag{48}$$

Theorem 6.1 (Finite caliper certificate)**.** *If S has at least two distinct points, then S is contained in a closed unit equilateral triangle if and only if there are distinct $p, q \in S$ and $\varepsilon \in \{-1, 1\}$ for which*

$$W_S\big(\varepsilon \mathsf{J}(q-p)\big) \le \frac{\sqrt{3}}{2} \|p - q\|. \tag{49}$$

If S consists of one point, it is contained in equilateral triangles of arbitrarily small positive side length.

For $S_x = \{O, A, B, x\}$, every clause of (49) *is a finite polynomial clause of degree at most four in the coordinates of A, B, x; at most $12 \cdot 4^3$ clauses are required, and repeated point labels merely delete zero-length pairs.*

Proof. Apply the enclosure gauge of Proposition 5.5 to the polygonal hull of S. A minimizing normal is perpendicular to a hull edge $[p, q]$; after a cyclic rotation of the three normals it is $\varepsilon\mathsf{J}(q-p)/\|p-q\|$. Homogeneity gives (49). A nondegenerate segment is checked directly with its two normals.

For the polynomial expansion fix (p,q,ε), put $N=\varepsilon\mathsf{J}(q-p)$ and $N_k=\mathsf{R}^kN$, and choose support labels $z_0,z_1,z_2\in S_x$. The support cell is

$$\langle z_k,N_k\rangle\ge\langle z,N_k\rangle\quad(z\in S_x,\ k=0,1,2),\tag{50}$$

with $\|p-q\|^2>0$. Since $W_S(N)\ge 0$, the remaining test is

$$4\left(\sum_{k=0}^{2}\langle z_k,N_k\rangle\right)^2\le 3\|p-q\|^2.\tag{51}$$

The points are affine-linear in the parameters, so (50)–(51) have degree at most four. Six unordered pairs, two signs, and 4^3 support-label triples give the asserted finite exhaustive list. □

6.3. **Exact-one handoff geometry.**

Lemma 6.2 (Exact-one handoff chain)**.** *Assume the six V triangles cover ∂H and exactly one actual V triangle is supercritical. Rotate its index to 4. The strict handoffs supplied by Proposition 2.10 satisfy*

$$\xi_3<\xi_2<\xi_1<\xi_0<\xi_5<\xi_4.$$

With

$$a=1-\xi_3,\qquad b=\xi_4,\qquad p=1-b,\qquad q=1-a,$$

one has

$$0<a,b<1,\qquad a+b>1,\qquad a^2+ab+b^2<1,\tag{52}$$

and every selected V triangle obeys

$$a_i\ge p,\qquad b_i\ge q.$$

Proof. Every selected V triangle other than 4 is strictly nonsupercritical, so $\xi_i<\xi_{i-1}$ for $i\neq 4$, which is precisely (6.2). In particular ξ_3 is the global minimum and ξ_4 the global maximum. The two anchors of V triangle 4 lie in its open triangle; their mutual squared distance is a^2+ab+b^2, strictly below the diameter squared of the closed unit triangle. This proves (52). Finally every $\xi_j\in[\xi_3,\xi_4]$, so $1-\xi_{i-1}\ge 1-\xi_4=p$ and $\xi_i\ge\xi_3=q$. □

6.4. **The strict-supercritical AB-union.** Let e_A,e_B be unit vectors with angle 120°, put

$$O=0,\qquad A=ae_A,\qquad B=be_B,\qquad C=\{ue_A+ve_B:u,v\ge 0\},$$

and define

$$\mathcal{R}(a,b)=C\cap\bigcup\{T:T\text{ is a closed unit equilateral triangle and }O,A,B\in T\}.\tag{53}$$

The finite description in Theorem 6.1 applies to the four-point set $\{O,A,B,x\}$ and makes membership in (53) an exact finite semialgebraic condition of degree at most four.

The strict supercritical branch admits a simpler closed form. In the next statement write $x=ue_A+ve_B$ and $\|x\|^2=u^2+v^2-uv$.

Theorem 6.3 (Strict supercritical AB-union)**.** *Assume*

$$a+b>1, \qquad \rho = a^2+ab+b^2<1,$$

put $h=\sqrt{3}/2$, $D=\sqrt{4\rho-3}$, *and define*

$$\alpha = \frac{h(a+2b-aD)}{2\rho}, \qquad \beta = \frac{h(a-b+(a+b)D)}{2\rho},$$
$$\gamma = \frac{h(-a+b+(a+b)D)}{2\rho}, \qquad \delta = \frac{h(2a+b-bD)}{2\rho}.$$

Then all four coefficients are positive and

$$\mathcal{R}(a,b) = (C\cap\mathcal{D}_A\cap\mathcal{H}_A)\cup(C\cap\mathcal{D}_B\cap\mathcal{H}_B), \tag{54}$$

where

$$\begin{aligned}
\mathcal{D}_A &= \{(u,v) : (u-a)^2+v^2-(u-a)v\le 1\},\\
\mathcal{D}_B &= \{(u,v) : u^2+(v-b)^2-u(v-b)\le 1\},\\
\mathcal{H}_A &= \{(u,v) : \alpha(u-a)+\beta v\le 0\},\\
\mathcal{H}_B &= \{(u,v) : \gamma u+\delta(v-b)\le 0\}.
\end{aligned}$$

Its non-axis frontier consists, in counterclockwise order, of the A-circle, the A-line, the B-line, and the B-circle.

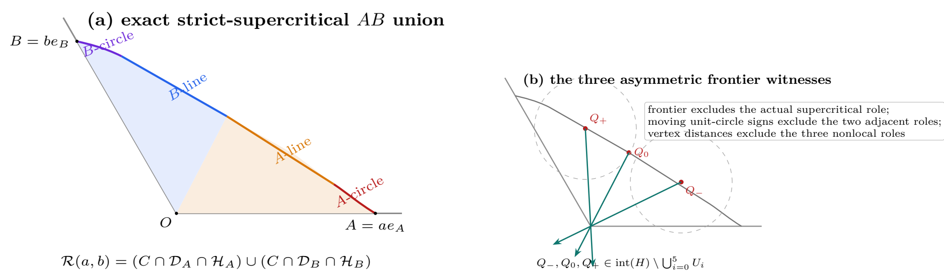


FIGURE 21. The strict-supercritical AB frontier and the three asymmetric witnesses. The source-conditioned union has, in counterclockwise order, an A-circle arc, an A-line segment, a B-line segment, and a B-circle arc. The points Q_-, Q_0, Q_+ are selected on this exact frontier and are then excluded from the six actual V roles by frontier, moving-circle, and vertex-distance arguments. The drawing is schematic; the labelled frontier order and witness roles are exact.

Proof. The identities

$$4\rho-3(a+b)^2=(a-b)^2, \qquad (a+b)^2D^2-(a-b)^2=4\rho((a+b)^2-1)$$

give $0<D<1$ and positivity of the four coefficients.

We apply the caliper criterion of Theorem 6.1. For a point $x=(u,v)$ in the relative interior of C but outside $\triangle OAB$, the four hull vertices occur in the order O,B,x,A. The two cone-axis hull edges cannot certify a unit triangle: their support sums, divided by h, are respectively

$$\max\{a,u\}+\max\{b,v-u\}, \qquad \max\{b,v\}+\max\{a,u-v\},$$

and are at least $a+b>1$. It remains to test the Ax and Bx calipers.

For the Ax caliper put $p=u-a$ and $r^2=p^2+v^2-pv$. Exact evaluation of the three supports gives

$$\frac{G_A}{h}=\frac{\max\{r^2,-ap+(a+b)v\}+N_A}{r},\qquad N_A=\begin{cases}0, & p\le 0,\\ ap, & 0\le p\le v,\\ (a+b)p-bv, & p\ge v.\end{cases}\tag{55}$$

No support label is omitted in these three sectors. If $p\le 0$, $G_A\le h$ is equivalent to

$$r\le 1,\qquad -ap+(a+b)v\le r.$$

The second inequality is the half-plane condition in (54), because

$$r^2-(-ap+(a+b)v)^2=\frac{(cp+dv)(c'p+d'v)}{4\rho},$$

where

$$c=a+2b-aD,\quad d=a-b+(a+b)D,\quad c'=a+2b+aD,\quad d'=a-b-(a+b)D,$$

and $c,c',d>0>d'$. Thus the low sector is exactly $\mathcal{D}_A\cap\mathcal{H}_A$. In the middle sector $0<p\le v$ one has $r\le v$, whereas the numerator in (55) is at least $(a+b)v>r$, so no fit is possible.

In the high sector $p\ge v$, the inequalities $G_A\le h$ imply

$$r^2+(a+b)p-bv\le r,\qquad bp+av\le r.\tag{56}$$

Write $x-A=ry(\theta)$, $0\le\theta\le 60^\circ$, with cone coordinates

$$y(\theta)=te_A+we_B,\quad t=\frac{2}{\sqrt{3}}\cos(\theta-30^\circ),\quad w=\frac{2}{\sqrt{3}}\sin\theta.$$

Then (56) gives

$$r\le f(\theta)=1-(a+b)t+bw,\qquad S(\theta)=bt+aw\le 1.$$

Let n_B be the unit vector with dual coordinates (γ,δ). The identities

$$b\gamma+(a+b)\delta=h,\quad (a+b)\gamma+a\delta=\frac{h}{2}(1+D),\quad b\delta-a\gamma=\frac{h}{2}(1-D)$$

show that the unique angle θ_B of this normal satisfies $S(\theta_B)=1$ and $f(\theta_B)=(1-D)/2$. The function S has at most one critical point, a maximum, so $S(\theta)\le 1$ implies $\theta\le\theta_B$. On the feasible part $f,f'\ge 0$; hence $f(\theta)\cos(\theta_B-\theta)$ is increasing. Consequently

$$\langle n_B,x-B\rangle\le a\gamma-b\delta+\frac{h}{2}(1-D)=0.$$

The diameter bound also gives $\|x-B\|\le 1$. Thus every high sector fit belongs to $\mathcal{D}_B\cap\mathcal{H}_B$. Reflection exchanges a,u with b,v and supplies the corresponding result for the Bx caliper. This proves (54) in the cone interior; the axes follow by taking limits, since both sides are closed.

For completeness, the line–circle junctions are obtained by solving the two displayed line and circle equations. The line normals satisfy

$$\alpha^2+\alpha\beta+\beta^2=h^2,\qquad \gamma^2+\gamma\delta+\delta^2=h^2,$$

and the line determinant is

$$\alpha\delta-\gamma\beta=\frac{3(1-D)(D+3)}{16\rho}>0.$$

The successive cone slopes are

$$+\infty,\quad \frac{2}{1-D},\quad \frac{a(1-D)+2b}{2a+b(1-D)},\quad \frac{1-D}{2},\quad 0;$$

the two middle comparisons reduce to $a(4-(1-D)^2)>0$ and $b(4-(1-D)^2)>0$. This proves the asserted four-piece order. $\square$

6.5. **Six directly forced radial points.** Put $h=\sqrt{3}/2$ and

$$s=p+q,\qquad m=\min(p,q),\qquad M=\max(p,q),\qquad \chi=s^4-s^2+pq.$$

The strict domain (52) gives

$$pq>(1-s)(2-s),\qquad m>1-s,\qquad s>1-m. \tag{57}$$

Let $c_*=c_{\max}(p,q)$ be the exact radial envelope of Proposition 5.6. On the present subcritical domain, equation (6.2) becomes

$$c_*=\begin{cases} C_L(m), & \chi\le 0,\\ \dfrac{2M}{1+\sqrt{4s^2-3}}, & \chi>0,\end{cases} \tag{58}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2,1]$ of

$$z^4-z^2+mz-m^2=0.$$

At $\chi=0$ the two expressions agree and equal s.

Lemma 6.4 (Direct radial forcing)**.** *One has*

$$0<c_*<1,\qquad c_*>1-m. \tag{59}$$

Assume that every U_i is Vd0 *and that $U_C,U_0,\dots,U_5$ cover H. Then, for every i,*

$$D_i=(1-c_*)V_i\in U_C.$$

Consequently U_C contains the centered closed disk

$$\mathcal{D}_\eta=\{x:\|x\|\le\eta\},\qquad \eta=h(1-c_*). \tag{60}$$

Proof. On the C_L branch, $C_L(m)<1$ because the defining polynomial is positive at 1; if $s<\sqrt{3}/2$, then $c_*\ge\sqrt{3}/2>s>1-m$, while if $s\ge\sqrt{3}/2$ its value at s is $\chi\le 0$, so $c_*\ge s>1-m$. On the other branch put $E=\sqrt{4s^2-3}$. The selector gives $sE>M-m$, whence $c_*<s<1$. Moreover, with

$$A_*=\frac{2M}{1-m}-1,$$

one has

$$A_*^2-E^2=\frac{4(1-s)\{(1-m)(1-2m)+m(2-m)(1-s)\}}{(1-m)^2}>0.$$

Thus $A_*>E$ and $c_*>1-m$. This proves (59).

Fix i. If $D_i\in U_i$, openness supplies $\varepsilon>0$ with $c_*+\varepsilon<1$ such that $(1-c_*-\varepsilon)V_i\in U_i$. The same triangle contains its two selected handoff anchors. Since their incoming and forward-reach lower bounds dominate p and q, convexity and passage to the closure give a closed unit triangle realizing $(p,q,c_*+\varepsilon)$. This contradicts the definition of $c_*=c_{\max}(p,q)$. Hence $D_i\notin U_i$.

If D_i belonged to U_{i-1} or U_{i+1}, a small plane ball inside that open triangle would meet r_i in a positive-length interval. This contradicts the Vd0 definition. For the remaining triangles put $r = 1 - c_* > 0$. Directly,

$$\|rV_i - V_{i\pm 2}\|^2 = 1 + r + r^2 > 1, \qquad \|rV_i - V_{i+3}\| = 1 + r > 1.$$

Since $V_j \in U_j$ and two points of an open unit triangle are at distance strictly below 1, these inequalities exclude D_i from the other three V triangles. Thus no V triangle contains D_i. The point lies in $\operatorname{int}(H)$ by (59), so the assumed cover forces $D_i \in U_C$.

The six points D_i form a regular hexagon of circumradius $1 - c_*$. Their convex hull contains its centered incircle of radius $h(1 - c_*)$, and convexity of U_C gives (60). $\square$

We next define the three asymmetric witnesses. At V_4 use local coordinates

$$P = V_4 + u(V_5 - V_4) + v(V_3 - V_4), \qquad \mathsf{q}(u, v) = u^2 + v^2 - uv.$$

The forward-reach lower bound is b and the backward-reach lower bound is a. Accordingly, set

$$\alpha = \frac{h(b + 2a - bD)}{2\rho}, \qquad \beta = \frac{h(b - a + (a + b)D)}{2\rho},$$
$$\gamma = \frac{h(-b + a + (a + b)D)}{2\rho}, \qquad \delta = \frac{h(2b + a - aD)}{2\rho},$$

where now $\rho = a^2 + ab + b^2$ and $D = \sqrt{4\rho - 3}$. The two line pieces of Theorem 6.3 are

$$L_-(\lambda) = (b - \beta\lambda, \alpha\lambda), \qquad L_+(\mu) = (\delta\mu, a - \gamma\mu).$$

They meet at $\lambda_* = \mu_* = 8h\rho/(3(D + 3))$ in

$$J = \left(\frac{2b + a - aD}{D + 3}, \frac{b + 2a - bD}{D + 3}\right). \tag{61}$$

Put

$$E_- = (1 - b, 2 - b), \qquad E_+ = (2 - a, 1 - a).$$

The actual adjacent handoffs have local centers

$$Z_5(t) = (1 + t, t), \qquad 1 - a \le t \le b, \quad \text{and} \quad Z_3(y) = (y, 1 + y), \qquad 1 - b \le y \le a.$$

The restrictions of the two unit-circle equations are

$$g_-(\lambda) = \frac{3}{4}\lambda^2 - 3(\alpha + b\beta)\lambda + 3b^2 - 3b + 2,$$
$$g_+(\mu) = \frac{3}{4}\mu^2 - 3(\delta + a\gamma)\mu + 3a^2 - 3a + 2. \tag{62}$$

Let λ_- and μ_+ be their first roots; explicitly,

$$\lambda_- = 2(\alpha + b\beta) - \frac{2}{3}\sqrt{9(\alpha + b\beta)^2 - 9b^2 + 9b - 6},$$
$$\mu_+ = 2(\delta + a\gamma) - \frac{2}{3}\sqrt{9(\delta + a\gamma)^2 - 9a^2 + 9a - 6}.$$

Lemma 6.5 (Moving adjacent-triangle signs)**.** *Throughout* (52)*,*

$$\lambda_* < \lambda_- < h^{-1}, \qquad \mu_* < \mu_+ < h^{-1}. \tag{63}$$

Moreover $J_u, J_v < 1/2$. The point $L_+(\mu_+)$ satisfies

$$(L_+(\mu_+))_u + (L_+(\mu_+))_v < 3 - 2a,$$

and the reflected bound holds for $L_-(\lambda_-)$*. Writing* $F(P,t) = \mathsf{q}(P - Z_5(t)) - 1$*, one has, for every* $t \in [1-a, b]$*,*

$$F(L_+(\mu_+), t) \geq 0, \qquad F(J,t) > 0, \qquad F(L_-(\lambda_-), t) > 0. \tag{64}$$

Under the reflection $(u,v,a,b) \leftrightarrow (v,u,b,a)$*, the analogous signs hold for every* $y \in [1-b, a]$*: the first-root point* $L_-(\lambda_-)$ *has nonnegative sign against* $Z_3(y)$*, and the other two witnesses have strictly positive sign.*

Proof. Put

$$U = a + b - 1, \qquad z = a - b, \qquad D^2 = z^2 + 6U + 3U^2.$$

Then $0 < U < 1/6$ and $z^2 < \Omega := 1 - 6U - 3U^2$. We give the fixed-line calculation for the circle centered at $E_+ = (2-a, 1-a)$; reflection gives the calculation for E_-. Put

$$F(P,t) = \mathsf{q}(P - (1+t, t)) - 1, \qquad t_0 = 1 - a,$$

so this circle is $F(P, t_0) = 0$. At the junction, exact expansion gives

$$(D+3)^2 F(J, t_0) = 3D(z^2 + Uz + 1 - 3U) + P_J,$$

where

$$P_J = z^4 + 5z^2 + 6Uz^2 + 3U^2z^2 + 9Uz + 3U + 15U^2 > 0;$$

indeed $z^2 + Uz + 1 - 3U \geq 1 - 3U - U^2/4 > 0$, and $5z^2 + 9Uz + 15U^2$ is positive definite. Thus $F(J, t_0) > 0$. Also

$$J_u + J_v = \frac{(1+U)(3-D)}{3+D} < 1.$$

Indeed this inequality is $3U < D(2+U)$, which follows from $D^2 \geq 3U(2+U)$ and $(2+U)^3 > 3U$. Since

$$\frac{\partial F}{\partial t}(P,t) = 1 + 2t - P_u - P_v,$$

$F(J,t)$ is strictly increasing for $t \geq 0$, and therefore is positive throughout $[1-a, b]$.

On the opposite line L_-, the derivative at the junction, after multiplication by a positive factor, has at $t = 1 - a$ the numerator

$$N_0 = D(9 + 3z^2 + 6Uz - 9U^2) + P_0,$$

where

$$\begin{aligned} P_0 = {} & 18U^3 + 6U^2z^2 + 63U^2 + 18Uz^2 + 18Uz + 54U \\ & + 2z^4 + 9z^2 + 9. \end{aligned}$$

The coefficient of D is at least $9 - 6U - 9U^2 > 0$, while $P_0 \geq 9 + 54U - 18U > 0$. Hence $N_0 > 0$. At the other endpoint $t = b$, the corresponding numerator is

$$N_1 = D(9 + 3z^2 + 6U - 3U^2) + P_1,$$

where

$$\begin{aligned} P_1 = {} & 9U^4 - 3U^3z + 45U^3 + 9U^2z^2 - 6U^2z + 72U^2 \\ & - Uz^3 + 21Uz^2 + 9Uz + 45U + 2z^4 + 9z^2 + 9. \end{aligned}$$

Its D-coefficient is positive and, using $|z| < 1$,

$$P_1 \geq 9 + 45U - 9U - U - 6U^2 - 3U^3 > 0.$$

The junction derivative is affine in t, so it is positive throughout $[1-a, b]$. Since the restriction of F to L_- is a convex quadratic, is positive at the junction, and is increasing there, it has no intersection on the opposite line side for any actual adjacent handoff.

On the correct line L_+, the value at its outer endpoint h^{-1} has, after multiplication by a positive factor, numerator

$$A_* = P_2 - 4DU(1 + U + z),$$

where

$$\begin{aligned} P_2 = {} & 3U^4 + 6U^3 z + 6U^3 + 4U^2 z^2 + 12U^2 z + 12U^2 \\ & + 2Uz^3 + 6Uz^2 + 2Uz + 6U + z^4 + 2z^2 - 3. \end{aligned}$$

For fixed U, replace the odd powers of z by $x = |z|$. The resulting polynomial $P_2^+(U, x)$ satisfies

$$\partial_x P_2^+ = 2(3U^3 + 4U^2 x + 6U^2 + 3Ux^2 + 6Ux + U + 2x^3 + 2x) > 0,$$

and therefore its supremum for $z^2 < \Omega$ occurs at $x = \sqrt{\Omega}$. There

$$P_2^+(U, \sqrt{\Omega}) = 4U(U + \sqrt{\Omega} - 3) < 0.$$

As $1 + U + z = 2a > 0$, this gives $A_* < 0$. The positive junction value and negative endpoint value force the first root strictly between them, proving the L_+ half of (63); reflection proves the other half.

For the coordinate bounds, the exact identities

$$D^2 - (3U - z)^2 = 6U(1 + z - U), \qquad D^2 - (3U + z)^2 = 6U(1 - z - U)$$

give $J_u, J_v < 1/2$. It remains to control the coordinate sum along L_+. For $E = L_+(h^{-1})$, the sign of $3 - 2a - E_u - E_v$ is the sign of

$$D(6U + 2z + 6) + B(U, z),$$

where

$$\begin{aligned} B(U, z) = {} & -9U^3 - 9U^2 z - 9U^2 - 3Uz^2 - 18Uz + 3U \\ & - 3z^3 + 3z^2 - 3z + 3. \end{aligned}$$

Here

$$B_z = -3(3z^2 + 2Uz + 6U - 2z + 3U^2 + 1) < 0,$$

because the quadratic in parentheses has discriminant $-32U^2 - 80U - 8 < 0$. Hence its minimum on $z^2 < \Omega$ is at $z = \sqrt{\Omega}$, where $B(U, \sqrt{\Omega}) = 6(1 - 3U - \sqrt{\Omega}) > 0$ because $(1 - 3U)^2 - \Omega = 12U^2$. Thus $E_u + E_v < 3 - 2a$. Also $J_u + J_v < 1 < 3 - 2a$, and the coordinate sum is affine on the line segment, so the same strict bound holds at the first root. Now

$$\frac{\partial F}{\partial t}(P, t) = 1 + 2t - P_u - P_v.$$

Since $F(L_+(\mu_+), 1 - a) = 0$, the coordinate-sum bound makes its t-derivative strictly positive for $t \geq 1 - a$. Together with the no-opposite-line result, this proves (64). Reflection gives both the bound on L_- and the complete $Z_3(y)$ statement. $\square$

Define the local and global witnesses by

$$Q_-^{\mathrm{loc}} = L_-(\lambda_-), \qquad Q_0^{\mathrm{loc}} = J, \qquad Q_+^{\mathrm{loc}} = L_+(\mu_+), \tag{65}$$

and by the displayed conversion from (u, v) to the plane.

Lemma 6.6 (Direct asymmetric forcing)**.** *Assume that $U_C, U_0, \dots, U_5$ cover H. Then*

$$Q_-, Q_0, Q_+ \in \operatorname{int}(H) \setminus \bigcup_{i=0}^{5} U_i,$$

and consequently all three witnesses belong to U_C.

Proof. The root order and positivity of the line coordinates give $0 < u, v < 1$ for all three points. Each belongs to $\mathcal{R}(b, a)$ and hence lies in a closed unit triangle containing V_4, so $\mathsf{q}(u, v) \le 1$; hence $|u - v| < 1$ and all three points lie in $\operatorname{int}(H)$.

The closure of U_4 is one of the triangles represented by $\mathcal{R}(b, a)$, since it contains V_4, X_3, X_4. All three witnesses lie on its exact non-axis frontier. If one belonged to the open triangle U_4, a sufficiently small plane ball about it would lie both in U_4 and in the interior of the corner cone, making the witness an interior point of $\mathcal{R}(b, a)$. This contradiction excludes U_4.

For the triangles U_0, U_1, U_2, their contained vertices have local coordinates

i	0	1	2
V_i	$(2,1)$	$(2,2)$	$(1,2)$.

For $P = (u, v)$,

$$\|P - (2,1)\|^2 - 1 = \mathsf{q}(u, v) + 2 - 3u, \quad \|P - (1,2)\|^2 - 1 = \mathsf{q}(u, v) + 2 - 3v.$$

The inequalities $J_u, J_v < 1/2$ and the line order give distance greater than one from V_0 for Q_-, Q_0 and from V_2 for Q_0, Q_+. For the remaining pair, the first-root circle and the sum bound in Lemma 6.5 give

$$\|Q_+ - V_0\|^2 - 1 = a(3 - a - (Q_+)_u - (Q_+)_v) > a^2,$$

and, by reflection, $\|Q_- - V_2\|^2 - 1 > b^2$. All three points are farther than one from $V_1 = (2, 2)$, because writing $u = 1 - \xi$, $v = 1 - \eta$ gives

$$\|P - V_1\|^2 = 1 + \xi + \eta + \xi^2 + \eta^2 - \xi\eta > 1.$$

Since two points in one open unit equilateral triangle have distance strictly less than one, these inequalities exclude U_0, U_1, U_2.

The actual handoff $X_5 = Z_5(\xi_5)$ belongs to U_5, and $1 - a < \xi_5 < b$ by (6.2). The three signs in (64) therefore put every witness at distance at least one from X_5, excluding U_5. Similarly, if $y = 1 - \xi_2$, then $1 - b < y < a$, $X_2 = Z_3(y) \in U_3$, and the reflected signs exclude U_3. Thus no V triangle contains any witness. The assumed cover now forces all three interior witnesses into U_C. ☐

The nine directly forced points are $D_0, \dots, D_5, Q_-, Q_0, Q_+$. Under a putative cover, Lemmas 6.4 and 6.6 therefore force the compact set

$$K_{\mathrm{wit}}(a, b) = \mathcal{D}_\eta \cup \{Q_-, Q_0, Q_+\} \subset U_C. \tag{66}$$

6.6. **Tangent reduction and exact mixed overlaps.** For the nontrivial regime $c_* > 2/3$, we first remove the square radicals in the first-root witnesses. Normalize $\bar{\alpha} = \alpha/h$, and similarly for the other three coefficients, and put

$$\xi = \lambda/h, \quad \zeta = \mu/h, \quad \xi_* = \zeta_* = \frac{8\rho}{3(D+3)}.$$

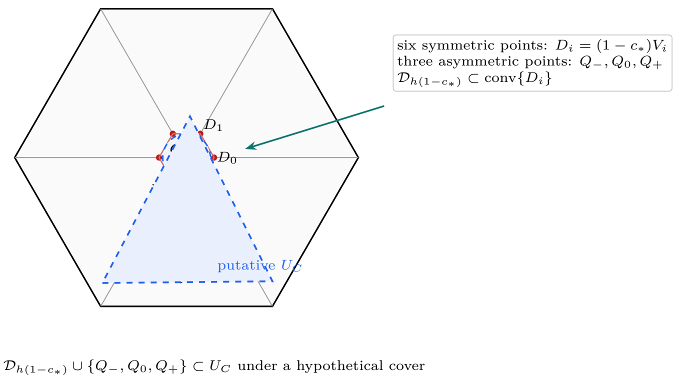


Figure 22. The zero-gap nine-point witness. The six symmetric radial points force a centered disk into the convex C triangle, while the three asymmetric frontier points supply the missing directional support. Marker shapes separate the two constructions: circles are radial witnesses and triangles are strict-supercritical AB-frontier witnesses. The displayed placement is illustrative; the point definitions are the exact ones in the preceding lemmas.

Let g_-, g_+ be the rescaled forms of (62), and take one Newton step at the junction:

$$\widehat{\xi}_- = \xi_* - \frac{g_-(\xi_*)}{g_-'(\xi_*)}, \qquad \widehat{\zeta}_+ = \zeta_* - \frac{g_+(\zeta_*)}{g_+'(\zeta_*)}.$$

Define, in local coordinates,

$$\begin{aligned} A &= \left(b - \frac{3}{4}\bar{\beta}\widehat{\xi}_-, \frac{3}{4}\bar{\alpha}\widehat{\xi}_-\right), \\ B &= J, \\ C &= \left(\frac{3}{4}\bar{\delta}\widehat{\zeta}_+, a - \frac{3}{4}\bar{\gamma}\widehat{\zeta}_+\right), \end{aligned}$$

and convert them to global vectors.

Lemma 6.7 (Newton inner reduction)**.** *The points above have coordinates rational in a, b, D and satisfy*

$$A \in (Q_0, Q_-), \qquad C \in (Q_0, Q_+).$$

Consequently

$$\widehat{K} = \mathrm{conv}(\mathcal{D}_\eta \cup \{A, B, C\}),$$

obeys $\widehat{K} \subseteq \mathrm{conv}(K_{\mathrm{wit}})$.

Proof. Each quadratic g_-, g_+ is strictly convex, positive, and decreasing at the junction. Its tangent zero therefore lies strictly between the junction and its first root. This gives the two open-segment inclusions. All displayed coefficients and Newton quotients are rational in a, b, D, and convexity gives the asserted containment. □

Put $W = C + RA$. For a disk $\mathbb{D}(X, r)$ let

$$I(X, r) = \{n : \|n\| = 1,\ \langle X, n\rangle + r \geq h\}.$$

6.7. **Ray order and four overlaps.**

Proposition 6.8 (Technical four-overlap verification)**.** *Assume* (52) *and* $c_* > 2/3$. *Then*

$$\|A\|, \|C\| > \eta,$$

the rays A, B, C, W, RA *occur in that cyclic order, and the four consecutive cap pairs*

$$I(A, 2\eta) \leftrightarrow I(B, 2\eta) \leftrightarrow I(C, 2\eta) \leftrightarrow I(W, \eta) \leftrightarrow I(RA, 2\eta)$$

overlap across their intervening short sectors.

Proof. Put $\tau = h - 2\eta = h(2c_* - 1)$. The rays of

$$A,\ B,\ C,\ W,\ RA \tag{67}$$

occur in this cyclic order from A to RA. Indeed the two line distances give

$$\frac{\operatorname{cross}(A, B)}{\|B - A\|} = \alpha(1 - b) + \beta > 0, \quad \frac{\operatorname{cross}(B, C)}{\|C - B\|} = \gamma + \delta(1 - a) > 0.$$

If $A = (-X, -Y)$ and $C = (-P, -Q)$ in the centered cone basis, then $0 < X, Y, P, Q < 1$, $X, Q > 1/2$, and

$$\frac{\operatorname{cross}(C, RA)}{h} = PX + (Q - P)Y > 0.$$

The last sign is immediate unless $Q < P$ and $X < Y$, in which case it is larger than $Q - P/2 > 0$. The ray of $W = C + RA$ lies between those of C and RA. The same coordinates give

$$\|A\|, \|C\| > h/2 > \eta. \tag{68}$$

For the two equal-radius overlaps it is enough to prove

$$\frac{\operatorname{cross}(A, B)}{\|B - A\|} \geq \tau, \qquad \frac{\operatorname{cross}(B, C)}{\|C - B\|} \geq \tau. \tag{69}$$

We now give the exact analytic verification. By reflection take $a \leq b$ and set $m = 1 - b$, $M = 1 - a$, $U = a + b - 1$, $s = m + M$, and $w = M - m$. The smaller of the two normalized left sides in (69) is

$$d = \frac{\mathcal{A} + U(2m - 1) + D(\mathcal{A} + U)}{2\rho}, \qquad \mathcal{A} = 1 - m + m^2. \tag{70}$$

On the C_L sheet, write $F_L(z) = z^4 - z^2 + mz - m^2$. This defining polynomial evaluated at $r = 1 - m/2 + m^2/2$ is

$$F_L(r) = \frac{m^2(m - 1)}{16}(m^5 - 3m^4 + 11m^3 - 17m^2 + 28m - 12) > 0.$$

Indeed the polynomial in parentheses is increasing on $(0, 1/2)$ and has value $-33/32$ at $1/2$; its derivative is $28 - 34m + m^2(5m^2 - 12m + 33) > 0$. Since $F_L(h) = -(m - h/2)^2 \leq 0$ and the selected root is unique, $2c_* - 1 \leq \mathcal{A}$. Put

$$X_0 = \mathcal{A} + U, \qquad Y_0 = 2\rho\mathcal{A} - \mathcal{A} - U(2m - 1).$$

Then

$$d - \mathcal{A} = \frac{D(\mathcal{A} + U) - Y_0}{2\rho},$$

and exact expansion gives

$$\begin{aligned} Y_0 &= 2(m^2-m+1)U^2+(2m^3-2m+3)U \\ &\quad +(m^2-m+1)(2m^2-2m+1)>0. \end{aligned}$$

Furthermore

$$D^2X_0^2-Y_0^2=4m\rho\Phi(U,m),$$

where

$$\begin{aligned} \Phi(U,m) &= (1-m)(m^2-m+2)U^2 \\ &\quad +(2-m^2)(m^2-m+1)U \\ &\quad -m(1-m)^2(m^2-m+1). \end{aligned}$$

In these variables

$$\chi=U^4-4U^3+5U^2-(m+2)U+m(1-m).$$

Since $U^2(U^2-4U+5)>0$, the selector $\chi\le 0$ implies $U>m(1-m)/(m+2)$; Φ is increasing in U and at that lower endpoint is

$$\frac{m^2(1-m)(m^4-2m^3+6m^2-6m+4)}{(m+2)^2}>0.$$

The last quartic equals $(m^2-m)^2+5m^2-6m+4>0$. Thus $d>\mathcal{A}\ge 2c_*-1$.

On the other radial sheet let $E=\sqrt{4s^2-3}$. Then $0\le w<sE$ and $c_*=(s+w)/(1+E)$. At fixed U, differentiation of (70) can be checked without an implicit estimate. Namely,

$$d=\frac{1+D}{2}-UB(D,w), \qquad B(D,w)=\frac{(1+U)(3+D)+w(1-D)}{D^2+3}.$$

Here $D^2=w^2+6U+3U^2$, so $0\le D'=w/D\le 1$, and

$$B_w=\frac{1-D}{D^2+3}\ge 0.$$

Writing $N=(1+U)(3+D)+w(1-D)$, direct differentiation gives

$$B_D=\frac{(1+U-w)(D^2+3)-2DN}{(D^2+3)^2}>-\frac{34}{27}.$$

Indeed the first term in the numerator is positive, while $1+U-w=2a>0$, $N<(7/6)4+1=17/3$, $D<1$, and $D^2+3\ge 3$. If $B_D<0$, then $B_DD'\ge B_D$; otherwise $B_DD'\ge 0$. Hence $B'=B_w+B_DD'>-34/27$, and $U<1/6$ yields

$$d'<\frac{1}{2}+\frac{1}{6}\frac{34}{27}=\frac{115}{162}<1,$$

whereas $(2c_*-1)'=2/(1+E)\ge 1$. Hence their difference decreases with w. At the boundary $w=sE$, where $\chi=0$ and $c_*=s$, the preceding sheet applies: this is an admissible point because $h<s<1$ and $D^2=4s^4-12s+9<1$. Indeed $s^4-3s+2=(s-1)(s^3+s^2+s-2)<0$ on this interval; the second factor is increasing and already equals $(7h-5)/4>0$ at $s=h$. The preceding sheet gives $d\ge 2s-1$. Therefore (69) holds on both sheets, and reflection supplies the other line.

It remains to check the two mixed overlaps. The exact rational envelopes, Gram reductions, and global positive-basis identities are recorded once in Appendix C. Theorem C.3 gives precisely the overlaps for $\mathbb{D}(C,2\eta)$ with $\mathbb{D}(W,\eta)$ and for $\mathbb{D}(W,\eta)$ with $\mathbb{D}(RA,2\eta)$.

Thus the four consecutive pairs of caps centered on the rays (67) overlap. □

6.8. **Support arcs and the enclosure conclusion.** Let R denote counterclockwise rotation through $2\pi/3$. Retain the points A, B, C, the vector $W = C + \mathsf{R}A$, the forced disk $\mathcal{D}_\eta$, and the exact witness set K_{wit} constructed above. For uniform notation put

$$P_- = A, \qquad P_0 = B, \qquad P_+ = C, \qquad P_* = W, \qquad r_* = \eta,$$

and

$$\widehat{K} = \operatorname{conv}\bigl(\mathcal{D}_\eta \cup \{A, B, C\}\bigr).$$

For $X \in \mathbb{R}^2$ and $r \geq 0$, write

$$\mathbb{B}(X, r) = \{Y : \|Y - X\| \leq r\},$$

and define the level-h, $h = \sqrt{3}/2$, support arc

$$\operatorname{Cap}(X, r) = \{n \in \mathbb{S}^1 : \langle X, n\rangle + r \geq h\}.$$

Lemma 6.9 (Cyclic support-arc covering)**.** *Let K be compact and put*

$$\Sigma_K = K + \mathsf{R}K + \mathsf{R}^2K.$$

Suppose Σ_K contains the full R-orbits of

$$\mathbb{B}(P_-, 2r_*), \quad \mathbb{B}(P_0, 2r_*), \quad \mathbb{B}(P_+, 2r_*), \quad \mathbb{B}(P_*, r_*).$$

Assume that the rays

$$P_-, P_0, P_+, P_*, \mathsf{R}P_-$$

occur in this cyclic order, that every corresponding support arc is connected with half-width less than $\pi/2$, and that each consecutive pair of arcs intersects across the intervening short angular sector. Then

$$\Lambda(K) \geq 1.$$

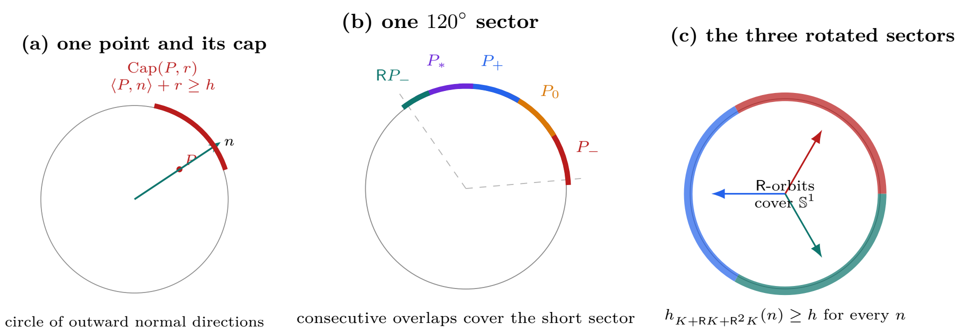


FIGURE 23. The support-cap conclusion for the zero-gap witness. A point and a disk radius determine an arc of normal directions in which their support is at least h. The four consecutive overlap checks cover the sector from the ray of P_- to that of $\mathsf{R}P_-$, and the three $120°$ rotations cover the full normal circle. The arc widths are schematic; the cyclic order and overlap pattern are the exact hypotheses of the support-arc lemma.

Proof. The four consecutive intersections and the cyclic order make the five arcs cover the sector from the ray of P_- to the ray of $\mathsf{R}P_-$. Their R-orbits cover $\mathbb{S}^1$. Hence for every unit vector n one of the displayed disks has support at least h in direction n, and therefore

$$h_{\Sigma_K}(n) \geq h.$$

Since

$$h_{\Sigma_K}(n) = \sum_{j=0}^{2} h_K(\mathsf{R}^j n),$$

Proposition 5.5 gives $\Lambda(K) \geq 1$. □

Theorem 6.10 (Witness enclosure)**.** *For every exact-one-supercritical parameter pair (a, b) constructed above,*

$$\Lambda(K_{\mathrm{wit}}(a, b)) \geq 1.$$

Proof. The centered disk alone gives

$$\Lambda(K_{\mathrm{wit}}) \geq 3(1 - c_*),$$

so the result follows when $c_* \leq 2/3$. Assume $c_* > 2/3$. Lemma 6.7 gives $\widehat{K} \subseteq \mathrm{conv}(K_{\mathrm{wit}})$. The Minkowski sum $\widehat{K} + \mathsf{R}\widehat{K} + \mathsf{R}^2\widehat{K}$ contains the disk orbits listed in Lemma 6.9. Proposition 6.8 supplies the ray order, radius bounds, and equal-radius intersections; Theorem C.3 supplies the two mixed intersections. Lemma 6.9 therefore gives $\Lambda(\widehat{K}) \geq 1$, and monotonicity yields $\Lambda(K_{\mathrm{wit}}) \geq 1$. □

Theorem 6.11 (Zero-gap obstruction)**.** *There is no cover of H by open unit equilateral triangles $U_C, U_0, \ldots, U_5$ such that every U_i is* Vd0*, the six vertex triangles cover ∂H, and exactly one V triangle is supercritical.*

Proof. The construction above forces the compact set K_{wit} into the open C triangle U_C, whereas Theorem 6.10 gives $\Lambda(K_{\mathrm{wit}}) \geq 1$. Every compact subset of an open unit equilateral triangle is contained in a closed equilateral triangle of side strictly below one, which would imply $\Lambda(K_{\mathrm{wit}}) < 1$. □

Proposition 6.12 (Cases excluded by the nine-point obstruction)**.** *Theorem 6.11 excludes every $N_{\mathrm{gap}} = 0$, $N_+ = 1$, all-*Vd0 *case, independently of the center class.*

Proof. Lemma 2.8 says that $N_{\mathrm{gap}} = 0$ is equivalent to coverage of ∂H by the six V triangles. The all-Vd0 and unique-supercritical hypotheses are exactly those of Theorem 6.11. □

7. Completion of the Proof

Proof of Theorem 1.1. Assume that seven open unit equilateral triangles cover H. By Proposition 2.14, the cover lies in exactly one row of Table 1.

First suppose $N_{\mathrm{gap}} = 0$. The $N_+ = 0$ row and the $N_+ = 1, d \geq 1$ row are excluded by Proposition 3.10. The $N_+ \geq 2$ row and the $N_+ = 1, d = 0, t \geq 1$ row are excluded by Proposition 4.5. The remaining $N_+ = 1, (d, t) = (0, 0)$ row is excluded by the explicit finite-enclosure theorem Proposition 6.12.

Now suppose $N_{\mathrm{gap}} \geq 1$. The rows assigned to trace length are excluded by Proposition 3.10. Every nonzero-gap Method 3 row is excluded by the reader-oriented finite-enclosure assembly, Proposition 5.39. In the final CE2 one-Vd1/Vd2 row, the

Method 1 complements and the finite-enclosure placements are assembled in Proposition 5.38; its replacement subcase recomputes the gap rank before selecting the closing strategy. Thus every zero- and nonzero-gap row is impossible, contradicting the assumed cover. □

Corollary 1.2 follows from Proposition 2.1.

Appendix A. Finite-Enclosure Calculation Details

This appendix contains the scalar and selector calculations suppressed from the reader-facing Method 3 chapter. The body retains every theorem interface and every geometric implication.

A.1. **Four direct high-radial outputs.**

Proof of Proposition 5.11. Since $c > 1/2$, any containing V triangle contains the radial midpoint. Therefore $a + b \leq 1$, and the supercritical cell of Proposition 5.6 is absent. On the cap $b = 1 - a$, the T-cell inequality becomes

$$M(c - M) \leq 0,$$

where $M = \max\{a, b\}$. Thus the linear cap is feasible exactly when $a \leq 1 - c$ or $a \geq c$.

Off the linear cap, a maximal point lies on $F_L = 0$ or $F_T = 0$. The selected L-cell equation has the two ordered roots $\ell(c), r(c)$; the selector $q \leq 0$ retains the smaller coordinate. The ordered halves of $F_T = 0$ give Q_+ and Q_-. Their equality boundary $a = b$ is $h(c)$ in the low-radial range, while their $q = 0$ boundaries are $(\ell(c), r(c))$ and $(r(c), \ell(c))$ in the high range. These conditions produce (6.66)–(6.67).

The formal other root in the Q_+ quadratic violates the geometric selector $c \leq 2\max\{a, b\}$ and is discarded. At the high-range junction, the selected T-cell point has value $r(c)$, whereas the adjacent selected L-cell point has value $\ell(c)$; this is the genuine downward jump stated above. $\square$

A.1.1. *Low-root and threshold calculations.* Let $c = 1 - e$ with $0 < e < 1 - \sqrt{3}/2$. The selected low root $\epsilon(e) = \ell(1 - e)$ satisfies the quadratic

$$p_e(z) = z^2 - (1 - e)z + (1 - e)^2 - (1 - e)^4 = 0. \tag{6.68}$$

Substitution gives

$$p_e(e) = e(1 - 3e + 4e^2 - e^3) > 0.$$

Since $e < (1 - e)/2$, the point e lies below the smaller root. For $U = 2e/(1 - 2e)$,

$$p_e(U) = -\frac{e^2}{(1 - 2e)^2}(4e^4 - 20e^3 + 37e^2 - 28e + 3).$$

The polynomial in parentheses is positive on the required interval, proving

$$e < \epsilon(e) < \frac{2e}{1 - 2e}.$$

For $e \leq 1/8$,

$$p_e(2e + 5e^2) = e^2(3e + 4)(8e - 1) \leq 0,$$

which proves (6.6).

Now let $A > \epsilon(e)$ and $C \geq 1 - e$. The linear branch of Proposition 5.11 is no longer available. On the selected constant and Q_- pieces, the outgoing reach is at most $\epsilon(e)$. On the Q_+ piece, the selected quadratic has its outgoing coordinate below the same root. Thus $B \leq \epsilon(e)$, proving the direct threshold Lemma 5.7 with no following-edge map.

A.1.2. *The strict-supercritical envelope.* Fix $0 \le c < 1/2$ and maximize the outgoing boundary reach over strict supercritical pairs. The maximum lies on the S-cell frontier and at the boundary where the incoming coordinate tends to the complementary value. Eliminating that coordinate from $F_S = 0$ gives

$$b^2 - cb + 2c - 1 = 0.$$

The larger selected root is

$$M_c^{\mathrm{sup}} = \frac{c + \sqrt{c^2 - 8c + 4}}{2}.$$

The corresponding equality state lies on the boundary $a + b = 1$, so it is not strictly supercritical. Therefore every actual strict-supercritical role has

$$B < M_c^{\mathrm{sup}}.$$

This strict nonattainment is the only supercritical output used in the T3-like and Vd1 rescuer proofs.

A.2. **Quarter radial envelope.**

Proof of Lemma 5.12. Only the L and T cells of Proposition 5.6 can occur. In the L cell, the selected radial root is the unique root of

$$P_{h_0}(c) = c^4 - c^2 + h_0 c - h_0^2.$$

At $c_0 = 1 - h_0/4$,

$$P_{h_0}(c_0) = \frac{h_0}{256}(h_0^3 - 16h_0^2 - 240h_0 + 128) > 0$$

for $0 < h_0 \le 1/2$. Since

$$P_{h_0}(\sqrt{3}/2) = -(h_0 - \sqrt{3}/4)^2 \le 0$$

and P_{h_0} is increasing on the selected root interval, the selected root is below c_0.

In the T cell put $s = p + h_0$ and $r = \sqrt{4s^2 - 3}$. Write

$$\gamma = \frac{p - h_0}{s} = wr, \qquad 0 \le w < 1.$$

Then

$$p = \frac{s(1 + wr)}{2}, \qquad h_0 = \frac{s(1 - wr)}{2},$$

and

$$c_T = \frac{s(1 + wr)}{1 + r}.$$

For fixed s, r, the function $c_T + h_0/4$ increases in w, so

$$c_T + \frac{h_0}{4} < \frac{s(9 - r)}{8}.$$

Using $4s^2 = r^2 + 3$,

$$256 - (r^2 + 3)(9 - r)^2 = (1 - r)(r^3 - 17r^2 + 67r + 13) \ge 0.$$

Hence $c_T + h_0/4 < 1$. The two cells are exhausted. $\square$

A.2.1. *Vd corner margins.* In the exact Vd1/Vd2 corner form, let the boundary reaches be a, b, let $t > 0$, and put $d = \sqrt{t^2 + t + 1}$. The own-radial reach is

$$c_0 = \frac{d - a - tb}{t + 1}.$$

If the role has positive support on the forward adjacent arm, its far endpoint there is

$$u_+ = \frac{d - a - tb - 1}{t}.$$

Since $d < t + 1$,

$$c_0 < 1 - \frac{a + tb}{t + 1} \leq 1 - \min\{a, b\}, \tag{6.70}$$

and

$$u_+ < 1 - b - \frac{a}{t} < 1 - b. \tag{6.71}$$

Thus lower boundary demands $a \geq A$, $b \geq H$ imply

$$c_0 < 1 - \min\{A, H\}, \qquad u_+ < 1 - H.$$

Reflection supplies the backward adjacent-arm statements. These two one-line inequalities are exactly the radial separations used in the one-Vd placement proof.

A.3. **CE1 reverse-path certificate.**

Proof of Proposition 5.26. The sign difference in (6.12) gives

$$RD > wA,$$

so the center exit on r_3 is $m = A/R$. Moreover

$$0 < A < P < \frac{1}{8}, \qquad 0 < D < \frac{R}{2}, \qquad 0 < m < \frac{w}{2} < \frac{1}{2}. \tag{6.15}$$

We first record

$$\epsilon(A) < X, \qquad h_0 > A. \tag{6.16}$$

Indeed, $A + wD < P$ and $D = R - X$ give $A < wX - \eta$. Convexity of

$$wX - \frac{X}{2(1 + X)}$$

on $R/2 < X < R$ gives

$$A < \frac{X}{2(1 + X)}.$$

Equation (6.5) yields $\epsilon(A) < X$. Also

$$2R(h_0 - A) = \eta + D + (1 - 2R)A > 0;$$

when $R > 1/2$, use $A < P = \eta E$ for the last sign.

Suppose, contrary to (6.14), that

$$A_0 \leq 1 - h_0. \tag{6.17}$$

The companion edge is V-gap-free. Thus

$$A_0 + B_5 \geq 1,$$

and (6.17) gives $B_5 \geq h_0$. Nonsupercriticality of T_5 and coverage of the next center-free edge imply

$$\boxed{B_4 \geq B_5 \geq h_0.} \tag{6.18}$$

The center-forced radial reach at T_4 is $1 - A$. Apply the exact local catalogue to the reflected boundary pair (B_4, A_4). If the local point is not on the selected Q_+ component, the constant, Q_-, and linear components together with (6.16) give

$$1 - A_4 \geq 1 - \epsilon(A) > 1 - X.$$

The center-free edges then carry this lower bound backward to $B_1 > 1 - X$, contradicting $A_1 \geq X$ and $A_1 + B_1 \leq 1$.

Hence the T_4 state is selected Q_+. Lemma 5.9 and (6.18) give

$$B_3 > L_1, \qquad L_1 = (2 - 4A)h_0 - (1 - 4A)A. \tag{6.19}$$

The selected-state condition also forces

$$\boxed{X > \frac{1}{2}.} \tag{6.20}$$

To see this, selectedness gives

$$h_0 \leq \epsilon(A) < \frac{2A}{1 - 2A}.$$

If $X \leq 1/2$, the lower CE1 inequality gives

$$2Rh_0 = \eta + A + D \geq w(R + A).$$

Consequently

$$f_R(A) < 0, \qquad f_R(z) = w(R + z)(1 - 2z) - 4Rz.$$

The quadratic is strictly concave and $f_R(0) > 0$. If $R \leq 1/2$, then

$$f_R(P) \geq \frac{1 - E}{2}(6E^3 - 2E^2 - 5E + 2) > 0.$$

If $R > 1/2$, then $A < A_* = w/2 - \eta$ and

$$f_R(A_*) = \frac{1 - E}{2}(E + 11R - 3ER - 5) > 0.$$

Concavity contradicts $f_R(A) < 0$ in both cases, proving (6.20). Thus

$$m < \frac{w}{2} < \frac{1}{4}. \tag{6.21}$$

If $B_3 \geq 1/2$, then $B_3 > 1 - X$ and the preceding immediate contradiction applies. Assume $B_3 < 1/2$. At T_3 the reflected incoming reach is $B_3 > h_0 > m$, while its center-forced radial reach is $1 - m$. If this state is not selected Q_+, the exact local catalogue gives

$$B_2 \geq 1 - B_3 > \frac{1}{2} > 1 - X,$$

and the remaining center-free edge again contradicts T_1.

Thus both T_4 and T_3 are selected Q_+. A second use of Lemma 5.9 gives

$$B_2 > (2 - 5m)B_3 - (1 - 5m)m.$$

Since $2 - 5m > 0$, (6.19) yields

$$\boxed{B_2 > L_2, \qquad L_2 = (2 - 5m)L_1 - (1 - 5m)m.} \tag{6.22}$$

We now prove the scalar estimate

$$\boxed{D < \frac{1}{10}, \qquad L_2 > 2D + 3D^2 > \epsilon(D), \qquad \epsilon(D) < X.} \tag{6.23}$$

The T_4 selected-state inequality and (6.6) give

$$D \le D_h := A(4R-1) + 10RA^2 - \eta. \tag{6.24}$$

If $D \ge 1/10$, then $A + wD < P$ gives

$$A < \overline{A} = \frac{w(5R-1)}{10}.$$

The right side of (6.24) is increasing in A, while $\eta > Rw/2$, so

$$D < U(R) := \overline{A}(4R-1) + 10R\overline{A}^2 - \frac{Rw}{2}.$$

Direct expansion gives

$$\frac{1}{10} - U(R) = -\frac{R}{10} P_4(R),$$

where

$$P_4(R) = 25R^4 - 60R^3 + 26R^2 + 22R - 14.$$

For $y = 2R - 1 \in (0,1)$,

$$16P_4(R) = 25y^4 - 20y^3 - 106y^2 + 124y - 39 \le -101y^2 + 124y - 39 < 0.$$

Hence $U(R) < 1/10$, a contradiction. This proves $D < 1/10$.

Put

$$\Psi(D) = L_2 - 2D - 3D^2, \qquad J(D) = L_2 - \frac{23}{10}D.$$

Since $D < 1/10$,

$$\Psi(D) - J(D) = 3D\left(\frac{1}{10} - D\right) > 0.$$

For fixed R, A, differentiation of (6.22) gives

$$J'(D) = \frac{S(R,A)}{10R^2},$$

where

$$S(R,A) = 100A^2 - (40R + 50)A + R(20 - 23R).$$

The nonempty interval $P - RA \le D \le D_h$ implies

$$A > \frac{4Rw}{25R-4} =: A_*.$$

On the relevant interval $\partial S/\partial A < -45$, and

$$S(R,A) < S(R,A_*) = -\frac{Rq_3(R)}{(25R-4)^2},$$

where

$$q_3(R) = 8775R^3 - 14260R^2 + 7928R - 1120.$$

For $y = 2R - 1$,

$$8q_3(R) = 8775y^3 - 2195y^2 + 997y + 3007 > 0.$$

Thus J decreases with D, so $J(D) \ge J(D_h)$.

At $D = D_h$, one has $h_0 = 2A + 5A^2$. If $L_{2,h}$ is the corresponding value, then

$$L_{2,h} - A(1 + 4R) = \frac{A}{R^2} Q(R, A),$$

where

$$Q(R, A) = 100A^3R - 40A^2R^2 - 30A^2R + 12AR^2 - 15AR + 5A - 4R^3 + 5R^2 - R.$$

This polynomial is concave in A on $0 \leq A \leq Rw/2$. Its endpoint values are

$$Q(R, 0) = Rw(4R - 1) > 0$$

and

$$Q\left(R, \frac{Rw}{2}\right) = \frac{Rw}{2}(25R^5 - 30R^4 + 20R^3 - 3R^2 - 7R + 3) > 0.$$

The last positivity follows after putting $y = 2R - 1$:

$$32p(R) = 25y^5 + 65y^4 + 90y^3 + 106y^2 - 35y + 5 > 0.$$

Hence $L_{2,h} > A(1 + 4R)$.

Finally $A < P = \eta E$, $A < Rw/2$, and $1/E > 1 + Rw/2$ give

$$\frac{D_h}{A} < C_* := 4R - 2 + 5R^2w - \frac{Rw}{2}.$$

Moreover

$$1 + 4R - \frac{23}{10}C_* = \frac{230R^3 - 253R^2 - 81R + 112}{20} > 0.$$

Thus $J(D_h) > 0$, so

$$L_2 > 2D + 3D^2.$$

Equation (6.6) gives $\epsilon(D) < 2D + 3D^2$, and the argument proving $\epsilon(A) < X$ also gives $\epsilon(D) < X$. This proves (6.23).

The center-forced radial reach at T_2 is $1 - D$. Since $B_2 > \epsilon(D)$, Lemma 5.7, applied after reflection, gives

$$A_2 \leq \epsilon(D).$$

Coverage of $e_{1,2}$ therefore forces

$$B_1 \geq 1 - A_2 \geq 1 - \epsilon(D) > 1 - X,$$

contradicting $A_1 \geq X$ and $A_1 + B_1 \leq 1$. Hence (6.17) is impossible. □

A.4. **T3-like rescuer inequalities.** With the notation of Proposition 5.31, put

$$x = \frac{aD}{R_0}, \qquad \theta = \frac{D - 1}{R_0}.$$

The midpoint constraints are

$$\frac{1 - 4\theta + \theta^2}{2(1 - 2\theta)} \leq x \leq \frac{1}{2} - \theta, \qquad 0 \leq \theta \leq 2 - \sqrt{3}.$$

For $s(z) = z(2 - z)/(1 + z)$, the equation $s(1 - M_c^{\mathrm{sup}}) = c = x + \theta$ shows that it is enough to prove $s(x) \leq x + \theta$. After multiplying by $1 + x$, this is

$$Q_\theta(x) = 2x^2 + (\theta - 1)x + \theta \geq 0.$$

For $0 \leq \theta \leq 1/5$, substitution of the left endpoint gives

$$Q_\theta(x) \geq \frac{\theta(1 - 5\theta + 11\theta^2 - \theta^3)}{2(1 - 2\theta)^2} \geq 0.$$

For $1/5 \le \theta \le 2-\sqrt{3}$, the vertex lies in the feasible interval and

$$Q_\theta(x) \ge \frac{10\theta - 1 - \theta^2}{8} > 0.$$

Thus $x \le 1 - M_c^{\sup}$. Since $D \ge R_0$,

$$a \le x, \qquad \frac{a}{a+1-u} = \frac{aD}{R_0} = x,$$

which proves the two inequalities used in the body.

A.5. **Detailed nonzero-gap terminal calculations.** This subsection supplies the calculations suppressed in the preceding case proofs. They are written as inequalities for actual reaches and radial witnesses. There is no formal chain notation.

A.5.1. *The disk–gap support calculation.* Let $c = c_{\max}(p,q)$, $m = \min\{p,q\}$, and $d = 1-c$. The regular radial hexagon has inradius hd. Choose the farther endpoint G of the complementary gap. Its norm is

$$\rho = \sqrt{1 - m + m^2}.$$

For a unit normal whose angle from G is θ, the three point supports are

$$\rho\cos\theta, \quad \rho\cos(\theta + 2\pi/3), \quad \rho\cos(\theta + 4\pi/3).$$

The disk replaces every one of these by at least hd. A minimizing orientation is either a disk-only orientation or a tie orientation. At a tie one projection equals hd. The other active projection is

$$\frac{-hd + \sqrt{3(\rho^2 - h^2 d^2)}}{2}.$$

Thus the support sum is at least h exactly when

$$3d(1-d) \ge m(1-m). \tag{6.51}$$

Since $d = 1-c$, this is (6.29).

We now derive the local inequality used in the high-c part. Work in a support orientation with angular parameter $t \in [0, \pi/3]$ and put

$$x = \cos\left|t - \frac{\pi}{6}\right|.$$

The support inequalities for the local kite $\operatorname{conv}\{0, pu, qv, c(u+v)\}$ imply

$$cx \le h \tag{6.52}$$

and, after taking the smaller boundary demand m,

$$mx + \frac{c}{2}\left(x + \sqrt{3}\sqrt{1-x^2}\right) \le h. \tag{6.53}$$

When $c > h$, the interval is $h \le x \le h/c$. The left side of (6.53) is concave. If $c > 1-m$, its value at $x = h$ is $h(m+c) > h$, so feasibility forces its value at the other endpoint to be at most h:

$$\frac{m}{c} + \frac{1}{2} + \sqrt{c^2 - \frac{3}{4}} \le 1.$$

Hence

$$m \le c\left(\frac{1}{2} - \sqrt{c^2 - \frac{3}{4}}\right). \tag{6.54}$$

Substitution of the right side into (6.51) gives

$$\begin{aligned}&3c(1-c) - c\left(\frac{1}{2} - \sqrt{c^2 - \frac{3}{4}}\right)\left[1 - c\left(\frac{1}{2} - \sqrt{c^2 - \frac{3}{4}}\right)\right]\\ &= \frac{c(1-c)}{2}\left(5 - 2c - 2c^2 + 2\sqrt{c^2 - \frac{3}{4}}\right) \geq 0.\end{aligned}$$

This proves the complementary-gap inequality directly from the four local support conditions.

A.5.2. *The CE2 one-gap threshold inequalities.* We prove the two inequalities in (6.39). Put

$$X = R - \delta, \qquad Q = \frac{\eta + \alpha + \delta}{2R}.$$

The right-trace inequality gives

$$WX = RW - W\delta = \eta + P - W\delta > \eta + \alpha. \tag{6.55}$$

The selected low root $\epsilon(\alpha)$ is the smaller root of

$$z^2 - (1-\alpha)z + (1-\alpha)^2 - (1-\alpha)^4 = 0. \tag{6.56}$$

It is enough to prove that X lies above that root. Since $0 < X < R$ and the quadratic opens upward, substitute the lower estimate in (6.55). Equivalently consider

$$\pi(q) = (\eta + q)(1 - 2q) - 2qW.$$

This is concave. At the endpoints of the admissible interval $0 \leq q \leq P$,

$$\pi(0) = \eta > 0,$$

while the identities $RW = \eta + P$ and $P = E\eta$ give

$$\pi(P) = \eta(\eta + 2ER^2) > 0.$$

Thus $\pi(\alpha) > 0$, which is precisely

$$\epsilon(\alpha) < X. \tag{6.57}$$

For the second inequality put $T = \alpha + \delta$. Multiply

$$\alpha + W\delta < P, \qquad R\alpha + \delta < P$$

by W and R, respectively, and add. Since $W + R^2 = W^2 + R = E^2$,

$$E^2 T < P = E\eta, \qquad T < \frac{\eta}{E}. \tag{6.58}$$

Let $d = \min\{\alpha, \delta\}$. Then $d \leq T/2$. By (6.5),

$$\min\{\epsilon(\alpha), \epsilon(\delta)\} < \frac{2d}{1-2d} \leq \frac{T}{1-T}.$$

It remains to prove $T/(1-T) < Q$. This is equivalent to

$$2RT < (\eta + T)(1 - T).$$

Define

$$\chi(T) = (\eta + T)(1 - T) - 2RT.$$

The function is concave. At $T = 0$, $\chi(0) = \eta > 0$. At $T = \eta/E$, direct simplification gives

$$\chi\left(\frac{\eta}{E}\right) = \frac{\eta}{E^2}(E - R)^2 > 0.$$

By (6.58), T lies strictly between these endpoints. Hence

$$\boxed{\min\{\epsilon(\alpha), \epsilon(\delta)\} < Q.} \tag{6.59}$$

This proves the CE2 dichotomy without a composed reach map.

A.5.3. *The CE2 short-ray root separation.* We give the remaining details of Theorem 5.20. The identities

$$RW = \eta + P, \qquad P = E\eta$$

and the CE2 inequalities imply

$$Wq = W(R - \delta) > \eta + \alpha,$$
$$Rp = R(W - \alpha) > \eta + \delta.$$

Thus

$$p, q > \eta + e. \tag{6.60}$$

Let $M = \max\{R, W\}$. Since $\alpha, \delta \geq e$,

$$e < \frac{P}{1 + M}.$$

Moreover

$$(1 + M)^2 - 3E^2 = (2M - 1)(2 - M) \geq 0,$$

so

$$\eta > \sqrt{3}e. \tag{6.61}$$

The universal bound

$$P \leq \frac{2\sqrt{3} - 3}{4}, \qquad 1 + M \geq \frac{3}{2}$$

gives

$$e < \frac{2\sqrt{3} - 3}{6} = e_*.$$

For $c = 1 - e$, the roots of $f_c(t) = c^4 - c^2 + ct - t^2$ are $t_\pm$ from the proof above. At $t_0 = (1 + \sqrt{3})e$,

$$f_c(t_0) = eB(e),$$

with

$$B(e) = e^3 - 4e^2 - 3\sqrt{3}e + \sqrt{3} - 1.$$

Since

$$B'(e) = 3e^2 - 8e - 3\sqrt{3} < 0$$

on $[0, e_*]$, one has

$$B(e) \geq B(e_*) = \frac{302\sqrt{3} - 501}{72} > 0.$$

Therefore $t_- < t_0 < p, q$. Also

$$p + q = 1 - \alpha - \delta \leq 1 - 2e = c - e.$$

If, say, $q \leq p$, then $q > t_-$ and

$$p < c - e - t_- < c - t_- = t_+.$$

The same holds with p, q interchanged. Hence

$$t_- < p, q < t_+,$$

which proves $f_c(p), f_c(q) > 0$. In the exact local finite-caliper formula at radial demand c, the four side candidates are

$$p + c, \quad q + c, \quad \frac{c^2}{\sqrt{p^2 - pc + c^2}}, \quad \frac{c^2}{\sqrt{q^2 - qc + c^2}}.$$

The first two exceed one because $p, q > e$, and the last two exceed one exactly because $f_c(p), f_c(q) > 0$. This completes the direct short-ray proof.

A.5.4. *Detailed adjacent Vd residual estimate.* We justify (6.48). In the adjacent placement define the residuals ρ_R, ρ_L as in Case 1 of Proposition 5.38. The signed CE2 domain first gives

$$\alpha + \delta < \frac{1}{6} \min\{R, W\}. \tag{6.62}$$

The positive residual ρ_L has one of two forms.

If

$$\rho_L = W - \alpha,$$

then

$$4\delta + \alpha \leq 4(\alpha + \delta) < \frac{2W}{3} < W,$$

so $4\delta < \rho_L$.

Assume instead

$$\rho_L = 1 - A_0.$$

The other residual cannot equal $1 - B_0$, since (6.47) would imply $A_0 + B_0 > 3/2$, contrary to the endpoint-distance bound $A_0 + B_0 \leq 2/\sqrt{3}$. Hence

$$\rho_R = 1 - (W + \delta), \qquad B_0 \geq y := \frac{k}{R}.$$

The two points of T_0 with reaches $1 - \rho_L$ and y have distance at most one. The diameter-transfer inequality gives

$$\rho_L > \frac{y}{2} > \frac{\eta}{2R}. \tag{6.63}$$

Since $\rho_R + \rho_L < 1/2$,

$$\delta > R - \frac{1}{2} + \frac{\eta}{2R} =: J(R). \tag{6.64}$$

Using $\eta/R = W/(1 + E)$, exact differentiation gives

$$4E(1 + E)^2 J'(R) = 2E(1 + E)(1 + 2E) - W(2R - 1) > 0.$$

Thus J is strictly increasing. Since $J(3/8) = 1/24$ and (6.62) gives $\delta < 1/24$, equation (6.64) forces $R < 3/8$. Then

$$(2 - 3R)^2 - E^2 = (3 - 8R)(1 - R) > 0,$$

so $E < 2 - 3R$ and

$$\frac{\eta}{R} = \frac{W}{1 + E} > \frac{1}{3}.$$

Together with (6.63),

$$\rho_L > \frac{1}{6} > 4\delta.$$

This proves (6.48) in both residual cases.

A.5.5. *Detailed nonadjacent Vd separation.* Let $x = k/W$ and $y = k/R$ be the two center near endpoints. Let B, U be the farthest already-covered extents on the corresponding edges. We prove

$$B \le M_0(x), \qquad U \le M_0(y). \tag{6.65}$$

By symmetry it is enough to treat B. The residual component formula says that B is either $W + \delta$ or the supercritical endpoint B_0. If $B = W + \delta$, the center contains boundary points with reaches x, B, so the diameter-one inequality gives $B \le M_0(x)$. If $B = B_0$ and $A_0 < x$, then the opposite covered extent is $U = A_0$; equation (6.47) would again force $A_0 + B_0 > 3/2$. Hence $A_0 \ge x$, and monotonicity of M_0 gives

$$B_0 \le M_0(A_0) \le M_0(x).$$

This proves (6.65).

The elementary identity

$$1 - M_0(z) > \frac{z}{2} \qquad (z > 0)$$

then gives

$$\rho_R > \frac{x}{2}, \qquad \rho_L > \frac{y}{2}.$$

The common center endpoint-slack bound is

$$\alpha + \delta < \frac{1}{2}\min\{x, y\}.$$

Thus (6.49) follows. The exact Vd corner form gives the strict own-radial bound used in Case 2.

A.5.6. *Detailed Vd1 rescue inequalities.* Use the Vd1 corner parameter $t \ge 1$, let $d = \sqrt{t^2 + t + 1}$, and write the supported interval as $[c, u]$. The endpoint relations are

$$c = \frac{t(1-b)}{t+1}, \qquad u = \frac{d - a - tb - 1}{t}.$$

The midpoint condition gives

$$a + tb \le d - 1 - \frac{t}{2}.$$

Eliminating $tb = t - c(t+1)$ gives

$$a \le F(t, c) := d - 1 - \frac{3t}{2} + c(t+1).$$

For fixed $c \le 1/2$,

$$\frac{\partial F}{\partial t} = \frac{2t+1}{2\sqrt{t^2+t+1}} - \frac{3}{2} + c < 0,$$

so

$$a \le F(1, c) = \sqrt{3} - \frac{5}{2} + 2c =: L(c).$$

The inequality $2L(c) \le 1 - M_c^{\sup}$ is equivalent, after squaring, to

$$Q(c) = 20c^2 + (18\sqrt{3} - 52)c + 47 - 24\sqrt{3} \ge 0.$$

The polynomial decreases on $[0, 1/2]$ and $Q(1/2) = 26 - 15\sqrt{3} > 0$. Hence

$$a \le 1 - M_c^{\sup}.$$

Put $\varepsilon = 1 - u$. Direct substitution gives

$$\varepsilon = \varepsilon_0 + \frac{a}{t}, \qquad \varepsilon_0 = \frac{2t + 1 - c(t+1) - d}{t} > 0.$$

The function

$$z \longmapsto \frac{z}{z + \varepsilon_0 + z/t}$$

is increasing. Since

$$\varepsilon_0 + \frac{F(t,c)}{t} = \frac{1}{2},$$

we obtain

$$\frac{a}{a + 1 - u} \leq \frac{F(t,c)}{F(t,c) + 1/2} \leq 2L(c) \leq 1 - M_c^{\sup}.$$

This proves the two direct Vd1 rescue inequalities in (6.50).

A.6. **Casewise witness audit.** We now record the finite set used in each combinatorial row, including the open-endpoint issues. This makes the Strategy 3 ownership independent of the former reach-propagation chapter.

A.6.1. *One gap and no supercritical V role.* Normalize the only gap to

$$J = [B_0, 1 - A_1] = [\ell, r] \subset e_{0,1}.$$

On every other edge choose an open-overlap point with coordinate x_i. Because $A_i + B_i \leq 1$, one has

$$x_1 \leq r, \qquad x_i \leq x_{i-1} \quad (2 \leq i \leq 5), \qquad x_5 \geq \ell. \tag{6.72}$$

The endpoint inequalities remain valid when a gap or overlap degenerates to a singleton; the strict open choices are obtained by an arbitrarily small perturbation and the final enclosure number is continuous.

Put $p = 1 - r$ and $q = \ell$. For T_1, the two selected lower demands are p and $x_1 \geq q$. For $2 \leq i \leq 5$, they are $1 - x_{i-1} \geq p$ and $x_i \geq q$. At T_0, they are $1 - x_5 \geq p$ and q. Thus every local role dominates (p, q).

Let $c_* = c_{\max}(p, q)$. The own role cannot contain $(1 - c_*)V_i$ by the definition of $c_{\max}$. If an adjacent role is Vd0, it has no positive interval on r_i. If it is T3-like, its unique adjacent capacity is C_+ or C_- and is at most c_* by Lemma 5.15. The other roles are excluded by diameter. Hence

$$(1 - c_*)V_i \in U_C \qquad (0 \leq i \leq 5). \tag{6.73}$$

The six points form a regular hexagon of circumradius $1 - c_*$. The convex center role therefore contains its inball

$$\mathcal{D}_{h(1-c_*)}.$$

The gap segment is exactly $J(p, q)$. This proves the hypotheses of Theorem 5.19 without any abstract intermediate-set argument.

A.6.2. *The T3-like neighboring capacity.* We verify directly that the T3-like supported endpoint is the radical branch of Proposition 5.10. Use the translated T3-like normal form with boundary reaches a, b and shape parameter D. The support-side relation is

$$R_0^2 - DR_0 + D^2 = 1,$$

and the far endpoint on the forward neighboring ray is

$$u = 1 + a - \frac{R_0}{D}.$$

The boundary reaches satisfy

$$a + b = \frac{1}{D}.$$

Solving the quadratic for R_0/D gives

$$\frac{R_0}{D} = \frac{1 + \sqrt{4(a+b)^2 - 3}}{2}.$$

Therefore

$$\boxed{u = a + \frac{1}{2} - \sqrt{(a+b)^2 - \frac{3}{4}},} \tag{6.74}$$

which is exactly the final radical branch of $C_+(a, b)$. Thus no additional T3-specific radial function occurs in the one-gap proof.

A T3-like role is nonsupercritical. Hence its boundary sum is at most one, and (6.74) is bounded by

$$1 - \min\{a, b\} \le c_{\max}(p, q)$$

whenever (a, b) dominates the common pair (p, q). This is the explicit reason why one or two T3-like roles do not shrink the regular radial witness hexagon in Proposition 5.23.

A.6.3. *Two gaps.* For CE2, write the two center intervals as

$$I_L = \left[\frac{k}{W}, R + \alpha\right], \qquad I_R = \left[\frac{k}{R}, W + \delta\right].$$

If both contain gaps, then the incident endpoint roles give

$$A_1 \ge R - \delta = q, \qquad B_5 \ge W - \alpha = p.$$

Choose handoffs $x_1, \dots, x_4$ on the four intervening edges. The five nonsupercritical roles give

$$p < x_4 < x_3 < x_2 < x_1 < 1 - q. \tag{6.75}$$

Every one of $T_1, \dots, T_5$ therefore dominates (p, q). In particular, the two transverse witnesses

$$D_2 = (1 - c_{\max}(p, q))V_2, \qquad D_4 = (1 - c_{\max}(p, q))V_4$$

are excluded from all V roles. The center exits at distances δ and α on these two rays. Theorem 5.20 states that one witness lies strictly beyond its exit. This argument does not inspect $A_0 + B_0$, so it simultaneously handles $N_+ = 0$ and $N_+ = 1$.

A.6.4. *One gap, one supercritical Vd0 role.* In the all-Vd0 branch, define the six actual radial endpoints

$$P_i = (1 - C_i)V_i.$$

Because U_i is open, the endpoint does not belong to U_i. The adjacent roles have no positive support on r_i, and nonlocal roles are separated by distance. Hence $P_i \in U_C$.

Suppose a second open unit triangle U' contains

$$O, M_0, X(\ell), X(r), P_0, \ldots, P_5.$$

The fixed six V roles together with U' cover every radial arm: U' contains $[O, P_i]$, and U_i contains a relatively open interval from V_i past every point short of P_i. They also cover the perimeter, since the unique missing boundary component is the gap and U' contains its two endpoints and hence its full segment. Thus all direct midpoint and signed-center conclusions apply to U'.

If d_i' is the exit of U' on r_i, containment of P_i says

$$d_i' \geq 1 - C_i, \qquad C_i \geq 1 - d_i'. \tag{6.76}$$

This is the only connection between the finite set and the local reach calculation. In CE2, (6.76) supplies radial reaches $1-\alpha$ at T_4 and $1-\delta$ at T_2; the direct threshold alternatives of the proof of Proposition 5.29 contradict the terminal bound at T_0. In CE1, (6.76) supplies the reaches used by Proposition 5.26. No statement about a residual polygon is substituted for these actual endpoint facts.

For a singleton gap, replace $r = \ell$ by $r_n > \ell$ with $r_n \downarrow \ell$. The diameter inequalities and the unique-supercritical pattern persist for large n. The witness sets converge in the Hausdorff metric, and Λ is continuous by the support formula. Hence the positive-length proof passes to the singleton limit.

A.6.5. *The T3-like rescuer endpoint.* In Proposition 5.33, the finite point $P_T = (1-u)V_1$ is forced into the center for a stronger reason than a generic radial capacity. The T3-like open trace begins at the vertex side at c and ends at u, so its O-side endpoint is not in the open role. The adjacent supercritical role misses M_1 and therefore, by convexity, misses the entire O-side segment. All remaining roles are Vd0. Thus the segment from O to P_T is owned by the center.

The two inequalities (6.44) then compare the far boundary tail with the one free strict-supercritical envelope. They do not propagate a numerical state through the four middle roles. Instead, adding the four actual edge-cover inequalities gives one path budget:

$$(A_2 + B_2) + (A_3 + B_3) + (A_4 + B_4) + (A_5 + B_5) \geq 4 + h_T - B_1 > 4.$$

This direct sum is the full global contradiction.

A.6.6. *The Vd radial points.* In the adjacent Vd placement, define

$$q_2 = 1 - \delta.$$

The two roles capable of reaching the center interval on r_2 have exact far endpoints bounded by

$$u_{1\to 2} < q_2, \qquad C_2 < q_2.$$

Therefore any point of r_2 between the larger of these two endpoints and q_2 is missed by all seven roles. This is a literal one-point obstruction, not a perimeter reformulation.

In the nonadjacent placement, the point

$$D_\tau = \min\{\rho_R, \rho_L\} V_\tau$$

is beyond the Vd own-radial trace by (6.70), and beyond the center trace by (6.49). Its adjacent roles are Vd0 and its nonlocal roles are separated by diameter. This is again a literal uncovered point.

In the Vd1 rescue placement, the O-side endpoint of the supported interval plays exactly the same role as P_T in the T3-like case. Only the local normal-form proof of the two inequalities changes; the center-hiding and four-role boundary sum are identical.

A.6.7. *The corrected replacement placement.* When neither distinguished role is based at V_0 and the exceptional role is Vd1, the shared edge is center-free. The corrected replacement constructs one nonsupercritical Vd0 role in each of the two correct vertex charts. It preserves all six boundary and radial skeleton traces needed for coverage but does not claim to preserve the number of gaps. After replacement, recompute N'_{gap}.

If $N'_{\mathrm{gap}} = 0$, the six nonsupercritical Vd0 roles cover the perimeter, contradicting the strict boundary-overlap sum. If $N'_{\mathrm{gap}} = 1$, use the common radial disk and complementary gap. If $N'_{\mathrm{gap}} = 2$, use the two transverse short-ray witnesses. Hence the replacement terminal has no return to a reach-composition argument.

A.7. **Selector and singleton-gap audit.**

A.7.1. *Branchwise proof of common-pair domination.* We expand the neighboring-capacity comparison in Lemma 5.15. Assume first that $q \le p$. The own-radial demand $1 - q$ is admissible because the exact edge-normal orientation has side

$$q + \max\{p, 1 - q\} = 1.$$

Thus

$$c_{\max}(p, q) \ge 1 - q. \tag{6.73a}$$

For $C_+(p, q)$ there are three possibilities.

On the linear branch,

$$C_+(p, q) = 1 - q.$$

On the plateau branch, $C_+ = p(p)$, where the cubic root satisfies

$$f_p(1 - q) = (1 - q)^3 - (p + 2)(1 - q)^2 + 2(p + 1)(1 - q) - 1 \ge 0$$

throughout the selected interval. Since the cubic is increasing, $p(p) \le 1 - q$. On the radical branch, the capacity decreases from its selected transition value and therefore is no larger than the plateau. Hence

$$C_+(p, q) \le 1 - q.$$

The reflected calculation gives the same bound for C_- and treats the order $p \le q$. Combining with (6.73a) proves

$$C_\pm(p, q) \le c_{\max}(p, q).$$

The same argument is valid for any actual pair (a, b) with $a \ge p$ and $b \ge q$, because the family of triangles containing the larger anchors is a subfamily of the one containing the smaller anchors. Thus the relevant capacity is coordinatewise nonincreasing.

A.7.2. *Finite-caliper selector audit.* For a finite point set K, Proposition 5.5 reduces $\Lambda(K)$ to finitely many orientations. We spell out the reduction used for every diagram and for the anisotropic one-gap set. Let the convex hull vertices be $P_0, \dots, P_{r-1}$ in cyclic order. On an angular interval on which the three support points are fixed, write

$$H(\theta) = \sum_{j=0}^{2} \langle Q_j, n_{\theta+2\pi j/3} \rangle.$$

Then

$$H''(\theta) = -H(\theta).$$

The support sum is positive unless K is a singleton at the origin. Hence an interior critical point is a strict maximum. A minimum occurs when two support points tie in one normal direction. That normal is perpendicular to a hull edge P_iP_{i+1}. Therefore

$$\Lambda(K) = \min_{0 \le i < r} \frac{1}{h} \sum_{j=0}^{2} h_K(\mathsf{R}^j n_i), \tag{6.73b}$$

where n_i is either unit normal to P_iP_{i+1}.

This finite-caliper identity has two consequences. First, the enclosure number is unchanged by replacing a witness set by its convex hull. Second, no numerical angular minimization is hidden in the proof: every terminal can be checked at finitely many support contacts. The complementary-gap theorem uses this fact analytically; the zero-gap nine-point theorem uses it through the support-arc certificate in the final appendix.

A.7.3. *Terminal upper bound in the one-gap exact-one case.* We give the elementary derivation of (6.38). The active center trace begins at k/R, so the unique supercritical role has forward boundary reach

$$B_0 \ge \frac{k}{R}.$$

The center exit on r_0 is $1-k$. Since the finite set contains the actual radial endpoint $P_0 = (1 - C_0)V_0$, containment by the candidate center gives

$$C_0 \ge k.$$

The closure of T_0 therefore contains three points based at V_0 with selected reaches

$$A_0, \qquad \frac{k}{R}, \qquad k.$$

Discarding the radial point only weakens the diameter condition, and gives

$$A_0 \le M_0\left(\frac{k}{R}\right).$$

For $0 < z < 1$,

$$M_0(z) < 1 - \frac{z}{2}.$$

Indeed, after moving terms and squaring, this is equivalent to $3z^2 > 0$. With $z = k/R$,

$$A_0 < 1 - \frac{k}{2R} = 1 - Q.$$

This is the terminal capacity contradicted separately by the direct CE1 and CE2 reach arguments.

A.7.4. *Strictness at singleton gaps.* Suppose the V-gap on $e_{0,1}$ is the singleton $X(\ell)$. Although its length is zero, neither incident open V trace contains the point. Boundary locality therefore forces $X(\ell) \in U_C$. Choose $r_n > \ell$ decreasing to ℓ, and reduce the incoming reach at T_1 from $1-\ell$ to $1-r_n$. Every local diameter inequality remains feasible, and the strict supercritical pattern is unchanged for large n. The non-singleton-gap witness sets converge to the singleton witness set. By Lemma 5.3, the inequality $\Lambda \geq 1$ passes to the limit. Thus no proof in this section charges numerical gap length; every one uses the actual missing point.

A.7.5. *Why a common disk is insufficient for $N_+ = 1$.* The one-gap $N_+ = 1$ finite set retains all six actual radial endpoints rather than replacing them by one common radius. This is necessary. The boundary sequence has two ascents when the gap jump is included: the V-gap jump and the unique supercritical V triangle. They need not join the global minimum to the global maximum. A common lower pair can therefore move several radial witnesses toward O and can produce a disk-plus-gap set with enclosure number below one. The anisotropic set $P_0, \dots, P_5$ preserves the six actual radial constraints, and equations (6.76), (6.38), and the direct CE1/CE2 arguments use those constraints individually. This is the precise point at which the new proof differs from a relabeling of the former propagation calculation.

Appendix B. Trace-Exact AB-Envelope Atlas

B.1. **Actual-gap atlas.** The following figures illustrate the normalized nonzero-gap cases in Table 2. Each shaded trace-exact envelope is a dense union of valid source triangles; the exact proof object is the source-conditioned union in Subsection 5.2. The dashed equilateral triangle is the numerical minimum for the displayed finite witness set and is included only to explain the terminal geometry.

Appendix C. Exact Unequal-Radius Support-Arc Intersections for the Zero-Gap Terminal

The two mixed cap overlaps in the zero-gap Method 3 terminal are certified by exact symbolic arithmetic. This section gives the mathematical reduction and verification algorithms; the repository provenance manifests identify the accompanying code and sparse integer data. No floating-point arithmetic, interval arithmetic, adaptive subdivision, branch-and-bound, or unrecorded numerical tolerance is used.

C.1. **Certificate manifest.** The machine-readable provenance manifest identifies the six sparse-data shards and both exact verifiers by full Git blob; the source ledger records their role. They decode to one canonical sparse-polynomial transcript with SHA–256 digest

`dc46aaf263655d5159ecd3a81db72ee82477951d06172f4743b248df37209485`

The derivation verifier reconstructs the transcript from the witness formulas and compares every coefficient with the authenticated object. The positivity verifier derives the Bernstein coefficients from that object and checks their signs exactly.

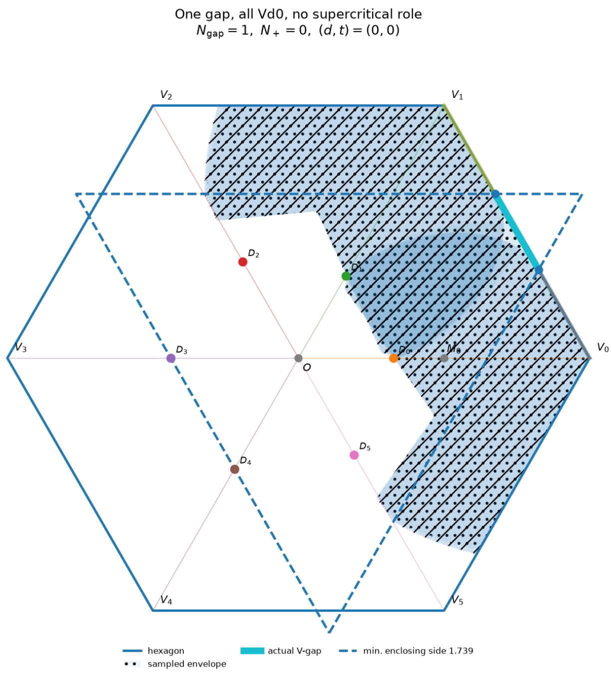
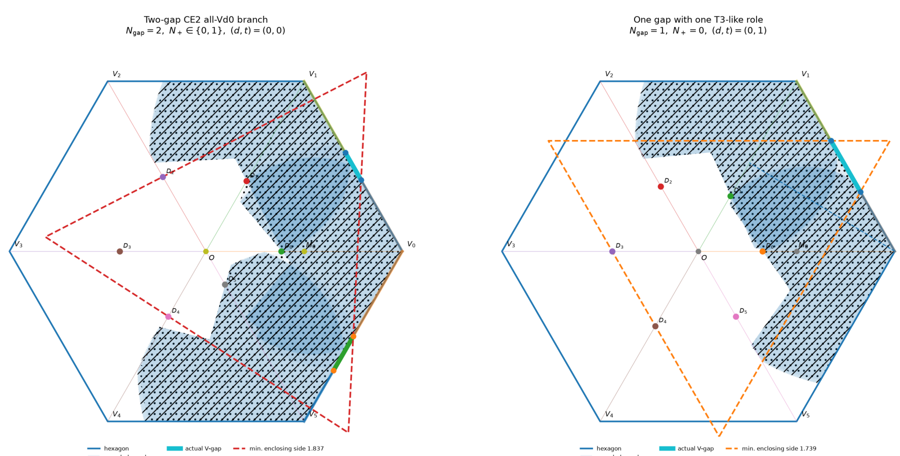


FIGURE 24. Trace-exact AB-envelope snapshot: one gap, all Vd0, no supercritical role. The zero-gap row is intentionally omitted from this actual-gap atlas and is illustrated by the original nine-point figure in Chapter 6.



(a) Two-gap CE2 all-Vd0 branch.   **(b)** One gap with one T3-like role.

FIGURE 25. Trace-exact AB-envelope snapshots: Two-gap CE2 all-Vd0 branch; One gap with one T3-like role.

C.2. **Rational radial upper envelopes.** Retain the notation of Chapter 6. Thus

$$p = 1 - b, \qquad q = 1 - a, \qquad s = p + q, \qquad m = \min\{p, q\}, \qquad M = \max\{p, q\},$$

$$\chi = s^4 - s^2 + mM, \qquad h = \frac{\sqrt{3}}{2}, \qquad c_* = c_{\max}(p, q).$$

The strict witness domain gives

$$mM > (1 - s)(2 - s), \qquad m > 1 - s, \qquad s > \frac{2}{3}. \tag{71}$$

Put

$$F_m(x) = x^4 - x^2 + mx - m^2, \qquad \kappa = F_m'(s) = 4s^3 - 2s + m.$$

Then

$$\kappa > 4s^3 - 3s + 1 = (s + 1)(2s - 1)^2 > 0.$$

Define

$$\bar{c}_L = s - \frac{\chi}{\kappa} \quad (\chi \leq 0), \qquad \bar{c}_T = s - \frac{\chi}{1 - m} \quad (\chi \geq 0). \tag{72}$$

Lemma C.1 (Branchwise radial upper bounds)**.** *On the corresponding radial cells,*

$$c_* \leq \bar{c}_L < 1, \qquad c_* \leq \bar{c}_T < 1.$$

At $\chi = 0$ both upper bounds equal $s = c_*$.

Proof. On the L cell, $F_m(s) = \chi \leq 0$ and $F_m(c_*) = 0$. For $x \geq s > 2/3$, $F_m''(x) > 0$, so convexity gives

$$0 \geq F_m(s) + F_m'(s)(c_* - s),$$

and hence $c_* \leq \bar{c}_L$. If $U = 1 - s$, then

$$\chi + U\kappa > U\{m + 1 - 4U + 8U^2 - 3U^3\} > 0$$

by (71); therefore $\bar{c}_L < 1$.

On the T cell put

$$E = \sqrt{4s^2 - 3}, \qquad m_- = \frac{s(1 - E)}{2}, \qquad m_+ = \frac{s(1 + E)}{2}.$$

Then

$$\chi = (m - m_-)(m_+ - m), \qquad s - c_* = \frac{2(m - m_-)}{1 + E}.$$

Consequently

$$\frac{\chi}{s - c_*} = \frac{1 + E}{2}(m_+ - m) \leq 1 - m,$$

which gives $c_* \leq \bar{c}_T$. The strict inequality $\chi > 0$ gives $\bar{c}_T < s < 1$. □

Reflect so that $z = a - b \geq 0$ and put $U = a + b - 1$. Then

$$s = 1 - U, \qquad m = \frac{1 - U - z}{2}, \qquad D^2 = z^2 + 6U + 3U^2 =: K(U, z),$$

and

$$\Xi(U, z) = 4\chi = 1 - 10U + 21U^2 - 16U^3 + 4U^4 - z^2,$$
$$G(U, z) = 2\kappa = 5 - 21U + 24U^2 - 8U^3 - z.$$

Thus

$$\bar{c}_L = 1 - U - \frac{\Xi(U, z)}{2G(U, z)}, \qquad \bar{c}_T = 1 - U - \frac{\Xi(U, z)}{2(1 + U + z)}. \tag{73}$$

C.3. **Gram reduction and the geometric implication.** On the active cell let $\bar{c}$ denote the corresponding expression in (73), and put

$$\bar{\eta} = h(1-\bar{c}), \qquad e = 1-\bar{c}, \qquad t = 2\bar{c}-1.$$

Lemma C.1 gives $0 < \bar{\eta} \le \eta = h(1-c_*)$. It therefore suffices to prove the mixed intersections for the smaller disks with radii $2\bar{\eta}, \bar{\eta}, 2\bar{\eta}$.

Represent a global vector as (x, hy) and put $C' = \mathsf{R}^{-1}C$. Define

$$N_A = \|A\|^2, \qquad N_C = \|C\|^2, \qquad \nu = \langle A, C'\rangle,$$
$$\Delta_0 = x_A y_{C'} - y_A x_{C'} > 0.$$

Thus the Euclidean cross product is $h\Delta_0$, and the Gram identity is

$$h^2\Delta_0^2 = N_A N_C - \nu^2. \tag{74}$$

Define

$$Q_A = \Delta_0^2 - \|tA - eC'\|^2, \qquad Q_C = \Delta_0^2 - \|tC' - eA\|^2. \tag{75}$$

Lemma C.2 (Residual signs imply the mixed cap intersections)**.** *If $Q_A, Q_C \ge 0$, then*

$$I(C, 2\bar{\eta}) \cap I(C + \mathsf{R}A, \bar{\eta}) \neq \varnothing,$$

and

$$I(C + \mathsf{R}A, \bar{\eta}) \cap I(\mathsf{R}A, 2\bar{\eta}) \neq \varnothing.$$

Consequently the same intersections hold with η in place of $\bar{\eta}$.

Proof. Rotate the first pair by R^{-1}. Its centers become C' and $A + C'$, so their difference is A. An outer common tangent normal n_A is chosen with

$$\langle A, n_A\rangle = \bar{\eta}.$$

Since $\|A\| > \bar{\eta}$ and $\Delta_0 > 0$, choose the tangent side for which

$$\langle C', n_A\rangle = \frac{\bar{\eta}\nu + \sqrt{N_A - \bar{\eta}^2}\, h\Delta_0}{N_A}.$$

The common support of the two disks is therefore

$$2\bar{\eta} + \frac{\bar{\eta}\nu + \sqrt{N_A - \bar{\eta}^2}\, h\Delta_0}{N_A}.$$

Let $\bar{\tau} = h - 2\bar{\eta} = ht$. Expanding (75) and using (74) gives

$$\bar{P}_A := (N_A - \bar{\eta}^2)h^2\Delta_0^2 - (\bar{\tau}N_A - \bar{\eta}\nu)^2 = h^2 N_A Q_A. \tag{76}$$

Thus $Q_A \ge 0$ implies

$$\sqrt{N_A - \bar{\eta}^2}\, h\Delta_0 \ge \bar{\tau}N_A - \bar{\eta}\nu,$$

and the common support is at least $2\bar{\eta} + \bar{\tau} = h$. Hence the two level-$h$ caps intersect. The same calculation with A and C' exchanged gives

$$\bar{P}_C := (N_C - \bar{\eta}^2)h^2\Delta_0^2 - (\bar{\tau}N_C - \bar{\eta}\nu)^2 = h^2 N_C Q_C, \tag{77}$$

and proves the second intersection. Enlarging the disk radii from $\bar{\eta}$ to η only enlarges the caps. $\square$

C.4. **Eight exact integer-polynomial signs.** The Newton witnesses A, C are rational functions of U, z, D. Substituting (73) into (75), clearing denominators, and reducing by

$$D^2 - K(U,z) = 0$$

gives, for $B \in \{L, T\}$ and $X \in \{A, C\}$,

$$\mathrm{num}(Q_{B,X}) = 32(K+3)^2\{A_{B,X}(U,z) + DB_{B,X}(U,z)\}, \tag{78}$$

where all eight core polynomials lie in $\mathbb{Z}[U,z]$. Every omitted denominator is strictly positive on the strict domain: its factors are positive constants, $(D+3)^4$, $G(U,z)^2$ or $(1+U+z)^2$, and squares of the two Newton-derivative numerators. Here $G = 2\kappa > 0$, and the fixed-line sign theorem proves both Newton derivatives strictly negative.

The derivation verifier constructs A, C, both rational envelopes, and all four residuals directly in the exact field $\mathbb{Q}(U,z,D)$. It performs the reduction modulo $D^2 - K$, removes the common positive factor, primitive-normalizes the eight integer polynomials, and compares their sparse term lists coefficient by coefficient with the authenticated transcript. Thus the data are checked against the geometric formulas rather than trusted as an independent precomputation.

C.5. **Three global positive-basis certificates.** Set

$$\mathcal{G}(U) = 1 - 6U - 3U^2, \qquad \mathcal{P}(U) = 1 - 10U + 21U^2 - 16U^3 + 4U^4,$$

$$U_\triangle = 1 - \frac{\sqrt{3}}{2}, \qquad U_{\max} = \frac{2}{\sqrt{3}} - 1.$$

The complete reflected domain is the union of the three closed cells

$$\begin{aligned} T:&\quad 0 \le U \le U_\triangle, \quad 0 \le z^2 \le \mathcal{P}(U),\\ L_0:&\quad 0 \le U \le U_\triangle, \quad \mathcal{P}(U) \le z^2 \le \mathcal{G}(U),\\ L_1:&\quad U_\triangle \le U \le U_{\max}, \quad 0 \le z^2 \le \mathcal{G}(U). \end{aligned}$$

For the T cell use

$$U = \frac{(x-1)(3-x)}{4x}, \qquad z = y\frac{9-x^4}{8x^2}, \qquad 1 \le x \le \sqrt{3}, \quad 0 \le y \le 1,$$

and then $x = 1 + (\sqrt{3}-1)r$. This maps $[0,1]^2$ onto the complete cell. After multiplication by positive powers of 2 and x, the four transformed core polynomials have global tensor Bernstein expansions with data

	bidegree	coefficient count	zero count
$A_{T,A}$	$(88, 22)$	2047	2
$B_{T,A}$	$(80, 20)$	1701	0
$A_{T,C}$	$(88, 22)$	2047	2
$B_{T,C}$	$(80, 20)$	1701	0.

Every nonzero coefficient is strictly positive.

For an L-cell polynomial write

$$F(U,z) = E_F(U,w) + zO_F(U,w), \qquad w = z^2,$$

and

$$R_F(U,w) = E_F(U,w)^2 - wO_F(U,w)^2.$$

The implications $E_F \ge 0$ and $R_F \ge 0$ give $E_F \ge \sqrt{w}\,|O_F|$, hence $F \ge 0$ for $z \ge 0$. Use the exact charts

$$U = U_\triangle r, \qquad w = \mathcal{P}(U) + s\{\mathcal{G}(U) - \mathcal{P}(U)\}$$

for L_0, and

$$U = U_{\triangle} + (U_{\max} - U_{\triangle})r, \qquad w = s\mathcal{G}(U)$$

for L_1. The positivity verifier forms one global Bernstein expansion for each of E_F, R_F on each square. The authenticated transcript records the bidegrees and coefficient counts of all sixteen expansions; every coefficient is strictly positive.

The verifier implements polynomial addition, multiplication, substitution, and power-to-Bernstein conversion over exact integers. Coefficients in $\mathbb{Q}(\sqrt{3})$ are represented as rational pairs and signed by exact rational comparisons. If $P(r,s) = \sum a_{ij} r^i s^j$ has bidegree (m,n), its exact tensor Bernstein coefficients are

$$b_{k\ell} = \sum_{i \le k} \sum_{j \le \ell} a_{ij} \frac{\binom{k}{i}}{\binom{m}{i}} \frac{\binom{\ell}{j}}{\binom{n}{j}}.$$

This is precisely the conversion implemented by the verifier.

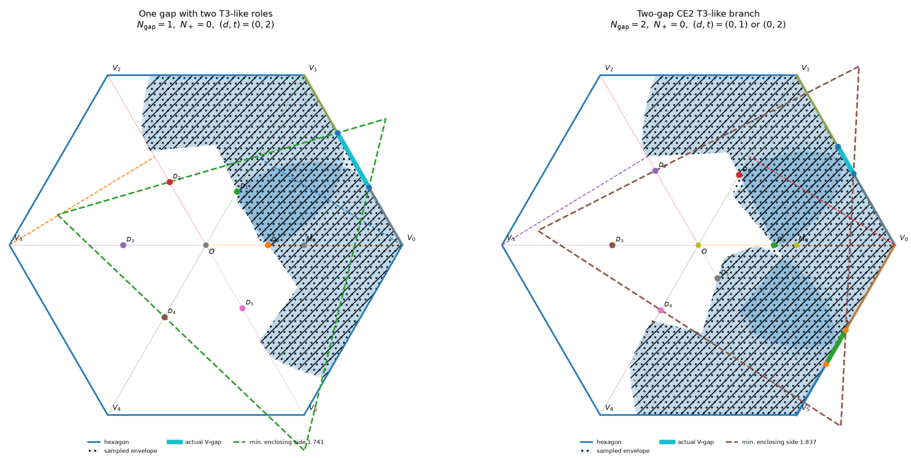
Theorem C.3 (Exact mixed-overlap certificate)**.** *On the complete strict witness domain with $c_* > 2/3$, all eight polynomials in* (78) *are nonnegative. Hence $Q_A, Q_C \ge 0$, and the two mixed cap intersections required by Proposition 6.8 hold.*

Proof. The T chart proves the four T-cell core signs by nonnegative Bernstein coefficients. On each L chart the positive Bernstein certificates for E_F and R_F prove the four L-cell signs. Since $D \ge 0$, (78) gives $Q_A, Q_C \ge 0$ for $z \ge 0$; reflection supplies $z \le 0$. Lemma C.2 then gives the two cap intersections. □

Remark C.4 (Reproduction). The authenticated data and the two verifiers constitute the exact electronic certificate used by the proof. Reproduce it from its package directory by executing

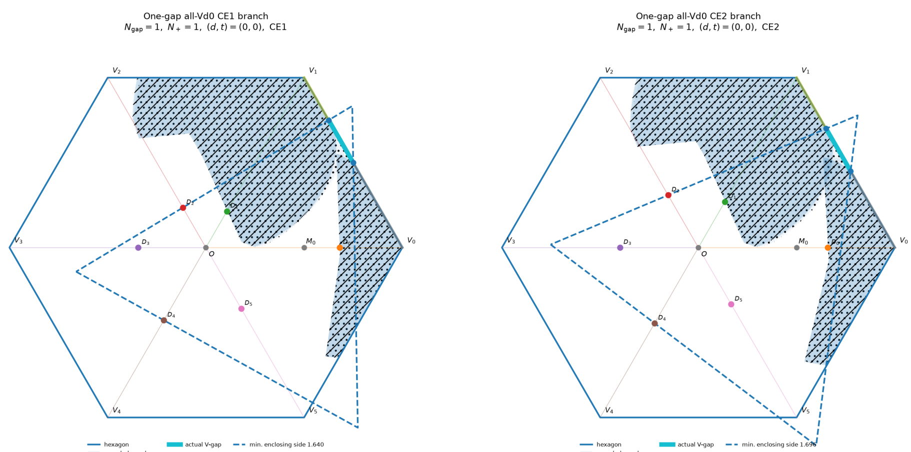
```
verify_mixed_overlap_core_derivation.py
verify_global_core_positivity.py
```

Both commands use exact arithmetic and must reproduce the transcript digest above and the asserted sign checks.



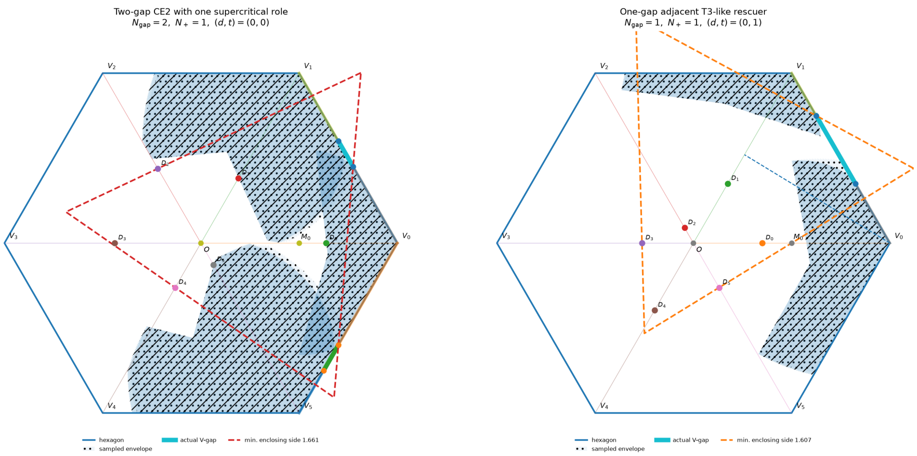
(a) One gap with two T3-like roles. **(b)** Two-gap CE2 T3-like branch.

FIGURE 26. Trace-exact AB-envelope snapshots: One gap with two T3-like roles; Two-gap CE2 T3-like branch.



(a) One-gap all-Vd0 CE1 branch. **(b)** One-gap all-Vd0 CE2 branch.

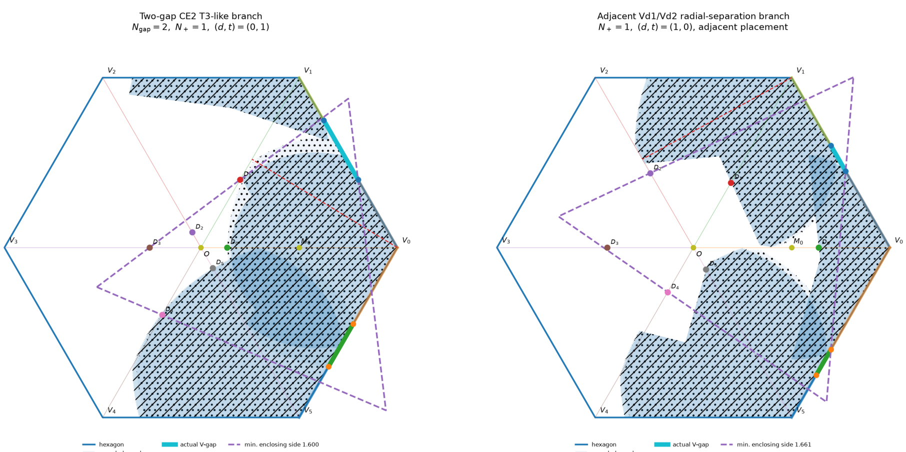
FIGURE 27. Trace-exact AB-envelope snapshots: One-gap all-Vd0 CE1 branch; One-gap all-Vd0 CE2 branch.



(a) Two-gap CE2 with one supercritical role.

(b) One-gap adjacent T3-like rescuer.

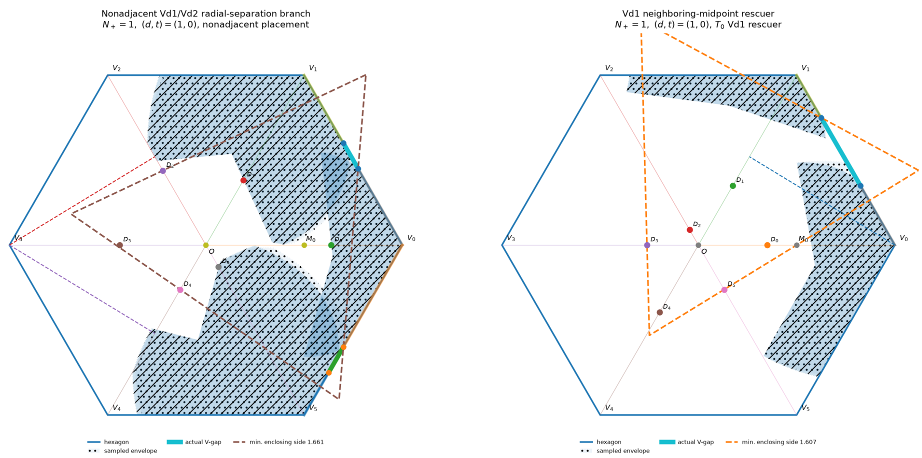
FIGURE 28. Trace-exact AB-envelope snapshots: Two-gap CE2 with one supercritical role; One-gap adjacent T3-like rescuer.



(a) Two-gap CE2 T3-like branch.

(b) Adjacent Vd1/Vd2 radial-separation branch.

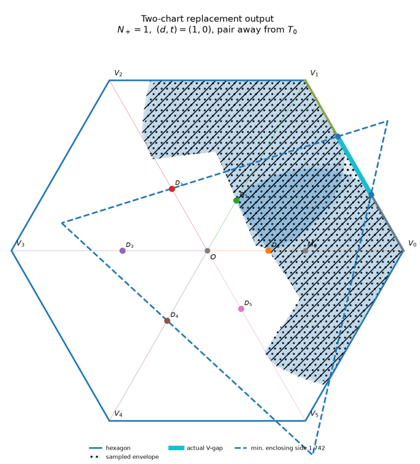
FIGURE 29. Trace-exact AB-envelope snapshots: Two-gap CE2 T3-like branch; Adjacent Vd1/Vd2 radial-separation branch.



(a) Nonadjacent Vd1/Vd2 radial-separation branch.

(b) Vd1 neighboring-midpoint rescuer.

FIGURE 30. Trace-exact AB-envelope snapshots: Nonadjacent Vd1/Vd2 radial-separation branch; Vd1 neighboring-midpoint rescuer.



(a) Two-chart replacement output.

FIGURE 31. Trace-exact AB-envelope snapshots: Two-chart replacement output.